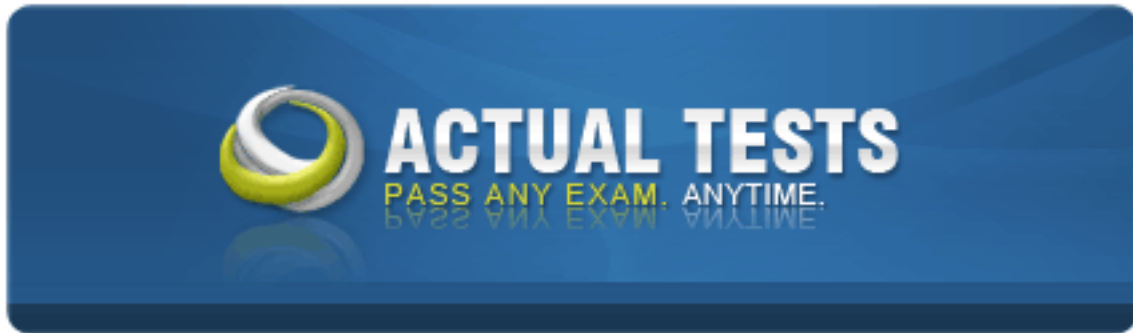


Cisco 642-618



Deploying Cisco ASA Firewall Solutions (FIREWALL)

V2.0

Version: 6.0

QUESTION NO: 1

On the Cisco ASA, tcp-map can be applied to a traffic class using which MPF CLI configuration command?

- A. inspect
- B. sysopt connection
- C. tcp-options
- D. parameters
- E. set connection advanced-options

Answer: E

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_tcpnorm.html

QUESTION NO: 2

By default, which traffic can pass through a Cisco ASA that is operating in transparent mode without explicitly allowing it using an ACL?

- A. ARP
- B. BPDU
- C. CDP
- D. OSPF multicasts
- E. DHCP

Answer: A

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/fwmode.html>

QUESTION NO: 3

When enabling a Cisco ASA to send syslog messages to a syslog server, which syslog level will produce the most messages?

- A. notifications
- B. informational
- C. alerts

- D. emergencies
- E. errors
- F. debugging

Answer: F

Explanation:

QUESTION NO: 4

Refer to the exhibit.

```
ASA-5510# show conn
54764 in use, 54764 most used
TCP outside 172.16.1.118:26093 inside 10.1.1.50:80, idle 0:00:23, bytes 0, flags aB
TCP outside 172.16.5.19:23598 inside 10.1.1.50:80, idle 0:00:13, bytes 0, flags aB
TCP outside 192.168.1.202:32729 inside 10.1.1.50:80, idle 0:00:25, bytes 0, flags aB
TCP outside 192.168.2.20:56481 inside 10.1.1.50:80, idle 0:00:29, bytes 0, flags aB
TCP outside 192.168.3.205:18073 inside 10.1.1.50:80, idle 0:00:02, bytes 0, flags aB
TCP outside 172.16.2.63:51503 inside 10.1.1.50:80, idle 0:00:03, bytes 0, flags aB
TCP outside 172.16.18.60:47733 inside 10.1.1.50:80, idle 0:00:27, bytes 0, flags aB
TCP outside 192.168.1.202:20773 inside 10.1.1.50:80, idle 0:00:02, bytes 0, flags aB
TCP outside 192.168.4.192:23112 inside 10.1.1.50:80, idle 0:00:06, bytes 0, flags aB
TCP outside 172.16.25.60:47733 inside 10.1.1.50:80, idle 0:00:27, bytes 0, flags aB
!<output omitted>
```

Flags: A - awaiting inside ACK to SYN, a - awaiting outside ACK to SYN,
 B - initial SYN from outside, b - TCP state-bypass or nailed, C - CTIOBE media,
 D - DNS, d - dump, E - outside back connection, F - outside FIN, f - inside FIN,
 G - group, g - MGCP, H - H.323, h - H.225.0, I - inbound data,
 i - incomplete, J - GTP, j - GTP data, K - GTP t3-response
 k - Skinny media, M - SMTP data, m - SIP media, n - GUP
 O - outbound data, P - inside back connection, p - Phone-proxy TFTP connection,
 q - SQL*Net data, R - outside acknowledged FIN,
 R - UDP SUNRPC, r - inside acknowledged FIN, S - awaiting inside SYN,
 s - awaiting outside SYN, T - SIP, t - SIP transient, U - up,
 V - VPN orphan, W - WAAS,
 X - inspected by service module

What can be determined about the connection status?

- A. The output is showing normal activity to the inside 10.1.1.50 web server.
- B. Many HTTP connections to the 10.1.1.50 web server have successfully completed the three-way TCP handshake.
- C. Many embryonic connections are made from random sources to the 10.1.1.50 web server.
- D. The 10.1.1.50 host is triggering SYN flood attacks against random hosts on the outside.
- E. The 10.1.1.50 web server is terminating all the incoming HTTP connections.

Answer: C

Explanation:

QUESTION NO: 5

What mechanism is used on the Cisco ASA to map IP addresses to domain names that are contained in the botnet traffic filter dynamic database or local blacklist?

- A. HTTP inspection
- B. DNS inspection and snooping
- C. WebACL
- D. dynamic botnet database fetches (updates)
- E. static blacklist
- F. static whitelist

Answer: B

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_botnet.html

QUESTION NO: 6

Refer to the exhibit.

```
class-map http
  match port tcp eq 21
class-map ftp
  match port tcp eq 21
policy-map test
  class http
    inspect http
  class ftp
    inspect ftp
```

Which statement about the policy map named test is true?

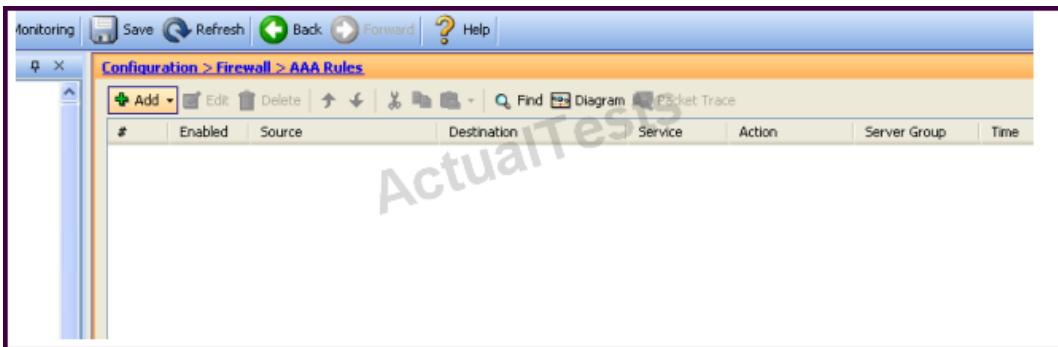
- A. Only HTTP inspection will be applied to the TCP port 21 traffic.
- B. Only FTP inspection will be applied to the TCP port 21 traffic.
- C. both HTTP and FTP inspections will be applied to the TCP port 21 traffic.
- D. No inspection will be applied to the TCP port 21 traffic, because the http class map configuration conflicts with the ftp class map.
- E. All FTP traffic will be denied, because the FTP traffic will fail the HTTP inspection.

Answer: B

Explanation:

QUESTION NO: 7

Refer to the exhibit.



Which Cisco ASA feature can be configured using this Cisco ASDM screen?

- A. Cisco ASA command authorization using TACACS+
- B. AAA accounting to track serial, ssh, and telnet connections to the Cisco ASA
- C. Exec Shell access authorization using AAA
- D. cut-thru proxy
- E. AAA authentication policy for Cisco ASDM access

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/asdm52/user/guide/aaarules.html>

And from

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/access_idfw.html#wp1324095

Configuring Cut-through Proxy Authentication

In an enterprise, some users log onto the network by using other authentication mechanisms, such as authenticating with a web portal (cut-through proxy) or by using a VPN. For example, users with a Macintosh and Linux client might log in a web portal (cut-through proxy) or by using a VPN. Therefore, you must configure the Identity Firewall to allow these types of authentication in connection with identity-based access policies.

The ASA designates users logging in through a web portal (cut-through proxy) as belonging to the

ActiveDirectory domain with which they authenticated. The ASA designates users logging in through a VPN as belonging to the LOCAL domain unless the VPN is authenticated by LDAP with Active Directory, then the Identity Firewall can associate the users with their Active Directory domain. The ASA reports users logging in through VPN authentication or a web portal (cut-through proxy) to the AD Agent, which distributes the user information to all registered ASA devices. Users can log in by using HTTP/HTTPS, FTP, Telnet, or SSH. When users log in with these authentication methods, the following guidelines apply:

- For HTTP/HTTPS traffic, an authentication window appears for unauthenticated users.
- For Telnet and FTP traffic, users must log in through the cut-through proxy and again to Telnet and FTP server.
- A user can specify an Active Directory domain while providing login credentials (in the format domain\username). The ASA automatically selects the associated AAA server group for the specified domain.
- If a user specifies an Active Directory domain while providing login credentials (in the format domain\username), the ASA parses the domain and uses it to select an authentication server from the AAA servers configured for the Identity Firewall. Only the username is passed to the AAA server.
- If the backslash (\) delimiter is not found in the log in credentials, the ASA does not parse a domain and authentication is conducted with the AAA server that corresponds to default domain configured for the Identity Firewall.
- If a default domain or a server group is not configured for that default domain, the ASA rejects the authentication.
- If the domain is not specified, the ASA selects the AAA server group for the default domain that is configured for the Identity Firewall.

QUESTION NO: 8

Refer to the exhibit.

```
failover
failover lan unit primary
failover lan interface MYFAILOVER GigabitEthernet0/2
failover interface ip MYFAILOVER 172.16.5.1 255.255.255.0 standby 172.16.5.10
failover link MYFAILOVER GigabitEthernet0/2
failover key cisco123
failover group 1
primary
preempt
failover group 2
secondary
preempt
```

Which command enables the stateful failover option?

- A. failover link MYFAILOVER GigabitEthernet0/2
- B. failover lan interface MYFAILOVER GigabitEthernet0/2
- C. failover interface ip MYFAILOVER 172.16.5.1 255.255.255.0 standby 172.16.5.10
- D. preempt
- E. failover group 1 primary
- F. failover lan unit primary

Answer: A

Explanation:

http://www.cisco.com/en/US/products/hw/vpndevc/ps2030/products_configuration_example09186a00807dac5f.shtml

QUESTION NO: 9

In which type of environment is the Cisco ASA MPF set connection advanced-options tcp-statebypass option the most useful?

- A. SIP proxy
- B. WCCP
- C. BGP peering through the Cisco ASA
- D. asymmetric traffic flow
- E. transparent firewall

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_tcpstatebypass.html

QUESTION NO: 10

Refer to the exhibit.

```
class-map type inspect ftp match-all ftp-cmd
  match request-command put
policy-map type inspect ftp ftp-insp
  class ftp-cmd
  reset
access-list ftp-acl extended permit tcp any any eq ftp
class-map ftp-cm
  match access-list ftp-acl
policy-map ftp-pm
  class ftp-cm
  inspect ftp strict ftp-insp
service-policy ftp-pm interface outside
```

Which statement about the MPF configuration is true?

- A. Any non-RFC complaint FTP traffic will go through additional deep FTP packet inspections.
- B. FTP traffic must conform to the FTP RFC, and the FTP connection will be dropped if the PUT command is used.
- C. Deep FTP packet inspections will be performed on all TCP inbound and outbound traffic on the outside interface.
- D. The ftp-pm policy-map type should be type inspect.
- E. Due to a configuration error, all FTP connections through the outside interface will not be permitted.

Answer: B

Explanation:

QUESTION NO: 11

Refer to the exhibit.

```
ASA# show local-host 10.1.1.99
```

```
Interface inside: 250 active, 250 maximum active, 0 denied
```

```
local host: <10.1.1.99>,
```

```
TCP connection count/limit = 146608/unlimited
```

```
TCP embryonic count = 146606
```

```
UDP connection count/limit = 0/unlimited
```

```
Xlate(s):
```

```
Global 209.165.201.21 Local 10.1.1.99
```

```
Conn(s):
```

```
TCP out 10.101.32.157:135 in 10.1.1.99:34580 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.103.108.191:135 in 10.1.1.99:8688 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.100.205.160:135 in 10.1.1.99:7774 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.101.182.19:135 in 10.1.1.99:39193 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.102.218.45:135 in 10.1.1.99:16462 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.100.21.120:135 in 10.1.1.99:30322 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.101.25.195:135 in 10.1.1.99:41116 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.103.17.219:135 in 10.1.1.99:59163 idle 0:01:43 Bytes 0 flags saA
```

```
TCP out 10.102.201.141:135 in 10.1.1.99:2978 idle 0:01:43 Bytes 0 flags saA
```

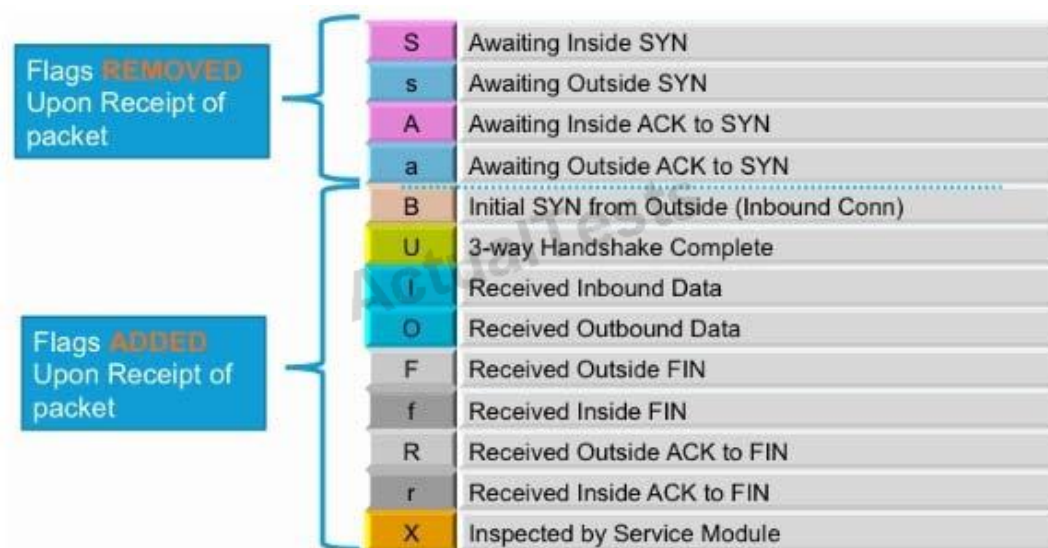
```
! <output omitted>
```

What is a reasonable conclusion?

- A. The maximum number of TCP connections that the 10.1.1.99 host can establish will be 146608.
- B. All the connections from the 10.1.1.99 have completed the TCP three-way handshake.
- C. The 10.1.1.99 hosts are generating a vast number of outgoing connections, probably due to a virus.
- D. The 10.1.1.99 host on the inside is under a SYN flood attack.
- E. The 10.1.1.99 host operations on the inside look normal.

Answer: C

Explanation:



C:\Documents and Settings\user-nw\Desktop\1.JPG

Flags: A - awaiting inside ACK to SYN, a - awaiting outside ACK to SYN,
 B - initial SYN from outside, b - TCP state-bypass or nailed, C - CTIOBE media,
 D - DNS, d - dump, E - outside back connection, F - outside FIN, f - inside FIN,
 G - group, g - MGCP, H - H.323, h - H.225.0, I - inbound data,
 i - incomplete, J - GTP, j - GTP data, K - GTP t3-response
 k - Skinny media, M - SMTP data, m - SIP media, n - GUP
 O - outbound data, P - inside back connection, p - Phone-proxy TFTP connection,
 q - SQL*Net data, R - outside acknowledged FIN,
 R - UDP SUNRPC, r - inside acknowledged FIN, S - awaiting inside SYN,
 s - awaiting outside SYN, T - SIP, t - SIP transient, U - up,
 V - VPN orphan, W - WAAS,
 X - inspected by service module

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 12

By default, how does the Cisco ASA authenticate itself to the Cisco ASDM users?

- A. The administrator validates the Cisco ASA by examining the factory built-in identity certificate thumbprint of the Cisco ASA.
- B. The Cisco ASA automatically creates and uses a persistent self-signed X.509 certificate to authenticate itself to the administrator.
- C. The Cisco ASA automatically creates a self-signed X.509 certificate on each reboot to authenticate itself to the administrator.
- D. The Cisco ASA and the administrator use a mutual password to authenticate each other.
- E. The Cisco ASA authenticates itself to the administrator using a one-time password.

Answer: C

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_configuration_example09186a00808efbd2.shtml

QUESTION NO: 13

When will a Cisco ASA that is operating in transparent firewall mode perform a routing table lookup instead of a MAC address table lookup to determine the outgoing interface of a packet?

- A. if multiple context mode is configured
- B. if the destination MAC address is unknown
- C. if the destination is more than a hop away from the Cisco ASA

- D. if NAT is configured
- E. if dynamic ARP inspection is configured

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/fwmode.html#wp1214750>

MAC Address vs. Route Lookups

When the ASA runs in transparent mode, the outgoing interface of a packet is determined by performing a MAC address lookup instead of a route lookup.

Route lookups, however, are necessary for the following traffic types:

- Traffic originating on the ASA—For example, if your syslog server is located on a remote network, you must use a static route so the ASA can reach that subnet.
- Voice over IP (VoIP) traffic with inspection enabled, and the endpoint is at least one hop away from the ASA—
For example, if you use the transparent firewall between a CCM and an H.323 gateway, and there is a router between the transparent firewall and the H.323 gateway, then you need to add a static route on the ASA for the H.323 gateway for successful call completion.
- VoIP or DNS traffic with NAT and inspection enabled—To successfully translate the IP address inside VoIP and DNS packets, the ASA needs to perform a route lookup. Unless the host is on a directly-connected network, then you need to add a static route on the ASA for the real host address that is embedded in the packet.

QUESTION NO: 14

Which flag not shown in the output of the show conn command is used to indicate that an initial SYN packet is from the outside (lower security-level interface)?

```
ASA5520# show conn
29 in use, 63 most used
TCP out 10.10.49.10:23 in 10.1.1.15:1026 idle 0:00:22 bytes 1774 flags UIO
<output omitted>
```

- A. B
- B. D

- C. b
- D. A
- E. a
- F. i
- G. l
- H. O

Answer: A

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_tech_note09186a0080bcad00.shtml

Originally answer from dump was A meaning B or initial SYN from outside but B is not shown in the output.

The question used to read "... Which flag shown in the output of the show conn command is used to indicate that an initial SYN packet is from the outside (lower security-level interface)?

TCP Connection Flag Values

Flags REMOVED Upon Receipt of packet	S	Awaiting Inside SYN
	s	Awaiting Outside SYN
	A	Awaiting Inside ACK to SYN
	a	Awaiting Outside ACK to SYN
Flags ADDED Upon Receipt of packet	B	Initial SYN from Outside (Inbound Conn)
	U	3-way Handshake Complete
	I	Received Inbound Data
	O	Received Outbound Data
	F	Received Outside FIN
	f	Received Inside FIN
	R	Received Outside ACK to FIN
	r	Received Inside ACK to FIN
	X	Inspected by Service Module

C:\Documents and Settings\user-nwz\Desktop\1.JPG

Flags: A - awaiting inside ACK to SYN, a - awaiting outside ACK to SYN,
 B - initial SYN from outside, b - TCP state-bypass or nailed, C - CTIOBE media,
 D - DNS, d - dump, E - outside back connection, F - outside FIN, f - inside FIN,
 G - group, g - MGCP, H - H.323, h - H.225.0, I - inbound data,
 i - incomplete, J - GTP, j - GTP data, K - GTP t3-response
 k - Skinny media, M - SMTP data, m - SIP media, n - GUP
 O - outbound data, P - inside back connection, p - Phone-proxy TFTP connection,
 q - SQL*Net data, R - outside acknowledged FIN,
 R - UDP SUNRPC, r - inside acknowledged FIN, S - awaiting inside SYN,
 s - awaiting outside SYN, T - SIP, t - SIP transient, U - up,
 V - VPN orphan, W - WAAS,
 X - inspected by service module

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 15

Which statement about the default ACL logging behavior of the Cisco ASA is true?

- A.** The Cisco ASA generates system message 106023 for each denied packet when a deny ACE is configured.
- B.** The Cisco ASA generates system message 106023 for each packet that matched an ACE.
- C.** The Cisco ASA generates system message 106100 only for the first packet that matched an ACE.
- D.** The Cisco ASA generates system message 106100 for each packet that matched an ACE.
- E.** No ACL logging is enabled by default.

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/acl_logging.html#wp1076483

To configure logging for an ACE, enter the following command:

Command	Purpose
<pre>access-list access_list_name [extended] {deny permit}...[log [[level] [interval secs] disable default]]</pre> Example: <pre>hostname(config)# access-list outside-acl permit ip host 1.1.1.1 any log 7 interval 600</pre>	Configures logging for an ACE. The access-list <i>access_list_name</i> syntax specifies the access list for which you want to configure logging. The extended option adds an ACE. The deny keyword denies a packet if the conditions are matched. Some features do not allow deny ACEs, such as NAT. (See the command documentation for each feature that uses an access list for more information.) The permit keyword permits a packet if the conditions are matched. If you enter the log option without any arguments, you enable system log message 106100 at the default level (6) and for the default interval (300 seconds). See the following options: • level—A severity level between 0 and 7. The default is 6. • interval secs—The time interval in seconds between system messages, from 1 to 600. The default is 300. This value is also used as the timeout value for deleting an inactive flow. • disable—Disables all access list logging. • default—Enables logging to message 106023. This setting is the same as having no log option. (See the access-list command in the <i>Cisco Security Appliance Command Reference</i> for more information about

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 16

Which Cisco ASA feature enables the ASA to do these two things? 1) Act as a proxy for the server and generate a SYN-ACK response to the client SYN request. 2) When the Cisco ASA receives an ACK back from the client, the Cisco ASA authenticates the client and allows the connection to the server.

- A. TCP normalizer
- B. TCP state bypass
- C. TCP intercept
- D. basic threat detection

- E. advanced threat detection
- F. botnet traffic filter

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/conns_connlimits.html#wp1080734

TCP Intercept and Limiting Embryonic Connections

Limiting the number of embryonic connections protects you from a DoS attack. The ASA uses the per-clientlimits and the embryonic connection limit to trigger TCP Intercept, which protects inside systems from a DoSattack perpetrated by flooding an interface with TCP SYN packets. An embryonic connection is a connectionrequest that has not finished the necessary handshake between source and destination. TCP Intercept usesthe SYN cookies algorithm to prevent TCP SYN-flooding attacks. A SYN-flooding attack consists of a series ofSYN packets usually originating from spoofed IP addresses. The constant flood of SYN packets keeps theserver SYN queue full, which prevents it from servicing connection requests. When the embryonic connectionthreshold of a connection is crossed, the ASA acts as a proxy for the server and generates a SYN-ACKresponse to the client SYN request. When the ASA receives an ACK back from the client, it can thenauthenticate the client and allow the connection to the server.

QUESTION NO: 17

Which option is not supported when the Cisco ASA is operating in transparent mode and also is using multiple security contexts?

- A. NAT
- B. shared interface
- C. security context resource management
- D. Layer 7 inspections
- E. failover

Answer: B

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/contexts.html>

Unique Interfaces

If only one context is associated with the ingress interface, the ASA classifies the packet into that context. Intransparent firewall mode, unique interfaces for contexts are required, so this method is used to classifypackets at all times.

QUESTION NO: 18

Refer to the exhibit.

Context Name	Class	Interfaces	URL
admin	default	GigabitEthernet0/0 , GigabitEthernet0/1	disk0:/admin.cfg
*CTX	default	GigabitEthernet0/0 , GigabitEthernet0/2	disk0:/CTX.cfg
Total active Security Contexts: 2			

What does the * next to the CTX security context indicate?

- A. The CTX context is the active context on the Cisco ASA.
- B. The CTX context is the standby context on the Cisco ASA.
- C. The CTX context contains the system configurations.
- D. The CTX context has the admin role.

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa70/configuration/guide/mngcntxt.html#wp1107587>

Context Configurations

The security appliance includes a configuration for each context that identifies the security policy, interfaces, and almost all the options you can configure on a standalone device. You can store context configurations on the internal Flash memory or the external Flash memory card, or you can download them from a TFTP, FTP, or HTTP(S) server.

System Configuration

The system administrator adds and manages contexts by configuring each context configuration location, allocated interfaces, and other context operating parameters in the system configuration, which, like a singlemode configuration, is the startup configuration. The system configuration identifies basic settings for the security appliance. The system configuration does not include any network interfaces or network settings for itself; rather, when the system needs to access network resources (such as downloading the contexts from the server), it uses one of the contexts that is designated as the admin context. The system configuration does include a specialized failover interface for failover traffic only.

Admin Context Configuration The admin context is just like any other context, except that when a user logs in to the admin context, then that user has system administrator rights and can access the system and all other contexts. The admin context is not restricted in any way, and can be used as a regular context. However, because logging into the admin context grants you administrator privileges over all contexts, you might need to restrict access to the admin context to appropriate users. The admin context must reside on Flash memory, and not remotely.

If your system is already in multiple context mode, or if you convert from single mode, the admin context is created automatically as a file on the internal Flash memory called admin.cfg. This context is named "admin." If you do not want to use admin.cfg as the admin context, you can change the admin context.

QUESTION NO: 19

Which Cisco ASA feature is implemented by the ip verify reverse-path interface interface_name command?

- A. uRPF
- B. TCP intercept
- C. botnet traffic filter
- D. scanning threat detection
- E. IPS (IP audit)

Answer: A

Explanation:

<https://supportforums.cisco.com/thread/2070206>

Unicast RPF is disabled by default on the ASA unless you explicitly enable it on an interface. Since it is disabled by default on all interfaces, you will not see them in the configuration. Once you enable RPF for a specific interface, you will see that enabled in the configuration.

For example:

If you have 3 interfaces: inside, dmz and outside, and you enable it for inside only, then when you perform "sh run ip verify reverse-path", you will see the following:

ip verify reverse-path interface inside

OR/ you will see that in the running configuration as well. The other 2 interfaces that you haven't explicitly enabled will still be disabled by default, and will not show under the configuration.

QUESTION NO: 20

In one custom dynamic application, the inside client connects to an outside server using TCP port 4444 and negotiates return client traffic in the port range of 5000 to 5500. The server then starts streaming UDP data to the client on the negotiated port in the specified range. Which Cisco ASA feature or command supports this custom dynamic application?

- A. TCP normalizer
- B. TCP intercept
- C. ip verify command
- D. established command
- E. tcp-map and tcp-options commands
- F. set connection advanced-options command

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/int5505.html>

Establishedcommand—This command allows return connections from a lower security host to a higher security host if there is already an established connection from the higher level host to the lower level host.

For same security interfaces, you can configure established commands for both directions.

QUESTION NO: 21

A Cisco ASA is operating in transparent firewall mode, but the MAC address table of the Cisco ASA is always empty, which causes connectivity issues. What should you verify to troubleshoot this issue?

- A. if ARP inspection has been disabled
- B. if MAC learning has been disabled
- C. if NAT has been disabled
- D. if ARP traffic is explicitly allowed using EtherType ACL
- E. if BPDU traffic is explicitly allowed using EtherType ACL

Answer: B

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/fwmode.html#wp122483>
6

QUESTION NO: 22

When active/active failover is implemented on the Cisco ASA, how many failover groups are supported on the Cisco ASA?

- A. 1
- B. 2
- C. 1 failover group per configured security context
- D. 2 failover groups per configured security context

Answer: B

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_configuration_example09186a0080834058.shtml#act1

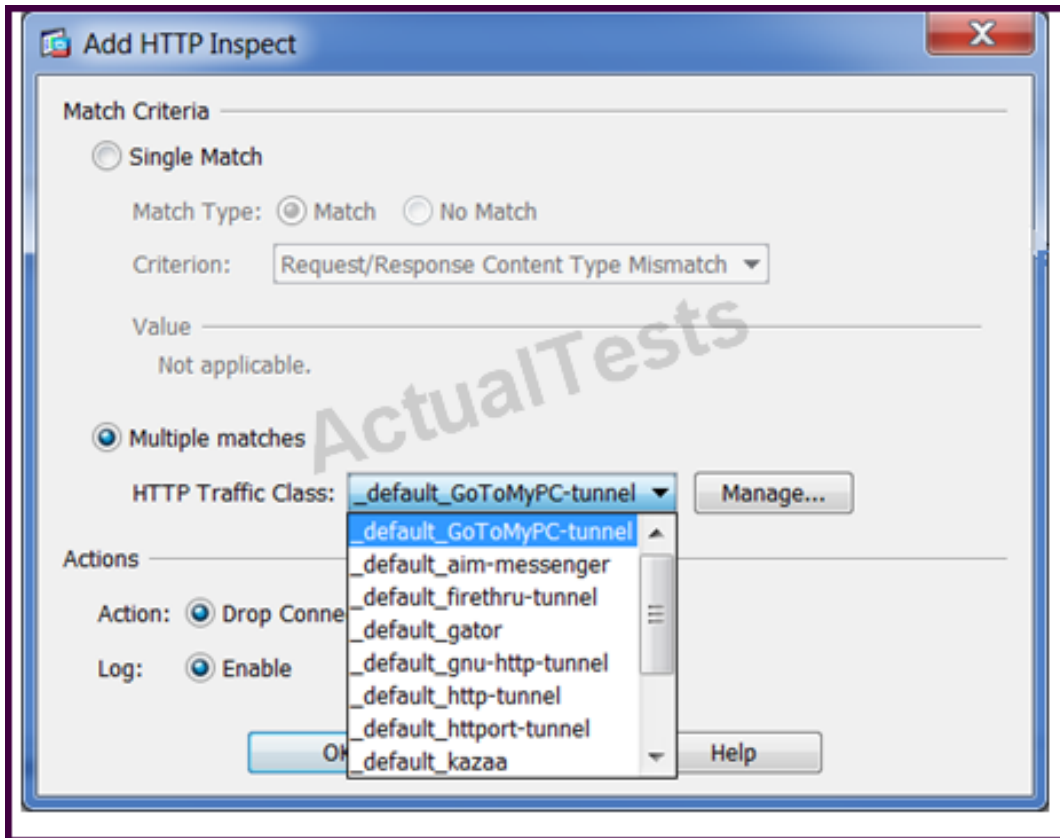
Active/Active Failover Overview

Active/Active failover is only available to security appliances in multiple context mode. In an Active/Active failover configuration, both security appliances can pass network traffic. In Active/Active failover, you divide the security contexts on the security appliance into failover groups. A failover group is simply a logical group of one or more security contexts. You can create a maximum of two failover groups on the security appliance. The admin context is always a member of failover group 1. Any unassigned security contexts are also members of failover group 1 by default. The failover group forms the base unit for failover in Active/Active failover. Interface failure monitoring, failover, and active/standby status are all attributes of a failover group rather than the unit. When an active failover group fails, it changes to the standby state while the standby failover group becomes active. The interfaces in the failover group that becomes active assume the MAC and IP addresses of the interfaces in the failover group that failed. The interfaces in the failover group that is now in the standby state take over the standby MAC and IP addresses.

Note: A failover group failing on a unit does not mean that the unit has failed. The unit may still have another failover group passing traffic on it.

QUESTION NO: 23

Refer to the exhibit.



What is the resulting CLI command?

- A. match request uri regex _default_GoToMyPC-tunnel
drop-connection log
- B. match regex _default_GoToMyPC-tunnel
drop-connection log
- C. class _default_GoToMyPC-tunnel
drop-connection log
- D. match class-map _default_GoToMyPC-tunnel
drop-connection log

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/inspect_basic.html

Step 6 To apply actions to matching traffic, perform the following steps.

- a. Specify the traffic on which you want to perform actions using one of the following methods:
Specify the DNS class map that you created in Step 3 by entering the following command:

```
hostname(config-pmap)# class class_map_name
```

```
hostname(config-pmap-c)#
```

Specify traffic directly in the policy map using one of the match commands described in Step 3. If you use amatch not command, then any traffic that does not match the criterion in the match not command has the action applied.

- b. Specify the action you want to perform on the matching traffic by entering the following

command:

```
hostname(config-pmap-c)# {[drop [send-protocol-error] |  
drop-connection [send-protocol-error]| mask | reset] [log] | rate-limit  
message_rate}
```

Not all options are available for each match or class command. See the CLI help or the Cisco ASA 5500 Series Command Reference for the exact options available.

The drop keyword drops all packets that match.

The send-protocol-error keyword sends a protocol error message.

The drop-connection keyword drops the packet and closes the connection.

The mask keyword masks out the matching portion of the packet.

The reset keyword drops the packet, closes the connection, and sends a TCP reset to the server and/or client.

The log keyword, which you can use alone or with one of the other keywords, sends a system log message

The rate-limit message_rate argument limits the rate of messages.

QUESTION NO: 24

Which Cisco ASA CLI command is used to enable HTTPS (Cisco ASDM) access from any inside host on the 10.1.16.0/20 subnet?

- A. http 10.1.16.0 0.0.0.0 inside
- B. http 10.1.16.0 0.0.15.255 inside
- C. http 10.1.16.0 255.255.240.0 inside
- D. http 10.1.16.0 255.255.255.255

Answer: C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa71/configuration/guide/mgaccess.html#wp1047288>

Allowing HTTPS Access for ASDM

To use ASDM, you need to enable the HTTPS server, and allow HTTPS connections to the security appliance.

All of these tasks are completed if you use the setup command. This section describes how to manually configure ASDM access.

The security appliance allows a maximum of 5 concurrent ASDM instances per context, if available, with a maximum of 32 ASDM instances between all contexts.

Note WebVPN and ASDM administration cannot be enabled on the same interface. If you enable WebVPN on an interface, then that interface cannot be used for ASDM.

To configure ASDM access, follow these steps:

Step 1 To identify the IP addresses from which the security appliance accepts HTTPS connections, enter the following command for each address or subnet:

```
hostname(config)# http source_IP_address mask source_interface
```

Step 2 To enable the HTTPS server, enter the following command:

```
hostname(config)# http server enable
```

QUESTION NO: 25

What is the first configuration step when using Cisco ASDM to configure a new Layer 3/4 inspection policy on the Cisco ASA?

- A. Create a new class map.
- B. Create a new policy map and apply actions to the traffic classes.
- C. Create a new service policy rule.
- D. Create the ACLs to be referenced by any of the new class maps.
- E. Disable the default global inspection policy.
- F. Create a new firewall access rule.

Answer: C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa80/asdm60/user/guide/svcrules.html#wp1161995>

Default Global Policy

By default, the configuration includes a policy that matches all default application inspection traffic and applies certain inspections to the traffic on all interfaces (a global policy). Not all inspections are enabled by default.

You can only apply one global policy, so if you want to alter the global policy, you need to either edit the default policy or disable it and apply a new one. (An interface policy overrides the global

policy.)

Service policies provide a consistent and flexible way to configure security appliance features. For example, you can use a service policy to create a timeout configuration that is specific to a particular TCP application, as opposed to one that applies to all TCP applications.

Configuring a service policy consists of adding one or more service policy rules per interface or for the global policy. For each rule, you identify the following elements:

1. Identify the interface to which you want to apply the rule, or identify the global policy.
2. Identify the traffic to which you want to apply actions. You can identify Layer 3 and 4 through traffic.
3. Apply actions to the traffic class. You can apply multiple actions for each traffic class.

QUESTION NO: 26

Which feature is not supported on the Cisco ASA 5505 with the Security Plus license?

- A. security contexts
- B. stateless active/standby failover
- C. transparent firewall
- D. threat detection
- E. traffic shaping

Answer: A

Explanation:

QUESTION NO: 27

Refer to the exhibit.

```
ASA# packet-tracer input inside tcp 10.0.0.1 1024 172.26.1.200 23

Phase: 1
Type: FLOW-LOOKUP
Subtype:
Result: ALLOW
Config:
Additional Information:
Found no matching flow, creating a new flow
Phase: 2
Type: ROUTE-LOOKUP
Subtype: input
Result: ALLOW
Config:
Additional Information:
in 0.0.0.0 0.0.0.0 outside
Phase: 3
Type: ACCESS-LIST
Subtype: log
Result: DROP
Config:
access-group DENY_OUT in interface inside
access-list DENY_OUT extended deny ip any any
Additional Information:
Result:
input-interface: inside
input-status: up
input-line-status: up
output-interface: outside
output-status: up
output-line-status: up
Action: drop
Drop-reason: (acl-drop) Flow is denied by configured rule
```

Which statement about the Telnet session from 10.0.0.1 to 172.26.1.200 is true?

- A. The Telnet session should be successful.
- B. The Telnet session should fail because the route lookup to the destination fails.
- C. The Telnet session should fail because the inside interface inbound access list will block it.
- D. The Telnet session should fail because no matching flow was found.
- E. The Telnet session should fail because inside NAT has not been configured.

Answer: C

Explanation:

QUESTION NO: 28

With Cisco ASA active/standby failover, by default, how many monitored interface failures will cause failover to occur?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Answer: A

Explanation:

Table 11-2 Failover Behavior for Active/Active Failover

Failure Event	Policy	Active Group Action	Standby Group Action	Notes
A unit experiences a power or software failure	Failover	Become standby Mark as failed	Become active Mark active as failed	When a unit in a failover pair fails, any active failover groups on that unit are marked as failed and become active on the peer unit.
Interface failure on active failover group above threshold	Failover	Mark active group as failed	Become active	None.
Interface failure on standby failover group above threshold	No failover	No action	Mark standby group as failed	When the standby failover group is marked as failed, then the active failover group will not attempt to fail over, even if the interface failure threshold is surpassed.
Formerly active failover group recovers	No failover	No action	No action	Unless configured with the preempt command, the failover groups remain active on their current unit.
Failover link failed at startup	No failover	Become active	Become active	If the failover link is down at startup, both failover groups on both units will become active.
Stateful Failover link failed	No failover	No action	No action	State information will become out of date, and sessions will be terminated if a failover occurs.
Failover link failed during operation	No failover	n/a	n/a	Each unit marks the failover interface as failed. You should restore the failover link as soon as possible because the unit cannot fail over to the standby unit while the failover link is down.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 29

Which statement about SNMP support on the Cisco ASA appliance is true?

- A.** The Cisco ASA appliance supports only SNMPv1 or SNMPv2c.
- B.** The Cisco ASA appliance supports read-only and read-write access.
- C.** The Cisco ASA appliance supports three built-in SNMPv3 groups in Cisco ASDM: Authentication and Encryption, Authentication Only, and No Authentication, No Encryption.
- D.** The Cisco ASA appliance can send SNMP traps to the network management station only using SNMPv2.

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/monitor_snmp.html#wp1042029

SNMP Version 3 Overview

SNMP Version 3 provides security enhancements that are not available in SNMP Version 1 or SNMP Version 2c. SNMP Versions 1 and 2c transmit data between the SNMP server and SNMP agent in clear text. SNMP Version 3 adds authentication and privacy options to secure protocol operations. In addition, this version controls access to the SNMP agent and MIB objects through the User-based Security Model (USM) and View-based Access Control Model (VACM). The ASA 5500 series ASAs also support the creation of SNMP groups and users, as well as hosts, which is required to enable transport authentication and encryption for secure SNMP communications.

Security Models

For configuration purposes, the authentication and privacy options are grouped together into security models.

Security models apply to users and groups, and are divided into the following three types:

- NoAuthPriv—No Authentication and No Privacy, which means that no security is applied to messages.
- AuthNoPriv—Authentication but No Privacy, which means that messages are authenticated.
- AuthPriv—Authentication and Privacy, which means that messages are authenticated and encrypted.

QUESTION NO: 30

Which command option/keyword in Cisco ASA 8.3 NAT configurations makes the NAT policy interface independent?

- A. interface
- B. all
- C. auto
- D. global
- E. any

Answer: E

Explanation:

<http://tunnelsup.com/2011/06/24/nat-for-cisco-asas-version-8-3/>

Using the “any” interface in the NAT statement

ASA 8.3 introduces the “any” interface when configuring NAT. For instance if you have a system on the DMZ that you wish to NAT not only to the outside interface, but to any interface you can use this command: `object network dmz-webserver host 192.168.1.23`
`nat (dmz,any) static 209.165.201.28`

This makes it so users on the inside can web to 209.165.201.28 and if traffic is routed to the firewall it will NAT it to the real IP in the DMZ.

QUESTION NO: 31

Refer to the exhibit.

! ASA 8.2

```
nat (inside) 1 192.168.1.0 255.255.255.0
global (outside) 1 interface
```

! ASA 8.3

```
object network inside-hosts
subnet 192.168.1.0 255.255.255.0
```

!

```
object network inside-hosts
```

_____ ?

Which corresponding Cisco ASA Software Version 8.3 command accomplishes the same Cisco ASA Software Version 8.2 NAT configuration?

- A. nat (any,any) dynamic interface
- B. nat (any,any) static interface
- C. nat (inside,outside) dynamic interface
- D. nat (inside,outside) static interface
- E. nat (outside,inside) dynamic interface
- F. nat (outside,inside) static interface

Answer: C

Explanation:

<http://tunnelsup.com/2011/06/24/nat-for-cisco-asas-version-8-3/>

Regular Dynamic PAT

To create a many-to-one NAT where the entire inside network is getting PAT'd to a single outside IP do the following.

Old 8.2 command:

```
nat (inside) 1 10.0.0.0 255.255.255.0 global (outside) 1 interface
```

New 8.3 equivalent command:

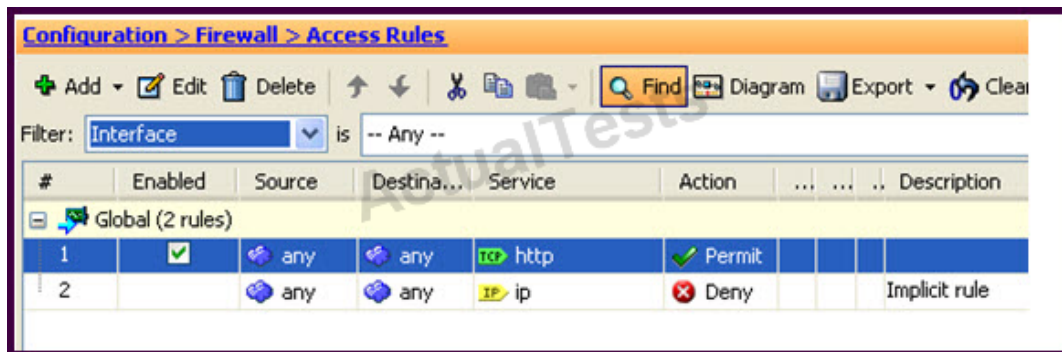
```
object network inside-net subnet 10.0.0.0 255.255.255.0
```

```
nat (inside,outside) dynamic interface
```

Note: the "interface" command is the 2nd interface in the nat statement, in this case the outside.

QUESTION NO: 32

Refer to the exhibit.



Which traffic is permitted on the inside interface without any interface ACLs configured?

- A. any IP traffic input to the inside interface
- B. any IP traffic input to the inside interface destined to any lower security level interfaces
- C. only HTTP traffic input to the inside interface
- D. only HTTP traffic output from the inside interface
- E. No input traffic is permitted on the inside interface.
- F. No output traffic is permitted on the inside interface.

Answer: C

Explanation:

QUESTION NO: 33

On Cisco ASA Software Version 8.4.1 and later, when you configure the Cisco ASA appliance in transparent firewall mode, how is the Cisco ASA management IP address configured?

- A. using the IP address global configuration command
- B. using the IP address GigabitEthernet 0/x interface configuration command
- C. using the IP address BVI x interface configuration command
- D. using the bridge-group global configuration command
- E. using the bridge-group GigabitEthernet 0/x interface configuration command
- F. using the bridge-group BVI x interface configuration command

Answer: C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa84/command/reference/i3.html#wp1898863>

Command History

Release	Modification
7.0(1)	For routed mode, this command was changed from a global configuration command to an interface configuration mode command.
8.4(1)	For transparent mode, bridge groups were introduced. You now set the IP address for the BVI , and not globally.

Usage Guidelines

In single context routed firewall mode, each interface address must be on a unique subnet. In multiple context mode, if this interface is on a shared interface, then each IP address must be unique but on the same subnet. If the interface is unique, this IP address can be used by other contexts if desired.

A transparent firewall does not participate in IP routing. The only IP configuration required for the ASA is to set the **BVI** address. This address is required because the ASA uses this address as the source address for traffic originating on the ASA, such as system messages or communications with AAA servers. You can also use this address for remote management access. This address must be on the same subnet as the upstream and downstream routers. For multiple context mode, set the management IP address within each context. For models that include a Management interface, you can also set an IP address for this interface for management purposes.

The standby IP address must be on the same subnet as the main IP address.

Examples

The following example sets the IP addresses and standby addresses of two interfaces:

```
hostname(config)# interface gigabitethernet0/2
hostname(config-if)# nameif inside
hostname(config-if)# security-level 100
hostname(config-if)# ip address 10.1.1.1 255.255.255.0 standby 10.1.1.2
hostname(config-if)# no shutdown
hostname(config-if)# interface gigabitethernet0/3
hostname(config-if)# nameif outside
hostname(config-if)# security-level 0
hostname(config-if)# ip address 10.1.2.1 255.255.255.0 standby 10.1.2.2
hostname(config-if)# no shutdown
```

C:\Documents and Settings\user-nw\Desktop\1.JPG

The following example sets the management address and standby address of bridge group 1:

```
hostname(config)# interface bvi 1
hostname(config-if)# ip address 10.1.3.1 255.255.255.0 standby 10.1.3.2
```

C:\Documents and Settings\user-nw\Desktop\1.JPG

QUESTION NO: 34

Which statement about Cisco ASA multicast routing support is true?

- A. The Cisco ASA appliance supports PIM dense mode, sparse mode, and BIDIR-PIM.
- B. The Cisco ASA appliance supports only stub multicast routing by forwarding IGMP messages from multicast receivers to the upstream multicast router.
- C. The Cisco ASA appliance supports DVMRP and PIM.
- D. The Cisco ASA appliance supports either stub multicast routing or PIM, but both cannot be enabled at the same time.
- E. The Cisco ASA appliance supports only IGMP v1.

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/route_multicast.html#wp1060775

Multicast routing is a bandwidth-conserving technology that reduces traffic by simultaneously delivering a singlestream of information to thousands of corporate recipients and homes. Applications that take advantage ofmulticast routing include videoconferencing, corporate communications, distance learning, and distribution ofsoftware, stock quotes, and news. Multicast routing protocols delivers source traffic to multiple receivers without adding any additional burden onthe source or the receivers while using the least network bandwidth of any competing technology. Multicastpackets are replicated in the network by Cisco routers enabled with Protocol Independent Multicast (PIM) andother supporting multicast protocols resulting in the most efficient delivery of data to multiple receivers possible.The ASA supports both stub multicast routing and PIM multicast routing. However, you cannot configure bothconcurrently on a single ASA.

QUESTION NO: 35

Which statement about access list operations on Cisco ASA Software Version 8.3 and later is true?

- A.** If the global and interface access lists are both configured, the global access list is matched first before the interface access lists.
- B.** Interface and global access lists can be applied in the input or output direction.
- C.** In the inbound access list on the outside interface that permits traffic to the inside interface, the destination IP address referenced is always the "mapped-ip" (translated) IP address of the inside host.
- D.** When adding an access list entry in the global access list using the Cisco ASDM Add Access Rule window, choosing "any" for Interface applies the access list entry globally.

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/access_rules.html#wp1083595

Using Global Access Rules

Global access rules allow you to apply a global rule to ingress traffic without the need to specify an interface to which the rule must be applied. Using global access rules provides the following benefits:

- When migrating to the ASA from a competitor appliance, you can maintain a global access rule policy instead of needing to apply an interface-specific policy on each interface.
 - Global access control policies are not replicated on each interface, so they save memory space.
 - Global access rules provides flexibility in defining a security policy. You do not need to specify which interface a packet comes in on, as long as it matches the source and destination IP addresses.
 - Global access rules use the same mtrie and stride tree as interface-specific access rules, so scalability and performance for global rules are the same as for interface-specific rules.
- You can configure global access rules in conjunction with interface access rules, in which case, the specific interface access rules are always processed before the general global access rules.

QUESTION NO: 36

Refer to the exhibit.

The screenshot shows the Cisco ASDM configuration page for NAT Rules. The breadcrumb trail is Configuration > Firewall > NAT Rules. Below the breadcrumb trail is a toolbar with buttons for Add, Edit, Delete, and other actions. The main table displays the NAT rule configuration:

Match Criteria: Original Packet						Action: Translated Packet		
#	Source Intf	Dest Intf	Source	Destination	Service	Source	Destination	Service
1	dmz	outside	any	any	any	200.200.200.200	outside (P)	-- Original --

Below the table, it says "Network Object" NAT (No rules).

Which Cisco ASA CLI nat command is generated based on this Cisco ASDM NAT configuration?

- A. nat (dmz, outside) 1 source static any any
- B. nat (dmz, outside) 1 source static any outside
- C. nat (dmz,outside) 1 source dynamic any interface
- D. nat (dmz, outside) 1 source dynamic any interface destination dynamic outside outside
- E. nat (dmz, outside) 1 source static any interface destination static any any
- F. nat (dmz, outside) 1 source dynamic any outside destination static any any

Answer: C

Explanation:

Pretty straight forward - like this example

<http://tunnelsup.com/2011/06/24/nat-for-cisco-asas-version-8-3/>

Regular Dynamic PAT

To create a many-to-one NAT where the entire inside network is getting PAT'd to a single outside IP do the following.

Old 8.2 command:

```
nat (inside) 1 10.0.0.0 255.255.255.0
```

```
global (outside) 1 interface
```

New 8.3 equivalent command:

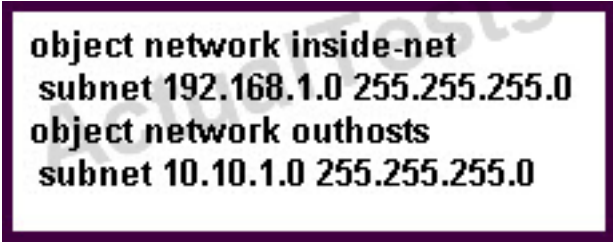
```
object network inside-netsubnet 10.0.0.0 255.255.255.0
```

```
nat (inside,outside) dynamic interface
```

Note: the "interface" command is the 2nd interface in the nat statement, in this case the outside.

QUESTION NO: 37

Refer to the exhibit.



```
object network inside-net  
  subnet 192.168.1.0 255.255.255.0  
object network outhosts  
  subnet 10.10.1.0 255.255.255.0
```

Which additional Cisco ASA Software Version 8.3 NAT configuration is needed to meet the following requirements?

When any host in the 192.168.1.0/24 subnet behind the inside interface accesses any destinations in the 10.10.1.0/24 subnet behind the outside interface, PAT them to the outside interface. Do not change the destination IP in the packet.

- A. nat (inside,outside) source static inside-net interface destination static outhosts outhosts
- B. nat (inside,outside) source dynamic inside-net interface destination static outhosts outhosts
- C. nat (outside,inside) source dynamic inside-net interface destination static outhosts outhosts
- D. nat (outside,inside) source static inside-net interface destination static outhosts outhosts
- E. nat (any, any) source dynamic inside-net interface destination static outhosts outhosts
- F. nat (any, any) source static inside-net interface destination static outhosts outhosts

Answer: B

Explanation:

QUESTION NO: 38

A Cisco ASA appliance running software version 8.4.1 has an active botnet traffic filter license with 1 month left on the time-based license. Which option describes the result if a new botnet traffic filter with a 1 year time-based license is activated also?

- A. The time-based license for the botnet traffic filter is valid only for another month.
- B. The time-based license for the botnet traffic filter is valid for another 12 months.
- C. The time-based license for the botnet traffic filter is valid for another 13 months.
- D. The new 1 year time-based license for the botnet traffic filter cannot be activated until the current botnet traffic filter license expires in a month.

Answer: C

Explanation:

http://www.cisco.com/en/US/prod/collateral/vpndevc/ps6032/ps6094/ps6120/product_bulletin_c25-593781.html

Time-based license stacking: Customers can extend time-based licenses such as Botnet Traffic Filter and SSLVPN Burst by applying multiple licenses.

QUESTION NO: 39

How many interfaces can a Cisco ASA bridge group support and how many bridge groups can a Cisco ASA appliance support?

- A. up to 2 interfaces per bridge group and up to 4 bridge groups per Cisco ASA appliance
- B. up to 2 interfaces per bridge group and up to 8 bridge groups per Cisco ASA appliance
- C. up to 4 interfaces per bridge group and up to 4 bridge groups per Cisco ASA appliance
- D. up to 4 interfaces per bridge group and up to 8 bridge groups per Cisco ASA appliance
- E. up to 8 interfaces per bridge group and up to 4 bridge groups per Cisco ASA appliance
- F. up to 8 interfaces per bridge group and up to 8 bridge groups per Cisco ASA appliance

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/asdm64/configuration_guide/interface_complete_transparent.html#wp1321327

Firewall Mode Guidelines

- You can configure up to 8 bridge groups in single mode or per context in multiple mode. Note that you must use at least 1 bridge group; data interfaces must belong to a bridge group.

- Each bridge group can include up to 4 interfaces.

QUESTION NO: 40

Which addresses are considered "ambiguous addresses" and are put on the greylist by the Cisco ASA botnet traffic filter feature?

- A. addresses that are unknown
- B. addresses that are on the greylist identified by the dynamic database
- C. addresses that are blacklisted by the dynamic database but also are identified by the static whitelist
- D. addresses that are associated with multiple domain names, but not all of these domain names are on the blacklist

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa83/asdm63/configuration_guide/protect_botnet.html

Botnet Traffic Filter Address Categories

Addresses monitored by the Botnet Traffic Filter include:

- Known malware addresses—These addresses are on the blacklist identified by the dynamic database and the static blacklist.
- Known allowed addresses—These addresses are on the whitelist. The whitelist is useful when an address is blacklisted by the dynamic database and also identified by the static whitelist.
- Ambiguous addresses—These addresses are associated with multiple domain names, but not all of these domain names are on the blacklist. These addresses are on the greylist.
- Unlisted addresses—These addresses are unknown, and not included on any list.

QUESTION NO: 41

For which purpose is the Cisco ASA CLI command `aaa authentication match` used?

- A. Enable authentication for SSH and Telnet connections to the Cisco ASA appliance.
- B. Enable authentication for console connections to the Cisco ASA appliance.
- C. Enable authentication for connections through the Cisco ASA appliance.
- D. Enable authentication for IPsec VPN connections to the Cisco ASA appliance.

- E. Enable authentication for SSL VPN connections to the Cisco ASA appliance.
- F. Enable authentication for Cisco ASDM connections to the Cisco ASA appliance.

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/access_fwaaa.html

Or

http://www.cisco.com/en/US/products/ps6120/products_tech_note09186a0080ba6110.shtml

To enable network access authentication, perform the following steps:

Step 1 Using the `aaa-server` command, identify your AAA servers. If you have already identified your AAA servers, continue to the next step.

Step 2 Using the `access-list` command, create an access list that identifies the source addresses and destination addresses of traffic you want to authenticate.

The permit ACEs mark matching traffic for authentication, while deny entries exclude matching traffic from authentication. Be sure to include the destination ports for either HTTP, HTTPS, Telnet, or FTP in the access list because the user must authenticate with one of these services before other services are allowed through the ASA.

Step 3 To configure authentication, enter the following command:

```
hostname(config)# aaa authentication match acl_name interface_name server_group
```

Where `acl_name` is the name of the access list you created; `interface_name` is the name of the interface as specified with the `nameif` command, and `server_group` is the AAA server group you created.

The following commands authenticate Telnet traffic from the outside interface to a particular server (209.165.201.5):

```
hostname(config)# aaa-server AuthInbound protocol tacacs+
hostname(config-aaa-server-group)# exit
hostname(config)# aaa-server AuthInbound (inside) host 10.1.1.1
hostname(config-aaa-server-host)# key TACPlusUauthKey
hostname(config-aaa-server-host)# exit
hostname(config)# access-list TELNET_AUTH extended permit tcp any host
209.165.201.5 eq telnet
```

QUESTION NO: 42

On the Cisco ASA Software Version 8.3 and later, which type of NAT configuration can be used to translate the source and destination IP addresses of the packet?

- A. auto NAT
- B. object NAT
- C. one-to-one NAT
- D. many-to-one NAT
- E. manual NAT
- F. identity NAT

Answer: E

Explanation:

<http://tunnelsup.com/2011/06/24/nat-for-cisco-asas-version-8-3/>

Manual NAT or Twice NAT or Policy NAT or Reverse NAT

The limitation that Auto NAT has is that it cannot take the destination into consideration when conducting its NAT. This also of course results in it not being able to alter the destination address either. To accomplish either of these tasks you must use "manual NAT".

All of these terms are identical: Manual NAT, Twice NAT, Policy NAT, Reverse NAT. Don't be confused by fancy mumbo jumbo.

http://www.cisco.com/en/US/docs/security/asa/asa83/asdm63/configuration_guide/nat_overview.html#wpixref64594

Main Differences Between Network Object NAT and Twice NAT

The main differences between these two NAT types are:

- How you define the real address.
 - Network object NAT—You define NAT as a parameter for a network object; the network object definition itself provides the real address. This method lets you easily add NAT to network objects. The objects can also be used in other parts of your configuration, for example, for access rules or even in twice NAT rules.
 - Twice NAT—You identify a network object or network object group for both the real and mapped addresses.
- In this case, NAT is not a parameter of the network object; the network object or group is a parameter of the NAT configuration. The ability to use a network object group for the real address means that twice NAT is more scalable.
- How source and destination NAT is implemented.

–Network object NAT— Each rule can apply to either the source or destination of a packet. So two rules might be used, one for the source IP address, and one for the destination IP address. These

two rules cannot be tied together to enforce a specific translation for a source/destination combination.

–Twice NAT—A single rule translates both the source and destination. A matching packet only matches the one rule, and further rules are not checked. Even if you do not configure the optional destination address for twice NAT, a matching packet still only matches one twice NAT rule. The source and destination are tied together, so you can enforce different translations depending on the source/destination combination. For example, sourceA/destinationA can have a different translation than sourceA/destinationB.

•Order of NAT Rules.

–Network object NAT—Automatically ordered in the NAT table.

–Twice NAT—Manually ordered in the NAT table (before or after network object NAT rules).

QUESTION NO: 43

Which option is one requirement before a Cisco ASA appliance can be upgraded from Cisco ASA Software Version 8.2 to 8.3?

- A. Remove all the pre 8.3 NAT configurations in the startup configuration.
- B. Upgrade the memory on the Cisco ASA appliance to meet the memory requirement of Cisco ASA Software Version 8.3.
- C. Request new Cisco ASA licenses to meet the 8.3 licensing requirement.
- D. Upgrade Cisco ASDM to version 6.2.
- E. Migrate interface ACL configurations to include interface and global ACLs.

Answer: B

Explanation:

http://www.cisco.com/en/US/prod/collateral/vpndevc/ps6032/ps6094/ps6120/product_bulletin_c25-586414.html

QUESTION NO: 44

Which statement about the Cisco ASA botnet traffic filter is true?

- A. The four threat levels are low, moderate, high, and very high.
- B. By default, the dynamic-filter drop blacklist interface outside command drops traffic with a threat

level of high or very high.

C. Static blacklist entries always have a very high threat level.

D. A static or dynamic blacklist entry always takes precedence over the static whitelist entry.

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_botnet.html

Information About the Static Database You can manually enter domain names or IP addresses (host or subnet) that you want to tag as bad names in a blacklist. Static blacklist entries are always designated with a Very High threat level. You can also enter names or IP addresses in a whitelist, so that names or addresses that appear on both the dynamic blacklist and the whitelist are identified only as whitelist addresses in syslog messages and reports. Note that you see syslog messages for whitelisted addresses even if the address is not also in the dynamic blacklist.

QUESTION NO: 45

Refer to the exhibit.

```
S 10.2.2.0 255.255.255.0 [1/0] via 172.16.1.10, dmz
S 10.3.3.0 255.255.255.0 [2/0] via 172.16.1.11, dmz
```

Which Cisco ASA CLI commands configure these static routes in the Cisco ASA routing table?

A. route dmz 10.2.2.0 0.0.0.255 172.16.1.10

route dmz 10.3.3.0 0.0.0.255 172.16.1.11

B. route dmz 10.2.2.0 0.0.0.255 172.16.1.10 1

route dmz 10.3.3.0 0.0.0.255 172.16.1.11 1

C. route dmz 10.2.2.0 0.0.0.255 172.16.1.10

route dmz 10.3.3.0 0.0.0.255 172.16.1.11 2

D. route dmz 10.2.2.0 255.255.255.0 172.16.1.10

route dmz 10.3.3.0 255.255.255.0 172.16.1.11

E. route dmz 10.2.2.0 255.255.255.0 172.16.1.10 1

route dmz 10.3.3.0 255.255.255.0 172.16.1.11 1

F. route dmz 10.2.2.0 255.255.255.0 172.16.1.10

route dmz 10.3.3.0 255.255.255.0 172.16.1.11 2

Answer: F

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/route_static.html#wp1121521

Command	Purpose
<pre>route if_name dest_ip mask gateway_ip [distance]</pre> Example: <pre>hostname(config)# route outside 10.10.10.0 255.255.255.0 192.168.1.1 [1]</pre>	This enables you to add a static route. The <i>dest_ip</i> and <i>mask</i> is the IP address for the destination network and the <i>gateway_ip</i> is the address of the next-hop router. The addresses you specify for the static route are the addresses that are in the packet before entering the ASA and performing NAT. The <i>distance</i> is the administrative distance for the route. The default is 1 if you do not specify a value. Administrative distance is a parameter used to compare routes among different routing protocols. The default administrative distance for static routes is 1, giving it precedence over routes discovered by dynamic routing protocols but not directly connect routes. The default administrative distance for routes discovered by OSPF is 110. If a static route has the same administrative distance as a dynamic route, the static routes take precedence. Connected routes always take precedence over static or dynamically discovered routes.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 46

Which statement about static or default route on the Cisco ASA appliance is true?

- A. The admin distance is 1 by default.
- B. From the show route output, the [120/3] indicates an admin distance of 3.
- C. A default route is specified using the 0.0.0.0 255.255.255.255 address/mask combination.
- D. The tunneled command option is used to enable route tracking.
- E. The interface-name parameter in the route command is an optional parameter if the static route points to the next-hop router IP address.

Answer: A

Explanation:

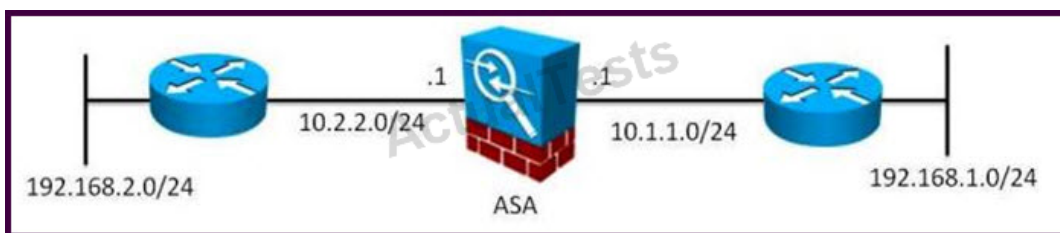
http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/route_static.html#wp1121521

Command	Purpose
<pre>route if_name dest_ip mask gateway_ip [distance]</pre> Example: <pre>hostname(config)# route outside 10.10.10.0 255.255.255.0 192.168.1.1 [1]</pre>	This enables you to add a static route. The <i>dest_ip</i> and <i>mask</i> is the IP address for the destination network and the <i>gateway_ip</i> is the address of the next-hop router. The addresses you specify for the static route are the addresses that are in the packet before entering the ASA and performing NAT. The <i>distance</i> is the administrative distance for the route. The default is 1 if you do not specify a value. Administrative distance is a parameter used to compare routes among different routing protocols. The default administrative distance for static routes is 1, giving it precedence over routes discovered by dynamic routing protocols but not directly connect routes. The default administrative distance for routes discovered by OSPF is 110. If a static route has the same administrative distance as a dynamic route, the static routes take precedence. Connected routes always take precedence over static or dynamically discovered routes.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 47

Refer to the exhibit.



Which Cisco ASA configuration has the minimum number of the required configuration commands to enable the Cisco ASA appliance to establish EIGRP neighborship with its two neighboring routers?

- A.** router eigrp 1
network 10.0.0.0 255.0.0.0
- B.** router eigrp 1
network 10.0.0.0 255.0.0.0
network 192.168.1.0 255.255.255.0

network 192.168.2.0 255.255.255.0

C. router eigrp 1

network 10.1.1.0 255.255.255.0

network 10.2.2.0 255.255.255.0

D. router eigrp 1

network 10.1.1.0 255.255.255.0

network 10.2.2.0 255.255.255.0

network 192.168.1.0 255.255.255.0

network 192.168.2.0 255.255.255.0

E. router eigrp 1

network 0.0.0.0 255.255.255.255

Answer: A

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_configuration_example09186a008086ebd2.shtml

EIGRP Configuration - the CLI configuration is very similar to the !Cisco IOS routerEIGRP configuration.

QUESTION NO: 48

Which configuration step is the first to enable PIM-SM on the Cisco ASA appliance?

A. Configure the static RP IP address.

B. Enable IGMP forwarding on the required interface(s).

C. Add the required static mroute(s).

D. Enable multicast routing globally on the Cisco ASA appliance.

E. Configure the Cisco ASA appliance to join the required multicast groups.

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/route_multicast.html#wp1060775

Enabling Multicast RoutingEnabling multicast routing lets the ASA forward multicast packets. Enabling multicast routing automaticallyenables PIM and IGMP on all interfaces.

To enable multicast routing, perform the following step:

Command	Purpose
<code>multicast-routing</code> Example: <code>hostname(config)# multicast-routing</code>	This step enables multicast routing. The number of entries in the multicast routing tables are limited by the amount of RAM on the system.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 49

Refer to the exhibit.

```
dynamic-filter updater-client enable
dynamic-filter use-database
dynamic-filter whitelist
address 10.10.10.10 255.255.255.255
dynamic-filter blacklist
address 192.168.1.1 255.255.255.255
dynamic-filter enable interface outside
dynamic-filter drop blacklist interface outside
!
policy-map global_policy
class inspection_default
inspect dns DNS_INSPECT_MAP
!
service-policy global_policy global
!
```

Which option describes the problem with this botnet traffic filter configuration on the Cisco ASA appliance?

- A. The traffic classification ACL is not defined.
- B. The use of the dynamic database is not enabled.
- C. DNS snooping is not enabled.
- D. The threat level range for the traffic to be dropped is not defined.
- E. The static black and white list entries should use domain name instead of IP address.

Answer: C

Explanation:

<https://supportforums.cisco.com/docs/DOC-8782>

Prerequisite

The ASA must be running minimum 8.2 code to be able to configure botnet feature.

Botnet license must be installed on the ASA

Limitations Step by Step Configuration

1. Enable DNS client on ASA
2. Enable dynamic traffic filtering (Botnet Traffic Filter).
3. Enable the Botnet Traffic Filter database update.
4. Classify the traffic that will be exempted and subjected.
5. Enable dynamic-filter classification on outside interface
6. Configure a class map and only match dns traffic
7. Enable DNS snooping on the external interface
8. Define local whitelists and/or blacklists if needed.

Never block addresses:

Manual Black List:

QUESTION NO: 50

In the default global policy, which traffic is matched for inspections by default?

- A. match any
- B. match default-inspection-traffic
- C. match access-list
- D. match port
- E. match class-default

Answer: B

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/inspect.html#wp1383691>

Default Inspection Policy

By default, the configuration includes a policy that matches all default application inspection traffic and applies inspection to the traffic on all interfaces (a global policy). Default application inspection traffic includes traffic to the default ports for each protocol. You can only apply one global policy, so if you want to alter the global policy, for example, to apply inspection to non-standard ports, or to add inspections that are not enabled by default, you need to either edit the default policy or disable it and apply a new one.

QUESTION NO: 51

Which option lists the main tasks in the correct order to configure a new Layer 3 and 4 inspection policy on the Cisco ASA appliance using the Cisco ASDM Configuration > Firewall > Service Policy Rules pane?

- A.** 1. Create a class map to identify which traffic to match.
2. Create a policy map and apply action(s) to the traffic class(es).
3. Apply the policy map to an interface or globally using a service policy.
- B.** 1. Create a service policy rule.
2. Identify which traffic to match.
3. Apply action(s) to the traffic.
- C.** 1. Create a Layer 3 and 4 type inspect policy map.
2. Create class map(s) within the policy map to identify which traffic to match.
3. Apply the policy map to an interface or globally using a service policy.
- D.** 1. Identify which traffic to match.
2. Apply action(s) to the traffic.
3. Create a policy map.
4. Apply the policy map to an interface or globally using a service policy.

Answer: B

Explanation:

http://www.cisco.com/en/US/docs/security/asdm/6_2/user/guide/inspctrl.html#wpxref87867

Choose Configuration > Firewall > Service Policy Rules.

Add or edit a service policy rule click the Protocol Inspection tab

In the Edit Service Policy Rule > Rule Actions dialog box,

Select each inspection type that you want to apply. You can predefine inspect maps in the Configuration >

Firewall > Objects > Inspect Maps pane

QUESTION NO: 52

By default, how does a Cisco ASA appliance process IP fragments?

- A.** Each fragment passes through the Cisco ASA appliance without any inspections.
- B.** Each fragment is blocked by the Cisco ASA appliance.
- C.** The Cisco ASA appliance verifies each fragment and performs virtual IP re-assembly before the full IP packet is forwarded out.

D. The Cisco ASA appliance forwards the packet out as soon as all of the fragments of the packet have been received.

Answer: C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa81/config/guide/intro.html>

Protecting from IP FragmentsThe adaptive security appliance provides IP fragment protection. This feature performs full reassembly of all ICMP error messages and virtual reassembly of the remaining IP fragments that are routed through the adaptive security appliance. Fragments that fail the security check are dropped and logged. Virtual reassembly cannot be disabled.

QUESTION NO: 53

Which additional active/standby failover feature was introduced in Cisco ASA Software Version 8.4?

- A. HTTP stateful failover
- B. OSPF and EIGRP routing protocol stateful failover
- C. SSL VPN stateful failover
- D. IPsec VPN stateful failover
- E. NAT stateful failover

Answer: B

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/ha_overview.html#wp1077551

Stateful Failover

When Stateful Failover is enabled, the active unit continually passes per-connection state information to the standby unit. After a failover occurs, the same connection information is available at the new active unit. Supported end-user applications are not required to reconnect to keep the same communication session.

In Version 8.4 and later, Stateful Failover participates in dynamic routing protocols, like OSPF and EIGRP, so routes that are learned through dynamic routing protocols on the active unit are maintained in a Routing Information Base (RIB) table on the standby unit. Upon a failover event, packets travel normally with minimal disruption to traffic because the Active secondary ASA initially has rules that mirror the primary ASA. Immediately after failover, the re-convergence timer starts on the newly Active unit. Then the epoch number for the RIB table increments. During re-

convergence, OSPF and EIGRP routes become updated with a new epochnumber. Once the timer is expired, stale route entries (determined by the epoch number) are removed from the table. The RIB then contains the newest routing protocol forwarding information on the newly Active unit.

QUESTION NO: 54

Which other match command is used with the match flow ip destination-address command within the class map configurations of the Cisco ASA MPF?

- A. match tunnel-group
- B. match access-list
- C. match default-inspection-traffic
- D. match port
- E. match dscp

Answer: A

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa71/configuration/guide/vpnggrp.html>

or

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_qos.html

QUESTION NO: 55

Which Cisco ASA configuration is used to configure the TCP intercept feature?

- A. a TCP map
- B. an access list
- C. the established command
- D. the set connection command with the embryonic-conn-max option
- E. a type inspect policy map

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/conns_connlimits.html#wp1080734

QUESTION NO: 56

Which configuration step (if any) is necessary to enable FTP inspection on TCP port 2121?

- A. None. FTP inspection is enabled by default using the global policy.
- B. Create a new class map to match TCP port 2121, then edit the global policy to inspect FTP for traffic matched by the new class map.
- C. Edit default-inspection-traffic to match FTP on port 2121.
- D. Add a new traffic class using the match protocol FTP option within the inspect_default class map.

Answer: B

Explanation:

QUESTION NO: 57

When the Cisco ASA appliance is processing packets, which action is performed first?

- A. Check if the packet is permitted or denied by the inbound interface ACL.
- B. Check if the packet is permitted or denied by the outbound interface ACL.
- C. Check if the packet is permitted or denied by the global ACL.
- D. Check if the packet matches an existing connection in the connection table.
- E. Check if the packet matches an inspection policy.
- F. Check if the packet matches a NAT rule.

Answer: D

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_tech_note09186a0080ba9d00.shtml

QUESTION NO: 58

Which Cisco ASA (8.4.1 and later) CLI command is the best command to use for troubleshooting SSH connectivity from the Cisco ASA appliance to the outside 192.168.1.1 server?

- A. telnet 192.168.1.1 22
- B. ssh -l username 192.168.1.1

- C. traceroute 192.168.1.1 22
- D. ping tcp 192.168.1.1 22
- E. packet-tracer input inside tcp 10.0.1.1 2043 192.168.4.1 ssh

Answer: D

Explanation:

QUESTION NO: 59

Refer to the exhibit.

```
ciscoasa(config)# ntp authenticate
ciscoasa(config)# ntp authentication-key 2 md5 secret
ciscoasa(config)# ntp trusted-key 2
ciscoasa(config)# ntp server 192.168.1.1
```

Which reason explains why the Cisco ASA appliance cannot establish an authenticated NTP session to the inside 192.168.1.1 NTP server?

- A. The ntp server 192.168.1.1 command is incomplete.
- B. The ntp source inside command is missing.
- C. The ntp access-group peer command and the ACL to permit 192.168.1.1 are missing.
- D. The trusted-key number should be 1 not 2.

Answer: A

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/basic.html#wp1067761>

```
hostname(config)# ntp server ip_address [key key_id] [source interface_name][prefer]
```

```
ntp server 192.168.1.1 2
```

QUESTION NO: 60

On which type of encrypted traffic can a Cisco ASA appliance running software version 8.4.1 perform application inspection and control?

- A. IPsec
- B. SSL
- C. IPsec or SSL
- D. Cisco Unified Communications
- E. Secure FTP

Answer: D

Explanation:

http://www.cisco.com/en/US/solutions/collateral/ns340/ns394/ns165/ns391/guide__c07-494658.html

QUESTION NO: 61

Where in the Cisco ASA appliance CLI are Active/Active Failover configuration parameters configured?

- A. admin context
- B. customer context
- C. system execution space
- D. within the system execution space and admin context
- E. within each customer context and admin context

Answer: C

Explanation:

System Execution Space

Unlike other contexts, the system execution space does not have any Layer 2 or Layer 3 interfaces or any network settings. Rather, it is mainly used to define the attributes of other security context attributes. Here are the three important attributes configured for each context in the system execution space:

Context name.

Location of context's startup configuration.

The configuration of each context is also known as a configlet. Interface allocation.

Additionally, many optional features, such as interface and boot parameters, can be configured within the system execution space. The important features that can be set up through the system execution space

Feature	Description
Interface	Sets up physical interfaces for speed and duplex. Interfaces can be enabled or disabled.
Banner	Specifies a login or session banner when connecting to the security appliance.
Boot	Specifies boot parameters to load proper image.
Activation key	Enables or disables security appliance features.
File management	Adds or deletes the security context configurations that are stored locally on the security appliance.
Firewall mode	Configures single- or multiple-mode firewall in the system execution space.
Failover	Sets the failover parameters to accommodate multiple physical security appliances.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

The system execution space configuration resides in the nonvolatile random-access memory (NVRAM) area of the security appliance, while the configurations for security contexts are stored either in local Flash memory or on a network storage server using one of the following protocols:

TFTP

FTP

HTTPS

HTTP

The system execution space designates one of the security contexts as the admin context, which is responsible for providing network access when the system needs to contact resources.

QUESTION NO: 62

With Cisco ASA active/active or active/standby stateful failover, which state information or table is not passed between the active and standby Cisco ASA by default?

- A. NAT translation table
- B. TCP connection states
- C. UDP connection states
- D. ARP table
- E. HTTP connection table

Answer: E

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/ha_overview.html#wp1078922

Stateful Failover

When Stateful Failover is enabled, the active unit continually passes per-connection state information to the standby unit. After a failover occurs, the same connection information is available at the new active unit. Supported end-user applications are not required to reconnect to keep the same communication session.

[Table 32-2](#) list the state information that is and is not passed to the standby unit when Stateful Failover is enabled.

Table 32-2 State Information

State Information Passed to Standby Unit	State Information Not Passed to Standby Unit
NAT translation table	The HTTP connection table (unless HTTP replication is enabled).
TCP connection states	The user authentication (uauth) table. Inspected protocols are subject to advanced TCP-state tracking, and the TCP state of these connections is not automatically replicated. While these connections are replicated to the standby unit, there is a best-effort attempt to re-establish a TCP state.
UDP connection states	The routing tables. After a failover occurs, some packets may be lost or routed out of the wrong interface (the default route) while the dynamic routing protocols rediscover routes.
The ARP table	State information for Security Service Modules.
The Layer 2 bridge table (when running in transparent firewall mode)	DHCP server address leases.
The HTTP connection states (if HTTP replication is enabled)	Stateful failover for phone proxy. When the active unit goes down, the call fails, media stops flowing, and the phone should unregister from the failed unit and reregister with the active unit. The call must be re-established.
The ISAKMP and IPsec SA table	—
GTP PDP connection database	—
SIP signalling sessions	—

C:\Documents and Settings\user-nwz\Desktop\1.JPG

The following WebVPN features are not supported with Stateful Failover:

- Smart Tunnels
- Port Forwarding
- Plugins
- Java Applets
- IPv6 clientless or Anyconnect sessions
- Citrix authentication (Citrix users must reauthenticate after failover)

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 63

Which Cisco ASA object group type offers the most flexibility for grouping different services

together based on arbitrary protocols?

- A. network
- B. ICMP
- C. protocol
- D. TCP-UDP
- E. service

Answer: E

Explanation:

QUESTION NO: 64

Using the default modular policy framework global configuration on the Cisco ASA, how does the Cisco ASA process outbound HTTP traffic?

- A. HTTP flows are not permitted through the Cisco ASA, because HTTP is not inspected by default.
- B. HTTP flows match the inspection_default traffic class and are inspected using HTTP inspection.
- C. HTTP outbound traffic is permitted, but all return HTTP traffic is denied.
- D. HTTP flows are statefully inspected using TCP stateful inspection.

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/intro_intro.html#wp1128055

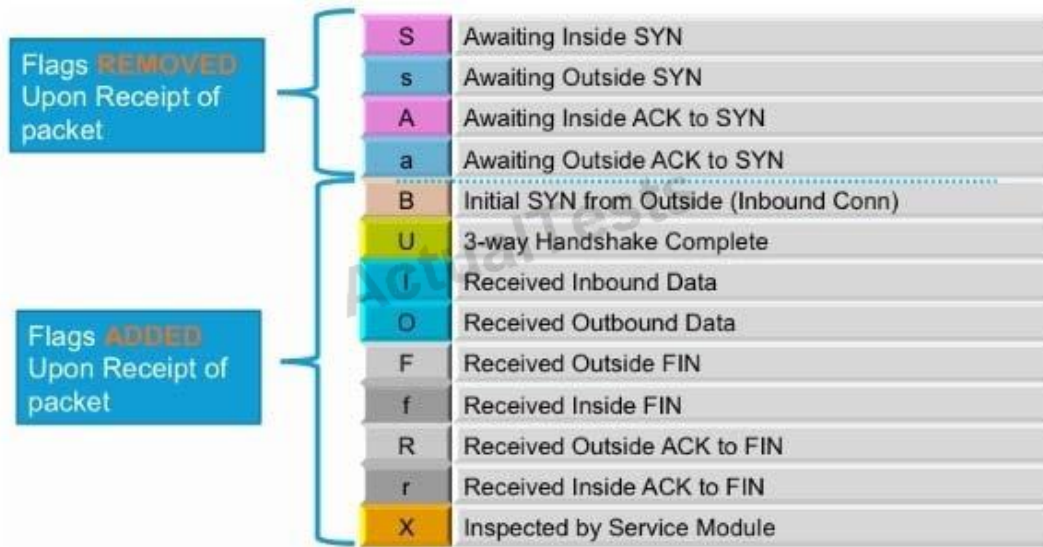
QUESTION NO: 65

Which flags should the show conn command normally show after a TCP connection has successfully been established from an inside host to an outside host?

- A. aB
- B. saA
- C. sIO
- D. AIO
- E. UIO
- F. F

Answer: E

Explanation:



C:\Documents and Settings\user-nwz\Desktop\1.JPG

Flags: A - awaiting inside ACK to SYN, a - awaiting outside ACK to SYN,
 B - initial SYN from outside, b - TCP state-bypass or nailed, C - CTIOBE media,
 D - DNS, d - dump, E - outside back connection, F - outside FIN, f - inside FIN,
 G - group, g - MGCP, H - H.323, h - H.225.0, I - inbound data,
 i - incomplete, J - GTP, j - GTP data, K - GTP t3-response
 k - Skinny media, M - SMTP data, m - SIP media, n - GUP
 O - outbound data, P - inside back connection, p - Phone-proxy TFTP connection,
 q - SQL*Net data, R - outside acknowledged FIN,
 R - UDP SUNRPC, r - inside acknowledged FIN, S - awaiting inside SYN,
 s - awaiting outside SYN, T - SIP, t - SIP transient, U - up,
 V - VPN orphan, W - WAAS,
 X - inspected by service module

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 66

Which Cisco ASA show command groups the xlates and connections information together in its output?

- A. show conn
- B. show conn detail
- C. show xlate
- D. show asp
- E. show local-host

Answer: E

Explanation:**show local-host**

To display the network states of local hosts, use the **show local-host** command in privileged EXEC mode.

show local-host [*ip_address*] [*detail*] [*all*] [*brief*] [*connection* {*tcp* <*start*>[-<*end*>] | *udp* <*start*>[-<*end*>] | *embryonic* <*start*>[-<*end*>]]

Syntax Description

all	(Optional) Includes local hosts connecting to the security appliance and from the security appliance.
detail	(Optional) Displays the detailed network states of local host information, including more information about active xlates and network connections.
ip_address	(Optional) Specifies the local host IP address.
brief	(Optional) Displays brief information on local hosts.
connection	(Optional) Displays three types of filters based on the number and type of connections: tcp, udp and embryonic. These filters can be used individually or jointly.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 67

When a Cisco ASA is configured in multiple context mode, within which configuration are the interfaces allocated to the security contexts?

- A. each security context
- B. system configuration
- C. admin context (context with the "admin" role)
- D. context startup configuration file (.cfg file)

Answer: B

Explanation:

<http://www.cisco.com/en/US/products/hw/vpndevc/ps2030/>

[products_configuration_example09186a00808d2b63.shtml](#)

In order to specify the interfaces that you can use in the context, enter the command appropriate for a physical interface or for one or more subinterfaces.

In order to allocate a physical interface, enter this command:

```
hostname(config-ctx)# allocate-interface <physical_interface> [mapped_name]
[visible | invisible]
```

Context Configurations

The security appliance includes a configuration for each context that identifies the security policy, interfaces, and almost all the options you can configure on a standalone device. You can store context configurations on the internal Flash memory or the external Flash memory card, or you can download them from a TFTP, FTP, or HTTP(S) server.

System Configuration

The system administrator adds and manages contexts by configuring each context configuration location, allocated interfaces, and other context operating parameters in the system configuration, which, like a singlemode configuration, is the startup configuration. The system configuration identifies basic settings for the security appliance. The system configuration does not include any network interfaces or network settings for itself; rather, when the system needs to access network resources (such as downloading the contexts from the server), it uses one of the contexts that is designated as the admin context. The system configuration does include a specialized failover interface for failover traffic only.

Admin Context Configuration

The admin context is just like any other context, except that when a user logs in to the admin context, then that user has system administrator rights and can access the system and all other contexts. The admin context is not restricted in any way, and can be used as a regular context. However, because logging into the admin context grants you administrator privileges over all contexts, you might need to restrict access to the admin context to appropriate users. The admin context must reside on Flash memory, and not remotely.

If your system is already in multiple context mode, or if you convert from single mode, the admin context is created automatically as a file on the internal Flash memory called admin.cfg. This context is named "admin." If you do not want to use admin.cfg as the admin context, you can change the admin context.

QUESTION NO: 68

When troubleshooting redundant interface operations on the Cisco ASA, which configuration should be verified?

- A. The name if configuration on the member physical interfaces are identical.
- B. The MAC address configuration on the member physical interfaces are identical.
- C. The active interface is sending periodic hellos to the standby interface.
- D. The IP address configuration on the logical redundant interface is correct.
- E. The duplex and speed configuration on the logical redundant interface are correct.

Answer: D

Explanation:

Concept

A logical redundant interface is a pair of an active and a standby physical interface. When the active interface fails, the standby interface becomes active. From firewall perspective this event is completely transparent and can be viewed as a single logical interface. We can use redundant interfaces to increase the security appliance reliability. This feature is separate from device-level failover, but you can configure redundant interfaces as well as failover if desired. We can configure up to 8 redundant interfaces.

Redundant interfaces are numbered from 1 to 8 and have the name redundant X. When adding physical interfaces to the redundant pair, please make sure there is no configuration on it and the interface is also in no shutdown state. This is just a precaution; the firewall will remove these settings when adding the physical interface to a new group. The logical redundant interface will take the MAC address of the first interface added to the group.

This MAC address is not changed with the member interface failures, but changes when you swap the order of the physical interfaces to the pair.

Once we have configured a redundant interface, we can assign it a name and a security level, followed by an IP address. The procedure is the same as with any interface in the system.

Configuration

-->

```
interface GigabitEthernet0/0
no nameif
no security-level
no ip address
!
interface GigabitEthernet0/1
no nameif
no security-level
no ip address
interface Redundant1
member-interface GigabitEthernet0/0
member-interface GigabitEthernet0/1
nameif outside
security-level 0
ip address 1.1.1.1 255.255.255.0
```

Verify

You can use the following command to verify--

-->

```
ciscoasa(config)# show interface redundant 1
```

Interface Redundant1 "outside", is up, line protocol is up
Hardware is i82546GB rev03, BW 1000 Mbps, DLY 10 usec
Auto-Duplex(Full-duplex), Auto-Speed(100 Mbps)
MAC address 5475.d0d4.9594, MTU 1500
IP address 1.1.1.1, subnet mask 255.255.255.0
27 packets input, 12330 bytes, 0 no buffer
Received 27 broadcasts, 0 runts, 0 giants
0 input errors, 0 CRC, 0 frame, 27 overrun, 0 ignored, 0 abort
10 L2 decode drops
1 packets output, 64 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 late collisions, 0 deferred
0 input reset drops, 0 output reset drops
input queue (curr/max packets): hardware (5/25) software (0/0)
output queue (curr/max packets): hardware (0/1) software (0/0)

Traffic Statistics for "outside":

17 packets input, 7478 bytes
1 packets output, 28 bytes
17 packets dropped
1 minute input rate 0 pkts/sec, 92 bytes/sec
1 minute output rate 0 pkts/sec, 0 bytes/sec
1 minute drop rate, 0 pkts/sec
5 minute input rate 0 pkts/sec, 0 bytes/sec
5 minute output rate 0 pkts/sec, 0 bytes/sec
5 minute drop rate, 0 pkts/sec

Redundancy Information:

Member GigabitEthernet0/0(Active), GigabitEthernet0/1
Last switchover at 23:13:03 UTC Dec 15 2011

QUESTION NO: 69

Which statement about the Cisco ASA 5505 configuration is true?

- A.** The IP address is configured under the physical interface (ethernet 0/0 to ethernet 0/7).
- B.** With the default factory configuration, the management interface (management 0/0) is configured with the 192.168.1.1/24 IP address.
- C.** With the default factory configuration, Cisco ASDM access is not enabled.
- D.** The switchport access vlan command can be used to assign the VLAN to each physical

interface (ethernet 0/0 to ethernet 0/7).

E. With the default factory configuration, both the inside and outside interface will use DHCP to acquire its IP address.

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/interface_start_5505.htm

|

	Command	Purpose
Step 1	<pre>interface ethernet0/port Example: hostname(config)# interface ethernet0/1</pre>	Specifies the switch port you want to configure, where <i>port</i> is 0 through 7.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 70

What is the correct regular expression to match HTTP requests whose URI is /welcome.jpg?

- A. ^/welcome.jpg
- B. ^/welcome\.
- C. ^*/welcome\.
- D. ^\welcome\.
- E. ^*/welcome\.

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/mpf.html#wp1101685>

^ Caret Specifies the beginning of a line.

\ Escape When used with a metacharacter, matches a literal character. For example, \[matches the left square bracket character

QUESTION NO: 71

Refer to the exhibit.

```
%ASA-2-106006: Deny inbound UDP from 10.1.1.1/520 to 224.0.0.9/520 on interface outside  
%ASA-2-106006: Deny inbound UDP from 192.168.1.1/520 to 224.0.0.9/520 on interface inside
```

A Cisco ASA in transparent firewall mode generates the log messages seen in the exhibit. What should be configured on the Cisco ASA to allow the denied traffic?

- A. extended ACL on the outside and inside interface to permit the multicast traffic
- B. EtherType ACL on the outside and inside interface to permit the multicast traffic
- C. stateful packet inspection
- D. static ARP mapping
- E. static MAC address mapping

Answer: A

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/mpf.html#wp1101685>

Allowing Broadcast and Multicast Traffic through the Transparent Firewall

In routed firewall mode, broadcast and multicast traffic is blocked even if you allow it in an access list, including unsupported dynamic routing protocols and DHCP (unless you configure DHCP relay). Transparent firewall mode can allow any IP traffic through. This feature is especially useful in multiple context mode, which does not allow dynamic routing, for example.

QUESTION NO: 72

With active/standby failover, what happens if the standby Cisco ASA does not receive three consecutive hello messages from the active Cisco ASA on the LAN failover interface?

- A. The standby ASA immediately becomes the active ASA.
- B. The standby ASA eventually becomes the active ASA after three times the hold-down timer interval expires.
- C. The standby ASA runs network activity tests, including ARP and ping, to determine if the active ASA has failed.
- D. The standby ASA sends additional hellos packets on all monitored interfaces, including the LAN failover interface, to determine if the active ASA has failed.
- E. Both ASAs go to the "unknown" state until the LAN interface becomes operational again.

Answer: D

Explanation:

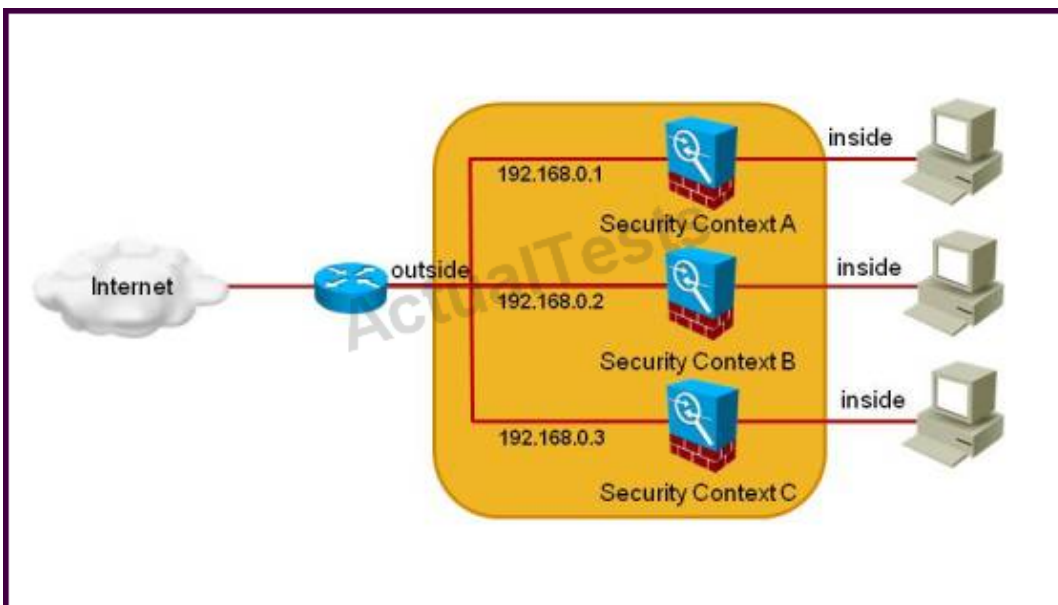
http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/ha_overview.html

Unit Health Monitoring The ASA determines the health of the other unit by monitoring the failover link. When a unit does not receive three consecutive hello messages on the failover link, the unit sends interface hello messages on each interface, including the failover interface, to validate whether or not the peer interface is responsive. The action that the ASA takes depends upon the response from the other unit. See the following possible actions:

- If the ASA receives a response on the failover interface, then it does not fail over.
- If the ASA does not receive a response on the failover link, but it does receive a response on another interface, then the unit does not failover. The failover link is marked as failed. You should restore the failover link as soon as possible because the unit cannot fail over to the standby while the failover link is down.
- If the ASA does not receive a response on any interface, then the standby unit switches to active mode and classifies the other unit as failed.

QUESTION NO: 73

Refer to the exhibit.



The Cisco ASA is dropping all the traffic that is sourced from the internet and is destined to any security context inside interface. Which configuration should be verified on the Cisco ASA to solve this problem?

- A.** The Cisco ASA has NAT control disabled on each security context.

- B. The Cisco ASA is using inside dynamic NAT on each security context.
- C. The Cisco ASA is using a unique MAC address on each security context outside interface.
- D. The Cisco ASA is using a unique dynamic routing protocol process on each security context.
- E. The Cisco ASA packet classifier is configured to use the outside physical interface to assign the packets to each security context.

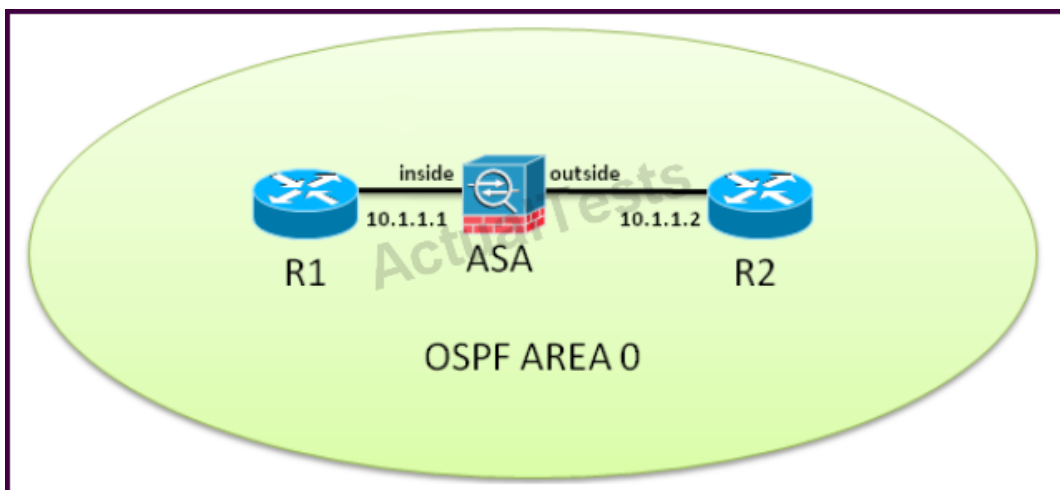
Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/mode_contexts.html#wp1134937

QUESTION NO: 74

Refer to the exhibit.



The Cisco ASA is operating in transparent mode. What is required on the Cisco ASA so that R1 and R2 can form OSPF neighbor adjacency?

- A. Map the R1 and R2 MAC address in the Cisco ASA MAC address table using the mac-address-table static if_name MAC_address command.
- B. Configure OSPF stateful packet inspection using MPF.
- C. Apply an EtherType ACL to the inside and outside interfaces to permit OSPF multicast traffic.
- D. Apply an extended ACL to the inside and outside interfaces to permit OSPF multicast traffic.
- E. Enable Advanced Application Inspection using MPF.

Answer: D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/mpf.html#wp1101685>

Allowing Broadcast and Multicast Traffic through the Transparent Firewall

In routed firewall mode, broadcast and multicast traffic is blocked even if you allow it in an access list, including unsupported dynamic routing protocols and DHCP (unless you configure DHCP relay). Transparent firewall mode can allow any IP traffic through. This feature is especially useful in multiple context mode, which does not allow dynamic routing, for example.

QUESTION NO: 75

On the Cisco ASA, where are the Layer 5-7 policy maps applied?

- A. inside the Layer 3-4 policy map
- B. inside the Layer 3-4 class map
- C. inside the Layer 5-7 class map
- D. inside the Layer 3-4 service policy
- E. inside the Layer 5-7 service policy

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/inspect_overview.html#wp1313159

QUESTION NO: 76

A Cisco ASA requires an additional feature license to enable which feature?

- A. transparent firewall
- B. cut-thru proxy
- C. threat detection
- D. botnet traffic filtering
- E. TCP normalizer

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa83/license_standalone/license_management/license.html#wp1450282

QUESTION NO: 77

With Cisco ASA active/standby failover, what is needed to enable subsecond failover?

- A. Use redundant interfaces.
- B. Enable the stateful failover interface between the primary and secondary Cisco ASA.
- C. Decrease the default unit failover polltime to 300 msec and the unit failover holdtime to 900 msec.
- D. Decrease the default number of monitored interfaces to 1.

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa83/asdm63/configuration_guide/ha_active_standby.html

Configuring the Unit and Interface Health Poll Times

The adaptive security appliance sends hello packets out of each data interface to monitor interface health. The appliance sends hello messages across the failover link to monitor unit health. If the adaptive security appliance does not receive a hello packet from the corresponding interface on the peer unit for over half of the hold time, then the additional interface testing begins. If a hello packet or a successful test result is not received within the specified hold time, the interface is marked as failed. Failover occurs if the number of failed interfaces meets the failover criteria.

Decreasing the poll and hold times enables the adaptive security appliance to detect and respond to interface failures more quickly, but may consume more system resources. Increasing the poll and hold times prevents the adaptive security appliance from failing over on networks with higher latency.

QUESTION NO: 78

Refer to the exhibit.

```
nat (inside,outside) source static command-option-1 command-option-2 destination
static command-option-3 command-option-4
```

Which command options represent the inside local address, inside global address, outside local address, and outside global address?

- A. 1 = outside local, 2 = outside global, 3 = inside global, 4 = inside local
- B. 1 = outside local, 2 = outside global, 3 = inside local, 4 = inside global
- C. 1 = outside global, 2 = outside local, 3 = inside global, 4 = inside local
- D. 1 = inside local, 2 = inside global, 3 = outside global, 4 = outside local
- E. 1 = inside local, 2 = inside global, 3 = outside local, 4 = outside global

Answer: D

Explanation:

Cisco twice NAT

```
nat [(real_ifc,mapped_ifc)] [line | {after-object [line]}] source static {nw_obj nw_obj | any any}
[destination static{mapped_obj | interface} real_obj] [service real_src_mapped_dest_svc_obj
mapped_src_real_dest_svc_obj]
[no-proxy-arp] [route-lookup] [inactive] [description desc]
```

Example:

```
hostname(config)# nat (inside,outside) source static MyInsNet MyInsNet destination static Server1
Server1
```

QUESTION NO: 79

On Cisco ASA Software Version 8.4.1 and later, when you configure the Cisco ASA appliance in transparent firewall mode, which configuration is mandatory?

- A. NAT
- B. static routes
- C. ARP inspections
- D. EtherType access-list
- E. bridge group(s)
- F. dynamic MAC address learning

Answer: E

Explanation:

Command History

Release	Modification
7.0(1)	For routed mode, this command was changed from a global configuration command to an interface configuration mode command.
8.4(1)	For transparent mode, bridge groups were introduced. You now set the IP address for the BVI, and not globally.

Usage Guidelines

In single context routed firewall mode, each interface address must be on a unique subnet. In multiple context mode, if this interface is on a shared interface, then each IP address must be unique but on the same subnet. If the interface is unique, this IP address can be used by other contexts if desired.

A transparent firewall does not participate in IP routing. The only IP configuration required for the ASA is to set the BVI address. This address is required because the ASA uses this address as the source address for traffic originating on the ASA, such as system messages or communications with AAA servers. You can also use this address for remote management access. This address must be on the same subnet as the upstream and downstream routers. For multiple context mode, set the management IP address within each context. For models that include a Management interface, you can also set an IP address for this interface for management purposes.

The standby IP address must be on the same subnet as the main IP address.

Examples

The following example sets the IP addresses and standby addresses of two interfaces:

```
hostname(config)# interface gigabitethernet0/2
hostname(config-if)# nameif inside
hostname(config-if)# security-level 100
hostname(config-if)# ip address 10.1.1.1 255.255.255.0 standby 10.1.1.2
hostname(config-if)# no shutdown
hostname(config-if)# interface gigabitethernet0/3
hostname(config-if)# nameif outside
hostname(config-if)# security-level 0
hostname(config-if)# ip address 10.1.2.1 255.255.255.0 standby 10.1.2.2
hostname(config-if)# no shutdown
```

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 80

Which access rule is disabled automatically after the global access list has been defined and applied?

- A. the implicit global deny ip any any access rule
- B. the implicit interface access rule that permits all IP traffic from high security level to low security level interfaces
- C. the implicit global access rule that permits all IP traffic from high security level to low security level interfaces
- D. the implicit deny ip any any rule on the global and interface access lists
- E. the implicit permit all IP traffic from high security level to low security level access rule on the global and interface access lists

Answer: B

Explanation:

http://www.cisco.com/en/US/docs/security/security_management/cisco_security_manager/

security_manager/4.3/user/guide/fwaccess.html

Understanding Device Specific Access Rule Behavior

If you do not create an access rule policy, the following is the default behavior based on the type of device, and what happens when you create an access rule:

- IOS devices—Permit all traffic through an interface.

When you create an access rule permitting source A to destination B without configuring TCP/UDP inspection on the inspection rule table, or configuring the established advanced option on the rule, the device permits any packet from A to B. However, for any returning packet from B to A, the packet is not allowed, unless there is a corresponding access rule permitting that packet. If you configure TCP/UDP inspection on the traffic in the inspection rule table, a rule permitting B to A is not needed in the access rule, as any returning packet from B to A automatically passes the device.

- ASA and PIX devices—Permit traffic from a higher-security interface to a lower-security interface. Otherwise, all traffic is denied.

If an access rule allows TCP/UDP traffic in one direction, the appliance automatically allows return traffic (you do not need to configure a corresponding rule for the return traffic), except for ICMP traffic, which does require a return rule (where you permit the reverse source and destination), or you must create an inspection rule for ICMP.

- FWSM devices—Deny all traffic entering an interface, permit all traffic leaving an interface. You must configure access rules to allow any traffic to enter the device.

QUESTION NO: 81

Which option can cause the interactive setup script not to work on a Cisco ASA 5520 appliance running software version 8.4.1?

- A. The clock has not been set on the Cisco ASA appliance using the clock set command.
- B. The HTTP server has not been enabled using the http server enable command.
- C. The domain name has not been configured using the domain-name command.
- D. The inside interface IP address has not been configured using the ip address command.
- E. The management 0/0 interface has not been configured as management-only and assigned a name using the nameif command.

Answer: E

Explanation:

<http://www.checkthenetwork.com/networksecurityCiscoASA1.asp>

shows need for nameif and

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/intparam.html>

shows management only The ASA 5510 and higher adaptive security appliance also includes the following type:

- management The management interface is a Fast Ethernet interface designed for management traffic only, and is specified as management 0/0. You can, however, use it for through traffic if desired (see the management-only command). In transparent firewall mode, you can use the management interface in addition to the two interfaces allowed for through traffic. You can also add subinterfaces to the management interface to provide management in each security context for multiple context mode.

Append the subinterface ID to the physical interface ID separated by a period (.).

In multiple context mode, enter the mapped name if one was assigned using the allocate-interface command.

For example, enter the following command:

```
hostname(config)# interface gigabitethernet0/1.1
```

Step 2 To name the interface, enter the following command:

```
hostname(config-if)# nameif name
```

The name is a text string up to 48 characters, and is not case-sensitive. You can change the name by reentering this command with a new value. Do not enter the no form, because that command causes all commands that refer to that name to be deleted.

Step 3 To set the security level, enter the following command:

```
hostname(config-if)# security-level number
```

Where number is an integer between 0 (lowest) and 100 (highest).

Step 4 (Optional) To set an interface to management-only mode, enter the following command:

```
hostname(config-if)# management-only
```

The ASA 5510 and higher adaptive security appliance includes a dedicated management interface called Management 0/0, which is meant to support traffic to the security appliance. However, you can configure any interface to be a management-only interface using the management-only command. Also, for Management 0/0, you can disable management-only mode so the interface can pass through traffic just like any other interface.

QUESTION NO: 82

Which statement about the Cisco ASA 5585-X appliance is true?

- A. The IPS SSP must be installed in slot 0 (bottom slot) and the firewall/VPN SSP must be installed in slot 1 (top slot).
- B. The IPS SSP operates independently. The firewall/VPN SSP is not necessary to support the IPS SSP.
- C. The ASA 5585-X appliance supports three types of SSP (the firewall/VPN SSP, the IPS SSP, and the CSC SSP).
- D. The ASA 5585-X appliance with the firewall/VPN SSP-60 has a maximum firewall throughput of 10 Gb/s.
- E. All IPS traffic (except the IPS management interface traffic) must flow through the firewall/VPN SSP first before it can be redirected to the IPS SSP.

Answer: E

Explanation:

http://www.cisco.com/en/US/docs/security/asa/quick_start/ips/ips_qsg.pdf

The IPS module runs a separate application from the ASA. The IPS module might include an external management interface so you can connect to the IPS module directly; if it does not have a management interface, you can connect to the IPS module through the ASA interface. Any other interfaces on the IPS module, if available for your model, are used for ASA traffic only. Traffic goes through the firewall checks before being forwarded to the IPS module.

QUESTION NO: 83

Which logging mechanism is configured using MPF and allows high-volume traffic-related events to be exported from the Cisco ASA appliance in a more efficient and scalable manner compared to classic syslog logging?

- A. SDEE
- B. Secure SYSLOG
- C. XML
- D. NSEL
- E. SNMPv3

Answer: D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/monitor_nsel.html#wp1111174

QUESTION NO: 84

Refer to the exhibit.

Match Criteria: Original Packet					Action: Translated Packet			
#	Source Intf	Dest Intf	Source	Destination	Service	Source	Destination	Service
"Network Object" NAT (No rules)								
1	inside	outside	insidenet	Partner-internal-subnets	any	insidenatted (D) outside	outnatted	-- Original --

Which option completes the CLI NAT configuration command to match the Cisco ASDM NAT configuration?

object network insidenatted

range 10.1.2.10 10.1.2.20

!

object network insidenet

range 172.16.1.10 172.16.1.100

!

object network outnatted

range 192.168.3.100 192.168.3.150

!

nat (inside,outside) after-auto 1 _____? _____

- A. source dynamic insidenet insidenatted destination static Partner-internal-subnets outnatted
- B. source dynamic insidenet insidenatted interface destination static Partner-internal-subnets outnatted
- C. source dynamic insidenet insidenatted destination static Partner-internal-subnets outnatted interface
- D. source dynamic insidenet interface destination static Partner-internal-subnets outnatted
- E. source dynamic insidenatted insidenet destination static Partner-internal-subnets outnatted
- F. source dynamic insidenatted interface destination static Partner-internal-subnets outnatted

Answer: B

Explanation:

QUESTION NO: 85

By default, not all services in the default inspection class are inspected. Which Cisco ASA CLI command do you use to determine which inspect actions are applied to the default inspection class?

- A. show policy-map global_policy
- B. show policy-map inspection_default
- C. show class-map inspection_default
- D. show class-map default-inspection-traffic
- E. show service-policy global

Answer: E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa81/command/ref/s7.html#wp1254424>

show service-policy

To display the service policy statistics, use the **show service-policy** command in privileged EXEC mode.

show service-policy [global | interface *intf*] [csc | inspect | ips | police | priority | shape]

show service-policy [global | interface *intf*] [set connection [details]]

show service-policy [global | interface *intf*] [flow protocol {host *src_host* | *src_ip* *src_mask*} [eq *src_port*] {host *dest_host* | *dest_ip* *dest_mask*} [eq *dest_port*]]

Syntax Description

csc	(Optional) Limits the output to policies that include the csc command.
dest_ip dest_mask	The destination IP address and netmask of the traffic flow.
details	(Optional) Displays per-client connection information, if a per-client connection limit is enabled.
eq dest_port	(Optional) The equals operator, requiring the destination port to match the port number that follows.
eq src_port	(Optional) The equals operator, requiring the source port to match the port number that follows.
flow protocol	(Optional) Specifies a traffic flow for which you want to see the policies that the security appliance would apply to the flow. The argument flow keyword specifies the flow in ip-5-tuple format. Valid values for the <i>protocol</i> argument are listed in the "Usage Guidelines" section.
global	(Optional) Limits output to the global policy, which applies to all interfaces.
host dest_host	The host destination IP address of the traffic flow.
host src_host	The host source IP address of the traffic flow.
icmp_control_message	(Optional) Specifies an ICMP control message of the traffic flow. Valid values for the <i>icmp_control_message</i> argument are listed in the "Usage Guidelines" section.
icmp_number	(Optional) Specifies the ICMP protocol number of the traffic flow.
inspect	(Optional) Limits the output to policies that include an inspect command.
interface intf	(Optional) Displays policies applied to the interface specified by the <i>intf</i> argument, where <i>intf</i> is the interface name given by the nameif command.
ips	Limits output to policies that include the ips command.
police	Limits output to policies that include the police command.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 86

Which Cisco ASDM 6.4.1 pane is used to enable the Cisco ASA appliance to perform TCP checksum verifications?

- A. Configuration > Firewall > Service Policy Rules

- B. Configuration > Firewall > Advanced > IP Audit > IP Audit Policy
- C. Configuration > Firewall > Advanced > IP Audit > IP Audit Signatures
- D. Configuration > Firewall > Advanced > TCP options
- E. Configuration > Firewall > Objects > TCP Maps
- F. Configuration > Firewall > Objects > Inspect Maps

Answer: E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/asdm52/release/notes/rn524.html>
shows:

Firewall Features	
TCP Normalization Enhancements	You can now configure TCP normalization actions for certain packet types. Previously, the default actions for these kinds of packets was to drop the packet. Now you can set the TCP normalizer to allow the packets. • TCP invalid ACK check (the invalid-ack command) • TCP packet sequence past window check (the seq-past-window command) • TCP SYN-ACK with data check (the synack-data command) You can also set the TCP out-of-order packet buffer timeout (the queue command timeout keyword). Previously, the timeout was 4 seconds. You can now set the timeout to another value. The default action for packets that exceed MSS has changed from drop to allow (the exceed-mss command). The following non-configurable actions have changed from drop to clear for these packet types: • Bad option length in TCP • TCP Window scale on non-SYN • Bad TCP window scale value • Bad TCP SACK ALLOW option In ASDM, see the Configuration > Global Objects > TCP Maps pane. <i>Also available in Version 8.0(4).</i>

C:\Documents and Settings\user-nwz\Desktop\1.JPG

<http://www.cisco.com/en/US/docs/security/asa/asa80/asdm60/user/guide/protect.html>

shows

- a. In the TCP Map Name field, enter a name.
- b. In the Queue Limit field, enter the maximum number of out-of-order packets, between 0 and 250.
- c. In the Reserved Bits area, click Clear and allow, Allow only, or Drop.
Allow only allows packets with the reserved bits in the TCP header.
Clear and allow clears the reserved bits in the TCP header and allows the packet.
Drop drops the packet with the reserved bits in the TCP header.

d. Check any of the following options:

- Clear Urgent Flag—Allows or clears the URG pointer through the security appliance.
- Drop Connection on Window Variation—Drops a connection that has changed its window size unexpectedly.
- Drop Packets that Exceed Maximum Segment Size—Allows or drops packets that exceed MSS set by peer.
- Check if transmitted data is the same as original—Enables and disables the retransmit data checks.
- Drop SYN Packets With Data—Allows or drops SYN packets with data.
- Enable TTL Evasion Protection—Enables or disables the TTL evasion protection offered by the securityappliance.

•Verify TCP Checksum—Enables and disables checksum verification.

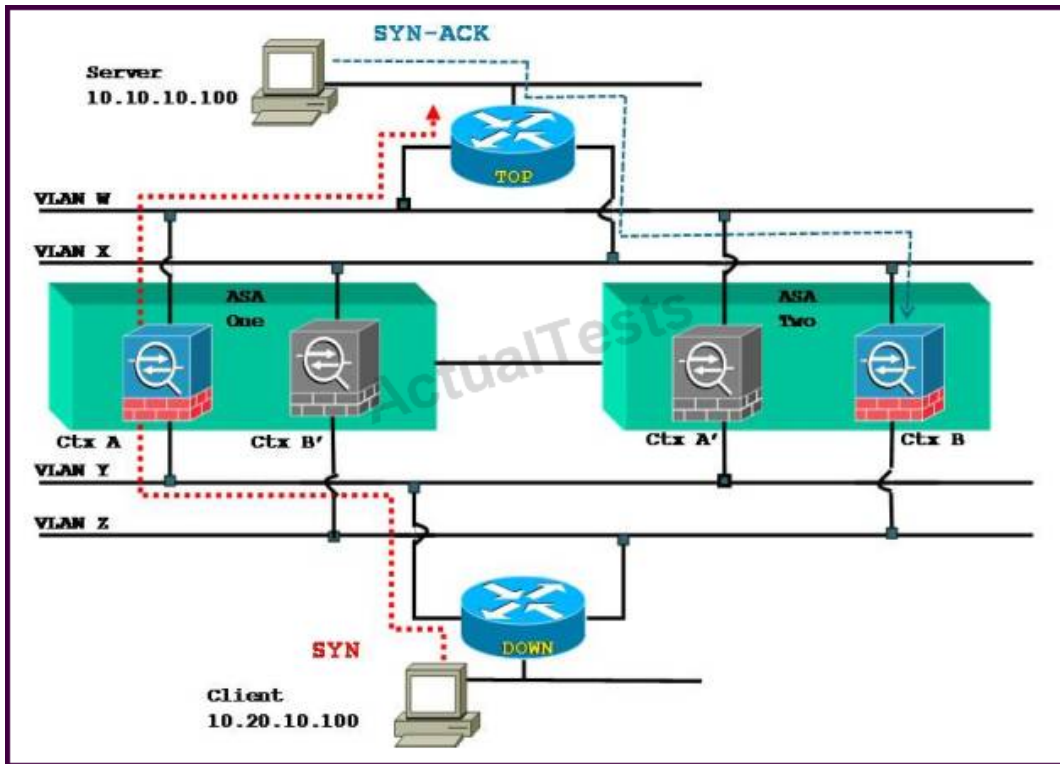
e. To set TCP options, check any of the following options:

- Clear Selective Ack—Lists whether the selective-ack TCP option is allowed or cleared.
- Clear TCP Timestamp—Lists whether the TCP timestamp option is allowed or cleared.
- Clear Window Scale—Lists whether the window scale timestamp option is allowed or cleared.
- Range—Lists the valid TCP options ranges, which should fall within 6-7 and 9-255. The lower bound shouldbe less than or equal to the upper bound.

f. Click OK.

QUESTION NO: 87

Refer to the exhibit.



Which two configurations are required on the Cisco ASAs so that the return traffic from the 10.10.10.100 outside server back to the 10.20.10.100 inside client can be rerouted from the Active Ctx B context in ASA Two to the Active Ctx A context in ASA One? (Choose two.)

- A. stateful active/active failover
- B. dynamic routing (EIGRP or OSPF or RIP)
- C. ASR-group
- D. no NAT-control
- E. policy-based routing
- F. TCP/UDP connections replication

Answer: A,C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/ha_active_active.html

Configuring Support for Asymmetrically Routed Packets When running in Active/Active failover, a unit may receive a return packet for a connection that originated through its peer unit. Because the ASA that receives the packet does not have any connection information for the packet, the packet is dropped. This most commonly occurs when the two ASAs in an Active/Active failover pair are connected to different service providers and the outbound connection does not use a NAT address.

You can prevent the return packets from being dropped using the `asr-group` command on interfaces where this is likely to occur. When an interface configured with the `asr-group` command receives a packet for which it has no session information, it checks the session information for the other interfaces that are in the same group. If it does not find a match, the packet is dropped. If it

finds a match, then one of the following actions occurs:

- If the incoming traffic originated on a peer unit, some or all of the layer 2 header is rewritten and the packet is redirected to the other unit. This redirection continues as long as the session is active.
- If the incoming traffic originated on a different interface on the same unit, some or all of the layer 2 header is rewritten and the packet is reinjected into the stream.

QUESTION NO: 88

Refer to the exhibit.

Name	Match Conditions		
	Match Type	Criterion	Value
_default_GoToMyPC-tunnel	AND	Request Arguments Request URI	R _default_GoToMyPC-tunnel R _default_GoToMyPC-tunnel
_default_aim-messenger		Request Header Field (host)	R _default_aim-messenger
_default_firethru-tunnel	AND	Request Header Field (host) Request URI	R _default_firethru-tunnel_1 R _default_firethru-tunnel_2
_default_gator		Request Header Field (user-agent)	R _default_gator
_default_gnu-http-tunnel	AND	Request Arguments Request URI	R _default_gnu-http-tunnel_ R _default_gnu-http-tunnel_
_default_http-tunnel		Request URI	R _default_http-tunnel
_default_httpport-tunnel		Request Header Field (host)	R _default_httpport-tunnel
_default_kazaa		Response Header Field Count (_default_... > 0	
_default_msn-messenger		Response Header Field (content-type)	R _default_msn-messenger
_default_shoutcast-tunneling-pr...		Request Header Field (_default_icy-meta...	R _default_shoutcast-tunneli
_default_windows-media-player...		Request Header Field (user-agent)	R _default_windows-media-pl
_default_yahoo-messenger		Request Body	R _default_yahoo-messenger

Which two statements about the class maps are true? (Choose two.)

- A. These class maps are referenced within the global policy by default for HTTP inspection.
- B. These class maps are all type inspect http class maps.
- C. These class maps classify traffic using regular expressions.
- D. These class maps are Layer 3/4 class maps.
- E. These class maps are used within the inspection_default class map for matching the default inspection traffic.

Answer: B,C

Explanation:

QUESTION NO: 89

Which three Cisco ASA configuration commands are used to enable the Cisco ASA to log only the debug output to syslog? (Choose three.)

- A. logging list test message 711001
- B. logging debug-trace
- C. logging trap debugging
- D. logging message 711001 level 7
- E. logging trap test

Answer: A,B,E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa80/command/reference/l2.html#wp1754683>

logging list

To create a logging list to use in other commands to specify messages by various criteria (logging level, event class, and message IDs), use the **logging list** command in global configuration mode. To remove the list, use the **no** form of this command.

logging list *name* [*level level* [*class event_class*] | *message start_id[-end_id]*]

no logging list *name*

Syntax Description

class <i>event_class</i>	(Optional) Sets the class of events for system log messages. For the level specified, only system log messages of the class specified are identified by the command. See the "Usage Guidelines" section for a list of classes.
level <i>level</i>	Sets the maximum level for system log messages. For example, if you set the level to 3, then the security appliance generates system log messages for levels 3, 2, 1, and 0. You can specify either the number or the name, as follows: • 0 or emergencies—System unusable. • 1 or alerts—Take immediate action. • 2 or critical—Critical condition. • 3 or errors—Error. • 4 or warnings—Warning. • 5 or notifications—Normal but significant condition. • 6 or informational—Information. • 7 or debugging—Debug messages, log FTP commands, and WWW URLs.
message <i>start_id[-end_id]</i>	Specifies a message ID or range of IDs. To look up the default level of a message, use the show logging command or see the <i>Cisco ASA 5500 Series System Log Messages</i> .
<i>name</i>	Sets the logging list name.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

logging debug-trace

To redirect debugging messages to logs as system log message 711001 issued at severity level 7, use the **logging debug-trace** command in global configuration mode. To stop sending debugging messages to logs, use the **no** form of this command.

logging debug-trace

no logging debug-trace

Syntax Description

This command has no arguments or keywords.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

logging trap

To specify which system log messages the security appliance sends to a syslog server, use the **logging trap** command in global configuration mode. To remove this command from the configuration, use the **no** form of this command.

logging trap [*logging_list* | *level*]

no logging trap

Syntax Description

<i>level</i>	Sets the maximum level for system log messages. For example, if you set the level to 3, then the security appliance generates system log messages for levels 3, 2, 1, and 0. You can specify either the number or the name, as follows: • 0 or emergencies—System unusable. • 1 or alerts—Take immediate action. • 2 or critical—Critical condition. • 3 or errors—Error. • 4 or warnings—Warning. • 5 or notifications—Normal but significant condition. • 6 or informational—Information. • 7 or debugging—Debug messages, log FTP commands, and WWW URLs.
<i>logging_list</i>	Specifies the list that identifies the messages to send to the syslog server. For information about creating lists, see the logging list command.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

http://www.cisco.com/en/US/docs/security/asa/asa84/asdm64/configuration_guide/monitor_syslog.html#wp1131130

Step 4 of sending syslog to external syslog server

Check the Send debug messages as syslogs check box to redirect all debugging trace output to system logs.

The syslog message does not appear on the console if this option is enabled. Therefore, to view debugging messages, you must have logging enabled at the console and have it configured as the destination for the debugging syslog message number and severity level. The syslog message number to use is 711001. The default severity level for this syslog message is debugging.

Logging list

Creates a logging list to use in other commands to specify messages by various criteria (logging level, eventclass, and message IDs).

QUESTION NO: 90

Which five options are valid logging destinations for the Cisco ASA? (Choose five.)

- A. AAA server
- B. Cisco ASDM

- C. buffer
- D. SNMP traps
- E. LDAP server
- F. email
- G. TCP-based secure syslog server

Answer: B,C,D,F,G

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/asdm64/configuration_guide/monitor_syslog.html#wp1131130

Choose the name of the logging destination to which you want to apply a filter. Available logging destinations are as follows:

- ASDM
- E-Mail
- Internal buffer
- SNMP server
- Syslog server also
- Telnet or SSH session
- Console port

QUESTION NO: 91

When configuring security contexts on the Cisco ASA, which three resource class limits can be set using a rate limit? (Choose three.)

- A. address translation rate
- B. Cisco ASDM session rate
- C. connections rate
- D. MAC-address learning rate (when in transparent mode)
- E. syslog messages rate
- F. stateful packet inspections rate

Answer: C,E,F

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/mngcntxt.html#wp1113880>

Table 6-1 lists the resource types and the limits. See also the show resource types command.

QUESTION NO: 92

Which two statements about Cisco ASA redundant interface configuration are true? (Choose two.)

- A. Each redundant interface can have up to four physical interfaces as its member.
- B. When the standby interface becomes active, the Cisco ASA sends gratuitous ARP out on the standby interface.
- C. Interface duplex and speed configurations are configured under the redundant interface.
- D. Redundant interfaces use MAC address-based load balancing to load share traffic across multiple physical interfaces.
- E. Each Cisco ASA supports up to eight redundant interfaces.

Answer: B,E

Explanation:

Configuring a Redundant Interface

A logical redundant interface pairs an active and a standby physical interface. When the active interface fails, the standby interface becomes active and starts passing traffic. You can configure a redundant interface to increase the security appliance reliability. This feature is separate from device-level failover, but you can configure redundant interfaces as well as failover if desired. You can configure up to 8 redundant interface pairs.

In Active/Standby failover, the active device uses the primary unit's MAC addresses. In the event of a failover, the secondary Cisco ASA becomes active and takes over the primary unit's MAC addresses, while the active device (now standby) takes over the standby unit's MAC addresses. Once the standby Cisco ASA becomes active, it sends out a gratuitous ARP on the network. A gratuitous ARP is an ARP request that the Cisco ASA sends out on the Ethernet networks with the source and destination IP addresses of the active IP addresses. The destination MAC address is the Ethernet broadcast address, i.e., ffff.ffff.ffff. All devices on the

Ethernet segment process this broadcast frame and update their ARP table with this information. Using gratuitous ARP, the Layer 2 devices, including bridges and switches, also update the Content Addressable Memory (CAM) table with the MAC address and the updated switch port information.

Using a virtual MAC address is recommended to avoid network disruptions. When a secondary Cisco ASA boots up before the primary Cisco ASA, it uses its physical MAC addresses as active

Layer 2 addresses.

However, when the primary Cisco ASA boots up, the secondary swaps the MAC addresses and uses the primary Cisco ASA's physical MAC addresses as active. With the virtual MAC address, Cisco ASA does not need to swap the MAC address.

When stateful failover is enabled, the active unit continually passes per-connection state information to the standby unit. After a failover occurs, the same connection information is available at the new active unit.

Supported end-user applications are not required to reconnect to keep the same communication session.

The state information passed to the standby unit includes these:

- The NAT translation table

- The TCP connection states

- The UDP connection states

- The ARP table

- The Layer 2 bridge table (when it runs in the transparent firewall mode)

- The HTTP connection states (if HTTP replication is enabled)

- The ISAKMP and IPsec SA table

- The GTP PDP connection database

The information that is not passed to the standby unit when stateful failover is enabled includes these:

- The HTTP connection table (unless HTTP replication is enabled)

- The user authentication (uauth) table

- The routing tables

- State information for security service modules

Note: If failover occurs within an active Cisco IP SoftPhone session, the call remains active because the call session state information is replicated to the standby unit. When the call is terminated, the IP SoftPhone client loses connection with the Call Manager. This occurs because there is no session information for the CTIQBEhang-up message on the standby unit. When the IP SoftPhone client does not receive a response back from the Call Manager within a certain time period, it considers the Call Manager unreachable and unregisters itself.

QUESTION NO: 93

The Cisco ASA must support dynamic routing and terminating VPN traffic. Which three Cisco ASA

options will not support these requirements? (Choose three.)

- A. transparent mode
- B. multiple context mode
- C. active/standby failover mode
- D. active/active failover mode
- E. routed mode
- F. no NAT-control

Answer: A,B,D

Explanation:

Dynamic routing (OSPF and RIP (in passive mode)) is supported by routed firewall.

Dynamic routing is NOT supported in Transparent UNLESS you can allow dynamic routing

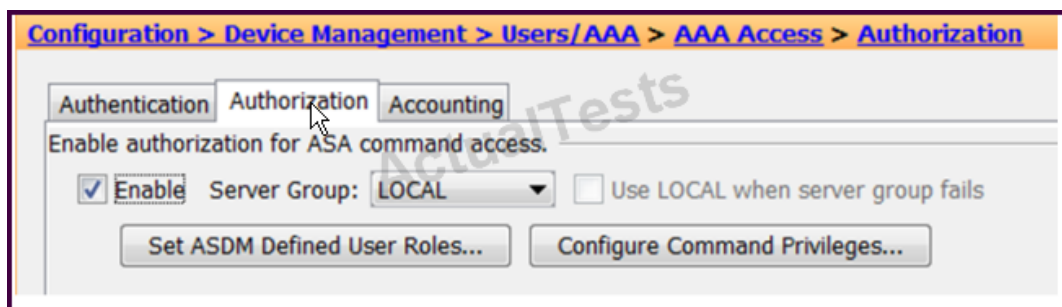
protocols through the ASA using an extended access list. Dynamic routing is NOT supported in

Multiple context mode

I HAD suggested ABE but note the NOT in question

QUESTION NO: 94

Refer to the exhibit.



Which two functions will the Set ASDM Defined User Roles perform? (Choose two.)

- A. enables role based privilege levels to most Cisco ASA commands
- B. enables the Cisco ASDM user to assign privilege levels manually to individual commands or groups of commands
- C. enables command authorization with a remote TACACS+ server
- D. enables three predefined user account privileges (Admin=Priv 15, Read Only=Priv 5, Monitor Only=Priv 3)

Answer: A,D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa80/asdm60/user/guide/devaccss.html>

- To use predefined user account privileges, click Set ASDM Defined User Roles.

The ASDM Defined User Roles Setup dialog box shows the commands and their levels. Click Yes to use the predefined user account privileges: Admin (privilege level 15, with full access to all CLI commands; Read Only (privilege level 5, with read-only access); and Monitor Only (privilege level 3, with access to the Monitoring section only).

- To manually configure command levels, click the Configure Command Privileges button.

The Command Privileges Setup dialog box appears. You can view all commands by choosing --All Modes-- from the Command Mode drop-down list, or you can choose a configuration mode to view the commands available in that mode. For example, if you choose context, you can view all commands available in context configuration mode. If a command can be entered in user EXEC/privileged EXEC mode as well as configuration mode, and the command performs different actions in each mode, you can set the privilege level for these modes separately.

The Variant column displays show, clear, or cmd. You can set the privilege only for the show, clear, or configure form of the command. The configure form of the command is typically the form that causes a configuration change, either as the unmodified command (without the show or clear prefix) or as the no form.

To change the level of a command, double-click it or click Edit. You can set the level between 0 and 15. You can only configure the privilege level of the main command. For example, you can configure the level of all aaa commands, but not the level of the aaa authentication command and the aaa authorization command separately.

To change the level of all shown commands, click Select All and then Edit.

Click OK to accept your changes.

QUESTION NO: 95

Which two statements about Cisco ASA failover troubleshooting are true? (Choose two.)

- A.** With active/active failover, failover link troubleshooting should be done in the system execution space.
- B.** With active/active failover, ASR groups must be enabled.
- C.** With active/active failover, user data passing interfaces troubleshooting should be done within the context execution space.
- D.** The failed interface threshold is set to 1. Using the show monitor-interface command, if one of the monitored interfaces on both the primary and secondary Cisco ASA appliances is in the unknown state, a failover should occur.
- E.** Syslog level 1 messages will be generated on the standby unit only if the logging standby command is used.

Answer: A,C

Explanation:

System Configuration

The system administrator adds and manages contexts by configuring each context configuration location, allocated interfaces, and other context operating parameters in the system configuration, which, like a singlemode configuration, is the startup configuration. The system configuration identifies basic settings for the security appliance. The system configuration does not include any network interfaces or network settings for itself; rather, when the system needs to access network resources (such as downloading the contexts from the server), it uses one of the contexts that is designated as the admin context. The system configuration does include a specialized failover interface for failover traffic only.

Context Configurations

The security appliance includes a configuration for each context that identifies the security policy, interfaces, and almost all the options you can configure on a standalone device. You can store context configurations on the internal Flash memory or the external Flash memory card, or you can download them from a TFTP, FTP, or HTTP(S) server.

QUESTION NO: 96

When troubleshooting a Cisco ASA that is operating in multiple context mode, which two verification steps should be performed if a user context does not pass user traffic? (Choose two.)

- A.** Verify the interface status in the system execution space.
- B.** Verify the mac-address-table on the Cisco ASA.
- C.** Verify that unique MAC addresses are configured if the contexts are using nonshared interfaces.
- D.** Verify the interface status in the user context.
- E.** Verify the resource classes configuration by accessing the admin context.

Answer: A,D

Explanation:

<http://www.ciscopress.com/articles/article.asp?p=426641>

Packet Flow in Multiple Mode When the packets traverse through the security appliance in multiple mode, they are classified and forwarded to the right context. The packets are then processed based on the configured security policies on a context.

TPacket Classification In multiple mode, the security appliance must classify the packets to find out

which context should operate on them. The packet classification is done at the ingress interface point that tags the packets using the source IP address, source port, destination IP address, destination port, and the interface or VLAN. The packet is processed based on the security policies configured in that context.

That said we need to note also that:

System Configuration

The system administrator adds and manages contexts by configuring each context configuration location, allocated interfaces, and other context operating parameters in the system configuration, which, like a singlemode configuration, is the startup configuration. The system configuration identifies basic settings for the security appliance. The system configuration does not include any network interfaces or network settings for itself; rather, when the system needs to access network resources (such as downloading the contexts from the server), it uses one of the contexts that is designated as the admin context. The system configuration does include a specialized failover interface for failover traffic only.

Context Configurations The security appliance includes a configuration for each context that identifies the security policy, interfaces, and almost all the options you can configure on a standalone device. You can store context configurations on the internal Flash memory or the external Flash memory card, or you can download them from a TFTP, FTP, or HTTP(S) server.

QUESTION NO: 97

Refer to the exhibit.

Add Public Server

Use this panel to define the server that you wish to expose to a public interface. You will need to specify the private interface and address of the server and the service to be exposed, and then the public interface, address and service that the server will be seen at.

Private Interface: inside

Private IP Address: 10.1.1.10

Private Service: tcp/http

Public Interface: outside

Public IP Address: 192.168.1.1

Options

☐ Specify Public Service if different from Private Service. This will enable the static PAT.

Public Service: (TCP or UDP service only)

OK Cancel Help

On Cisco ASA Software Version 8.3 and later, which two sets of CLI configuration commands result from this Cisco ASDM configuration? (Choose two.)

- A. nat (inside) 1 10.1.1.10
global (outside) 1 192.168.1.1
- B. nat (outside) 1 192.168.1.1
global (inside 1 10.1.1.10
- C. static(inside,outside) 192.168.1.1 10.1.1.10 netmask 255.255.255.255 tcp 0 0 udp 0
- D. static(inside,outside) tcp 192.168.1.1 80 10.1.1.10 80
- E. object network 192.168.1.1
nat (inside,outside) static 10.1.1.10
- F. object network 10.1.1.10
nat (inside,outside) static 192.168.1.1
- G. access-list outside_access_in line 1 extended permit tcp any object 10.1.1.10 eq http
access-group outside_access_in in interface outside
- H. access-list outside_access_in line 1 extended permit tcp any object 192.168.1.1 eq http
access-group outside_access_in in interface outside

Answer: F,G

Explanation:

QUESTION NO: 98

On the Cisco ASA Software Version 8.4.1, which three parameters can be configured using the set connection command within a policy map? (Choose three.)

- A. per-client TCP and/or UDP idle timeout
- B. per-client TCP and/or UDP maximum session time
- C. TCP sequence number randomization
- D. maximum number of simultaneous embryonic connections
- E. maximum number of simultaneous TCP and/or UDP connections
- F. fragments reassembly options

Answer: C,D,E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/command/reference/s1.html#wp1424045>

set connection

To specify connection limits within a policy map for a traffic class, use the **set connection** command in class configuration mode. To remove these specifications, thereby allowing unlimited connections, use the **no** form of this command.

set connection [[conn-max *n*] [embryonic-conn-max *n*] [per-client-embryonic-max *n*] [per-client-max *n*] [random-sequence-number {enable | disable}]]

no set connection [[conn-max *n*] [embryonic-conn-max *n*] [per-client-embryonic-max *n*] [per-client-max *n*] [random-sequence-number {enable | disable}]]

Syntax Description

conn-max <i>n</i>	Sets the maximum number of simultaneous TCP and/or UDP connections that are allowed, between 0 and 65535. The default is 0, which allows unlimited connections. For example, if two servers are configured to allow simultaneous TCP and/or UDP connections, the connection limit is applied to each configured server separately. When configured under a class, this keyword restricts the maximum number of simultaneous connections that are allowed for the entire class. In this case, one attack host can consume all the connections and leave none of the rest of the hosts matched in the access list under the class.
embryonic-conn-max <i>n</i>	Sets the maximum number of simultaneous embryonic connections allowed, between 0 and 65535. The default is 0, which allows unlimited connections.
per-client-embryonic-max <i>n</i>	Sets the maximum number of simultaneous embryonic connections allowed per client, between 0 and 65535. A client is defined as the host that sends the initial packet of a connection (that builds the new connection) through the adaptive security appliance. If an access-list is used with a class-map to match traffic for this feature, the embryonic limit is applied per-host, and not the cumulative embryonic connections of all clients that match the access list. The default is 0, which allows unlimited connections. This keyword is not available for management class maps.
per-client-max <i>n</i>	Sets the maximum number of simultaneous connections allowed per client, between 0 and 65535. A client is defined as the host that sends the initial packet of a connection (that builds the new connection) through the adaptive security appliance. If an access-list is used with a class-map to match traffic for this feature, the connection limit is applied per-host, and not the cumulative connections of all clients that match the access list. The default is 0, which allows unlimited connections. This keyword is not available for management class maps. When configured under a class, this keyword restricts the maximum number of simultaneous connections that are allowed for each host that is matched through an access list under the class.
random-sequence-number {enable disable}	Enables or disables TCP sequence number randomization. This keyword is not available for management class maps. See the " Usage Guidelines " section for more information.

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 99

On Cisco ASA Software Version 8.4.1, which four inspections are enabled by default in the global policy? (Choose four.)

- A. HTTP
- B. ESMTP
- C. SKINNY
- D. ICMP
- E. TFTP
- F. SIP

Answer: B,C,E,F

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/inspect_overview.html

QUESTION NO: 100

Which two statements about traffic shaping capability on the Cisco ASA appliance are true? (Choose two.)

- A. Traffic shaping can be applied to all outgoing traffic on a physical interface or, in the case of the Cisco ASA 5505 appliance, on a VLAN.
- B. Traffic shaping can be applied in the input or output direction.
- C. Traffic shaping can cause jitter and delay.
- D. You can configure traffic shaping and priority queuing on the same interface.
- E. With traffic shaping, when traffic exceeds the maximum rate, the security appliance drops the excess traffic.

Answer: A,C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/qos.html#wp1083655>

Information About Traffic Shaping Traffic shaping is used to match device and link speeds, thereby controlling packet loss, variable delay, and link saturation, which can cause jitter and delay.

- Traffic shaping must be applied to all outgoing traffic on a physical interface or in the case of the ASA5505, on a VLAN. You cannot configure traffic shaping for specific types of traffic.

- Traffic shaping is implemented when packets are ready to be transmitted on an interface, so the rate calculation is performed based on the actual size of a packet to be transmitted, including all the possible overhead such as the IPsec header and L2 header.

- The shaped traffic includes both through-the-box and from-the-box traffic.

- The shape rate calculation is based on the standard token bucket algorithm. The token bucket size is twice the Burst Size value. See the "What is a Token Bucket?" section.

- When burst traffic exceeds the specified shape rate, packets are queued and transmitted later.

Following are some characteristics regarding the shape queue (for information about hierarchical priority queuing, see the "Information About Priority Queuing" section):

- The queue size is calculated based on the shape rate. The queue can hold the equivalent of 200-millisecond worth of shape rate traffic, assuming a 1500-byte packet. The minimum queue size is 64.

- When the queue limit is reached, packets are tail-dropped.

- Certain critical keep-alive packets such as OSPF Hello packets are never dropped.

- The time interval is derived by $\text{time_interval} = \text{burst_size} / \text{average_rate}$. The larger the time interval is, the

bustier the shaped traffic might be, and the longer the link might be idle. The effect can be best understood using the following exaggerated example:

Average Rate = 1000000

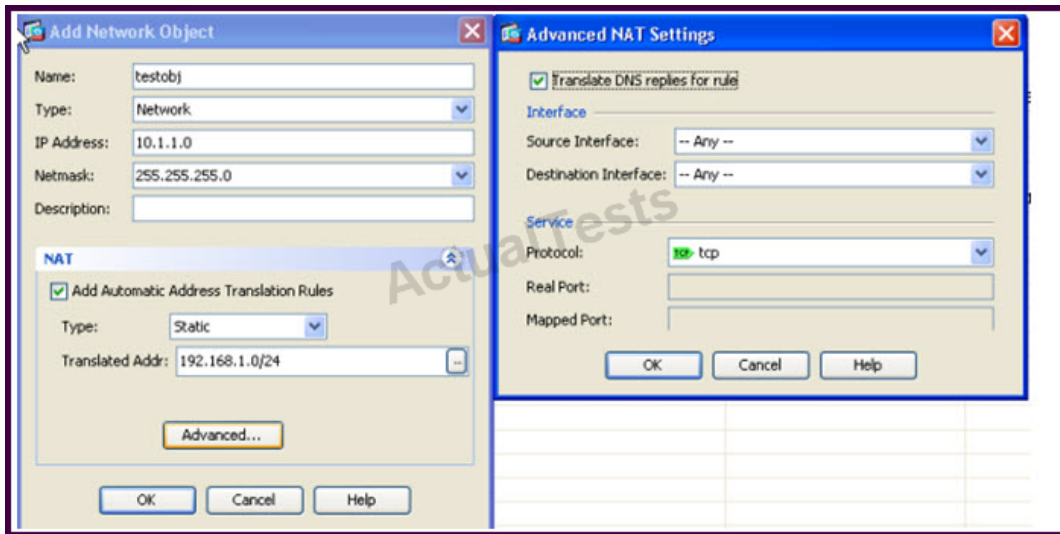
Burst Size = 1000000

In the above example, the time interval is 1 second, which means, 1 Mbps of traffic can be bursted out within the first 10 milliseconds of the 1-second interval on a 100 Mbps FE link and leave the remaining 990 milliseconds idle without being able to send any packets until the next time interval.

So if there is delay sensitive traffic such as voice traffic, the Burst Size should be reduced compared to the average rate so the time interval is reduced.

QUESTION NO: 101

Refer to the exhibit.



Which three CLI commands are generated by these Cisco ASDM configurations? (Choose three.)

- A. object-group network testobj
- B. object network testobj
- C. ip address 10.1.1.0 255.255.255.0
- D. subnet 10.1.1.0 255.255.255.0
- E. nat (any,any) static 192.168.1.0 dns
- F. nat (outside,inside) static 192.168.1.0 dns
- G. nat (inside,outside) static 192.168.1.0 dns
- H. nat (inside,any) static 192.168.1.0 dns
- I. nat (any,inside) static 192.168.1.0 dns

Answer: B,D,E

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/inspect_basic.html#wp1350877

When DNS inspection is enabled, DNS rewrite provides full support for NAT of DNS messages originating from any interface.

If a client on an inside network requests DNS resolution of an inside address from a DNS server

on an outside interface, the DNS A-record is translated correctly. If the DNS inspection engine is disabled, the A-record is not translated.

As long as DNS inspection remains enabled, you can configure DNS rewrite using the `alias`, `static`, or `nat` commands.

DNS Rewrite performs two functions:

- Translating a public address (the routable or "mapped" address) in a DNS reply to a private address (the "real" address) when the DNS client is on a private interface.
- Translating a private address to a public address when the DNS client is on the public interface.

QUESTION NO: 102

On Cisco ASA Software Version 8.3 and later, which two statements correctly describe the NAT table or NAT operations? (Choose two.)

- A.** The NAT table has four sections.
- B.** Manual NAT configurations are found in the first (top) and/or the last (bottom) section(s) of the NAT table.
- C.** Auto NAT also is referred to as Object NAT.
- D.** Auto NAT configurations are found only in the first (top) section of the NAT table.
- E.** The order of the NAT entries in the NAT table is not relevant to how the packets are matched against the NAT table.
- F.** Twice NAT is required for hosts on the inside to be accessible from the outside.

Answer: B,C

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa83/upgrading/migrating.html>

Table 3 New NAT Commands

New Commands	Configuration Mode	Syntax
Network Object NAT (Typically used for regular NAT configurations.)		
<code>nat dynamic</code>	Object network	<pre>object network name nat [(real_ifo,mapped_ifo)] dynamic {[mapped_inline_host_ip] [interface] [mapped_obj] [pat-pool mapped_obj [round-robin]] [interface]} [dns]</pre>
<code>nat static</code>	Object network	<pre>object network name nat [(real_ifo,mapped_ifo)] static {mapped_inline_ip mapped_obj interface} [dns service {tcp udp} real_port mapped_port] [no-proxy-arp] [route-lookup]</pre>
Twice NAT (Typically used for policy NAT configurations.)		
<code>nat source dynamic</code>	Global	<pre>nat [(real_ifo,mapped_ifo)] [line {after-object [line]}] source dynamic {real_obj any} {[mapped_obj] [pat-pool mapped_obj [round-robin]] [interface]} [destination static {mapped_obj interface} {real_obj any}] [service {mapped_dest_svo_obj real_dest_svo_obj} [dns] [unidirectional] [inactive] [description desc]]</pre>
<code>nat source static</code>	Global	<pre>nat [(real_ifo,mapped_ifo)] [line {after-object [line]}] source static {real_obj any} {mapped_obj interface any} [destination static {mapped_obj interface} {real_obj any}] [service {real_sro_mapped_dest_svo_obj any} mapped_sro_real_dest_svo_obj] [dns] [unidirectional [no-proxy-arp] [route-lookup]] [inactive] [description desc]]</pre>

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 103

The Cisco ASA software image has been erased from flash memory. Which two statements about the process to recover the Cisco ASA software image are true? (Choose two.)

- A. Access to the ROM monitor mode is required.
- B. The Cisco ASA appliance must have connectivity to the TFTP server where the Cisco ASA image is stored through the Management 0/0 interface.
- C. The copy tftp flash command is necessary to start the TFTP file transfer.
- D. The server command is necessary to set the TFTP server IP address.
- E. Cisco ASA password recovery must be enabled.

Answer: A,D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/admin_trouble.html

Using the ROM Monitor to Load a Software Image

To load a software image to an ASA from the ROM monitor mode using TFTP, perform the following steps:

Step 1 Connect to the ASA console port according to the instructions in the "Accessing the ApplianceCommand-Line Interface" section.

Step 2 Power off the ASA, then power it on.

Step 3 During startup, press the Escape key when you are prompted to enter ROMMON mode.

Step 4 In ROMMON mode, define the interface settings to the ASA, including the IP address, TFTP server

address, gateway address, software image file, and port, as follows:

```
rommon #1> ADDRESS=10.132.44.177
```

```
rommon #2> SERVER=10.129.0.30
```

```
rommon #3> GATEWAY=10.132.44.1
```

```
rommon #4> IMAGE=f1/asa840-232-k8.bin
```

```
rommon #5> PORT=Ethernet0/0
```

```
Ethernet0/0
```

Link is UP

MAC Address: 0012.d949.15b8

Note Be sure that the connection to the network already exists.

Step 5 To validate your settings, enter the set command.

rommon #6> set

ROMMON Variable Settings:

ADDRESS=10.132.44.177

SERVER=10.129.0.30

GATEWAY=10.132.44.1

PORT=Ethernet0/0

VLAN=untagged

IMAGE=f1/asa840-232-k8.bin

CONFIG=

LINKTIMEOUT=20

PKTTIMEOUT=4

RETRY=20

Step 6 Ping the TFTP server by entering the ping server command.

rommon #7> ping server

Sending 20, 100-byte ICMP Echoes to server 10.129.0.30, timeout is 4 seconds:

Success rate is 100 percent (20/20)

Step 7 Load the software image by entering the tftp command.

rommon #8> tftp

ROMMON Variable Settings:

ADDRESS=10.132.44.177

SERVER=10.129.0.30

GATEWAY=10.132.44.1

PORT=Ethernet0/0

VLAN=untagged

IMAGE=f1/asa840-232-k8.bin

CONFIG=

LINKTIMEOUT=20

PKTTIMEOUT=4

RETRY=20

tftp f1/asa840-232-k8.bin@10.129.0.30 via 10.132.44.1

Received 14450688 bytes

Launching TFTP Image...

Cisco ASA Security Appliance admin loader (3.0) #0: Mon Mar 5 16:00:07 MST 2011

Loading...

After the software image is successfully loaded, the ASA automatically exits ROMMON mode.

Step 8 To verify that the correct software image has been loaded into the ASA, check the version by entering the following command:

hostname# show version

QUESTION NO: 104

Which two Cisco ASA licensing features are correct with Cisco ASA Software Version 8.3 and later? (Choose two.)

- A. Identical licenses are not required on the primary and secondary Cisco ASA appliance.
- B. Cisco ASA appliances configured as failover pairs disregard the time-based activation keys.
- C. Time-based licenses are stackable in duration but not in capacity.
- D. A time-based license completely overrides the permanent license, ignoring all permanently licensed features until the time-based license is uninstalled.

Answer: A,C

Explanation:

http://www.cisco.com/en/US/prod/collateral/vpndevc/ps6032/ps6094/ps6120/product_bulletin_c25-593781.html

Time-based license stacking: Customers can extend time-based licenses such as Botnet Traffic Filter and SSLVPN Burst by applying multiple licenses.

Licensing of high-availability pairs: For several features, the requirement to deploy identical licenses on the standby unit in a high-availability pair has been removed. Security Plus licenses must still be purchased for both the Active and Standby units.

QUESTION NO: 105

Which four unicast or multicast routing protocols are supported by the Cisco ASA appliance? (Choose four.)

- A. RIP (v1 and v2)
- B. OSPF
- C. ISIS
- D. BGP
- E. EIGRP

- F. Bidirectional PIM
- G. MOSPF
- H. PIM dense mode

Answer: A,B,E,F

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/asdm64/configuration_guide/route_overview.html#wp1125708

•Enhanced Interior Gateway Routing Protocol (EIGRP)

Enhanced IGRP provides compatibility and seamless interoperability with IGRP routers. An automatic redistribution mechanism allows IGRP routes to be imported into Enhanced IGRP, and vice versa, so it is possible to add Enhanced IGRP gradually into an existing IGRP network. For more information on configuring EIGRP, see the chapter 'Configuring EIGRP'.

•Open Shortest Path First (OSPF)

Open Shortest Path First (OSPF) is a routing protocol developed for Internet Protocol (IP) networks by the interior gateway protocol (IGP) working group of the Internet Engineering Task Force (IETF). OSPF uses a link state algorithm in order to build and calculate the shortest path to all known destinations. Each router in an OSPF area contains an identical link-state database, which is a list of each of the router usable interfaces and reachable neighbors

For more information on configuring OSPF, see the chapter 'Configuring OSPF'.

•Routing Information ProtocolThe Routing Information Protocol (RIP) is a distance-vector protocol that uses hop count as its metric. RIP is widely used for routing traffic in the global Internet and is an interior gateway protocol (IGP), which means that it performs routing within a single autonomous system.

For more information on configuring RIP, see the chapter 'Configuring RIP'.

<http://www.cisco.com/en/US/docs/security/asa/asa81/config/guide/multicast.html#wp1060775>

Multicast Routing OverviewThe adaptive security appliance supports both stub multicast routing and PIM multicast routing. However, you cannot configure both concurrently on a single adaptive security appliance.

Stub multicast routing provides dynamic host registration and facilitates multicast routing. When configured for stub multicast routing, the adaptive security appliance acts as an IGMP proxy agent. Instead of fully participating in multicast routing, the adaptive security appliance forwards IGMP messages to an upstream multicast router, which sets up delivery of the multicast data. When configured for stub multicast routing, the adaptive security appliance cannot be configured for PIM.

The adaptive security appliance supports both PIM-SM and bi-directional PIM. PIM-SM is a multicast routing protocol that uses the underlying unicast routing information base or a separate multicast-capable routing information base. It builds unidirectional shared trees rooted at a single

Rendezvous Point per multicast group and optionally creates shortest-path trees per multicast source.

Bi-directional PIM is a variant of PIM-SM that builds bi-directional shared trees connecting multicast sources and receivers. Bi-directional trees are built using a DF election process operating on each link of the multicast topology. With the assistance of the DF, multicast data is forwarded from sources to the Rendezvous Point, and therefore along the shared tree to receivers, without requiring source-specific state. The DF election takes place during Rendezvous Point discovery and provides a default route to the Rendezvous Point.

QUESTION NO: 106

On Cisco ASA Software Version 8.4.1 and later, which three EtherChannel modes are supported? (Choose three.)

- A. active mode, which initiates LACP negotiation
- B. passive mode, which responds to LACP negotiation from the peer
- C. auto mode, which automatically responds to either PAgP or LACP negotiation from the peer
- D. on mode, which enables static port-channel mode
- E. off mode, which disables dynamic negotiation

Answer: A,B,D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/interface_start.html#wp1329030

Link Aggregation Control Protocol

The Link Aggregation Control Protocol (LACP) aggregates interfaces by exchanging the Link Aggregation Control Protocol Data Units (LACPDUs) between two network devices.

You can configure each physical interface in an EtherChannel to be:

- Active—Sends and receives LACP updates. An active EtherChannel can establish connectivity with either an active or a passive EtherChannel. You should use the active mode unless you need to minimize the amount of LACP traffic.
- Passive—Receives LACP updates. A passive EtherChannel can only establish connectivity with an active

EtherChannel.

- On—The EtherChannel is always on, and LACP is not used. An "on" EtherChannel can only establish a connection with another "on" EtherChannel.

LACP coordinates the automatic addition and deletion of links to the EtherChannel without user intervention. It also handles misconfigurations and checks that both ends of member interfaces are connected to the correct channel group. "On" mode cannot use standby interfaces in the channel group when an interface goes down, and the connectivity and configurations are not checked.

QUESTION NO: 107

Which two Cisco ASA configuration tasks are necessary to allow authenticated BGP sessions to pass through the Cisco ASA appliance? (Choose two.)

- A. Configure the Cisco ASA TCP normalizer to permit TCP option 19.
- B. Configure the Cisco ASA TCP Intercept to inspect the BGP packets (TCP port 179).
- C. Configure the Cisco ASA default global inspection policy to also statefully inspect the BGP flows.
- D. Configure the Cisco ASA TCP normalizer to disable TCP ISN randomization for the BGP flows.
- E. Configure TCP state bypass to allow the BGP flows.

Answer: A,D

Explanation:

1. The ASA strips TCP Option 19. This is used by Border Gateway Protocol (BGP) for authentication.

2. The ASA randomizes the TCP sequence numbers.

With Option 19 being stripped, BGP routers configured for authentication will not see credentials coming from their peer and thus will not establish the BGP neighbor.

First match the BGP Traffic.

```
access-list BGP extended permit tcp any eq bgp any
```

```
access-list BGP extended permit tcp any any eq bgp
```

Next create a TCP Map that allows Option 19.

```
tcp-map BGP
```

```
tcp-options range 19 19 allow
```

Now create a class-map to match the BGP ACL you created earlier.

```
class-map BGP
```

```
match access-list BGP
```

Finally, apply the class-map to the global policy:

```
policy-map global_policy
```

```
class BGP
```

```
set connection advanced-options BGP
```

Now for the second issue, while you are still in the policy-map configuration mode, you need to disable the random-sequence numbering.

set connection random-sequence-number disable

QUESTION NO: 108

Which two options show the required Cisco ASA command(s) to allow this scenario? (Choose two.)

An inside client on the 10.0.0.0/8 network connects to an outside server on the 172.16.0.0/16 network using TCP and the server port of 2001. The inside client negotiates a client port in the range between UDP ports 5000 to 5500. The outside server then can start sending UDP data to the inside client on the negotiated port within the specified UDP port range.

- A.** access-list INSIDE line 1 permit tcp 10.0.0.0 255.0.0.0 172.16.0.0 255.255.0.0 eq 2001
access-group INSIDE in interface inside
- B.** access-list INSIDE line 1 permit tcp 10.0.0.0 255.0.0.0 172.16.0.0 255.255.0.0 eq 2001
access-list INSIDE line 2 permit udp 10.0.0.0 255.0.0.0 172.16.0.0 255.255.0.0 eq established
access-group INSIDE in interface inside
- C.** access-list OUTSIDE line 1 permit tcp 172.16.0.0 255.255.0.0 eq 2001 10.0.0.0 255.0.0.0
access-list OUTSIDE line 2 permit udp 172.16.0.0 255.255.0.0 10.0.0.0 255.0.0.0 eq 5000-5500
access-group OUTSIDE in interface outside
- D.** access-list OUTSIDE line 1 permit tcp 172.16.0.0 255.255.0.0 eq 2001 10.0.0.0 255.0.0.0
access-list OUTSIDE line 2 permit udp 172.16.0.0 255.255.0.0 10.0.0.0 255.0.0.0 eq established
access-group OUTSIDE in interface outside
- E.** established tcp 2001 permit udp 5000-5500
- F.** established tcp 2001 permit from udp 5000-5500
- G.** established tcp 2001 permit to udp 5000-5500

Answer: A,G

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa72/command/reference/ef_72.html#wp1764664

Established command—This command allows return connections from a lower security host to a higher security host if there is already an established connection from the higher level host to the lower level host.

For same security interfaces, you can configure established commands for both directions.

QUESTION NO: 109

Which three actions can be applied to a traffic class within a type inspect policy map? (Choose three.)

- A. drop
- B. priority
- C. log
- D. pass
- E. inspect
- F. reset

Answer: A,C,F

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa80/configuration/guide/mpc.html>

```
hostname(config-pmap-c)# {[drop [send-protocol-error] |  
drop-connection [send-protocol-error]] mask | reset] [log] | rate-limit  
message_rate}
```

The drop keyword drops all packets that match.

The send-protocol-error keyword sends a protocol error message.

The drop-connection keyword drops the packet and closes the connection.

The mask keyword masks out the matching portion of the packet.

The reset keyword drops the packet, closes the connection, and sends a TCP reset to the server and/or client.

The log keyword, which you can use alone or with one of the other keywords, sends a system log message.

The rate-limit message_rate argument limits the rate of messages.

QUESTION NO: 110

On Cisco ASA Software Version 8.4 and later, which two options show the maximum number of active and standby ports that an EtherChannel can have? (Choose two.)

- A. 2 active ports
- B. 4 active ports

- C. 6 active ports
- D. 8 active ports
- E. 2 standby ports
- F. 4 standby ports
- G. 6 standby ports
- H. 8 standby ports

Answer: D,H

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/asdm64/configuration_guide/interface_start.pdf

Channel Group Interfaces

Each channel group can have eight active interfaces. Note that you can assign up to 16 interfaces to a channel group. While only eight interfaces can be active, the remaining interfaces can act as standby links in case of interface failure.

All interfaces in the channel group must be the same type and speed. The first interface added to the channel group determines the correct type and speed.

The EtherChannel aggregates the traffic across all the available active interfaces in the channel. The port is selected using a proprietary hash algorithm, based on source or destination MAC addresses, IP addresses, TCP and UDP port numbers and VLAN numbers.

QUESTION NO: 111

Which three types of class maps can be configured on the Cisco ASA appliance? (Choose three.)

- A. control-plane
- B. regex
- C. inspect
- D. access-control
- E. management
- F. stack

Answer: B,C,E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa80/configuration/guide/mpc.html>

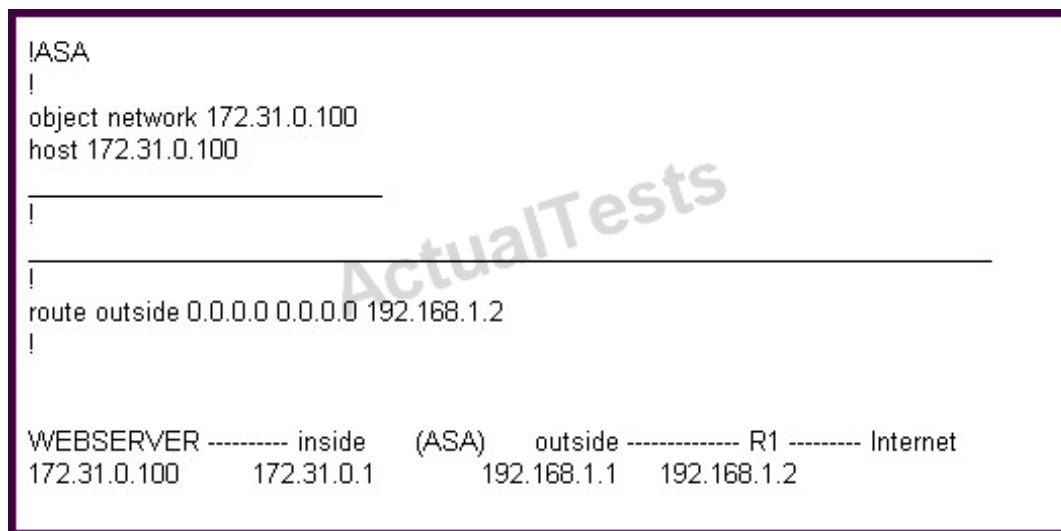
Maximum Class Maps The maximum number of class maps of all types is 255 in single mode or

per context in multiple mode. Classmaps include the following types:

- Layer 3/4 class maps (for through traffic and management traffic)
- Inspection class maps
- Regular expression class maps

QUESTION NO: 112

Refer to the partial Cisco ASA configuration and the network topology shown in the exhibit.



Which two Cisco ASA configuration commands are required so that any hosts on the Internet can HTTP to the WEBSERVER using the 192.168.1.100 IP address? (Choose two.)

- A. nat (inside,outside) static 192.168.1.100
- B. nat (inside,outside) static 172.31.0.100
- C. nat (inside,outside) static interface
- D. access-list outside_access_in extended permit tcp any object 172.31.0.100 eq http
- E. access-list outside_access_in extended permit tcp any object 192.168.1.100 eq http
- F. access-list outside_access_in extended permit tcp any object 192.168.1.1 eq http

Answer: A,D

Explanation:

QUESTION NO: 113

Which two statements about Cisco ASA 8.2 NAT configurations are true? (Choose two.)

- A. NAT operations can be implemented using the NAT, global, and static commands.
- B. If nat-control is enabled and a connection does not need a translation, then an identity NAT configuration is required.
- C. NAT configurations can use the any keyword as the input or output interface definition.
- D. The NAT table is read and processed from the top down until a translation rule is matched.
- E. Auto NAT links the translation to a network object.

Answer: A,B

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_configuration_example09186a008046f31a.shtml#IN1

QUESTION NO: 114

In which two directions are the Cisco ASA modular policy framework inspection policies applied? (Choose two.)

- A. in the ingress direction only when applied globally
- B. in the ingress direction only when applied on an interface
- C. in the egress direction only when applied globally
- D. in the egress direction only when applied on an interface
- E. bi-directionally when applied globally
- F. bi-directionally when applied on an interface

Answer: A,F

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa83/configuration/guide/mpf_service_policy.html#wp1162717

Feature Directionality

Actions are applied to traffic bidirectionally or unidirectionally depending on the feature. For features that are applied bidirectionally, all traffic that enters or exits the interface to which you apply the policy map is affected if the traffic matches the class map for both directions. When you use a global policy, all features are unidirectional; features that are normally bidirectional when applied to a single interface only apply to the ingress of each interface when applied globally. Because the policy is applied to all interfaces, the policy will be applied in both directions so bidirectionality in this case is redundant.

QUESTION NO: 115

Which three configurations are needed to enable SNMPv3 support on the Cisco ASA? (Choose three.)

- A. SNMPv3 Local EngineID
- B. SNMPv3 Remote EngineID
- C. SNMP Users
- D. SNMP Groups
- E. SNMP Community Strings
- F. SNMP Hosts

Answer: C,D,F

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa82/snmp/snmpv3_1.html

The adaptive security appliance requires that you configure the SNMP server group, the SNMP server user associated with the group, and the SNMP server host, which specifies the user for receiving SNMP traps.

To configure SNMP Version 3 operations, the required sequence of commands is as follows:

- snmp-server group
- snmp-server user
- snmp-server host

The following shows an example adaptive security appliance configuration:

```
hostname# snmp-server group authPriv v3 priv
hostname# snmp-server group authNoPriv v3 auth
hostname# snmp-server group noAuthNoPriv v3 noauth
```

QUESTION NO: 116

A customer is ordering a number of Cisco ASAs for their network. For the remote or home office, they are purchasing the Cisco ASA 5505. When ordering the licenses for their Cisco ASAs, which two licenses must they order that are "platform specific" to the Cisco ASA 5505? (Choose two.)

- A. AnyConnect Essentials license
- B. per-user Premium SSL VPN license
- C. VPN shared license
- D. internal user licenses

E. Security Plus license

Answer: D,E

Explanation:

Cisco ASA Model	ASA 5505 / Security Plus
	
Maximum Firewall Throughput ¹	150 Mbps
Maximum Firewall Throughput (Multi-Protocol)	-
Maximum Concurrent Threat Mitigation Throughput (Firewall + IPS Services)	75 Mbps with AIP SSC-5
Maximum Firewall Connections	10,000 /25,000
Maximum Firewall Connections/Second	4,000
Packets per second (64 byte)	85,000
Maximum 3DES/AES VPN Throughput ²	100 Mbps
Maximum Site-to-Site and IPsec IKEv1 Client VPN User Sessions	10/25
Maximum AnyConnect or Clientless VPN User Sessions	25
Bundled SSL VPN User Sessions	2
Security Contexts (Included, Maximum)	Not available
VLANs	3 (trunking disabled) /20

C:\Documents and Settings\user-nwz\Desktop\1.JPG

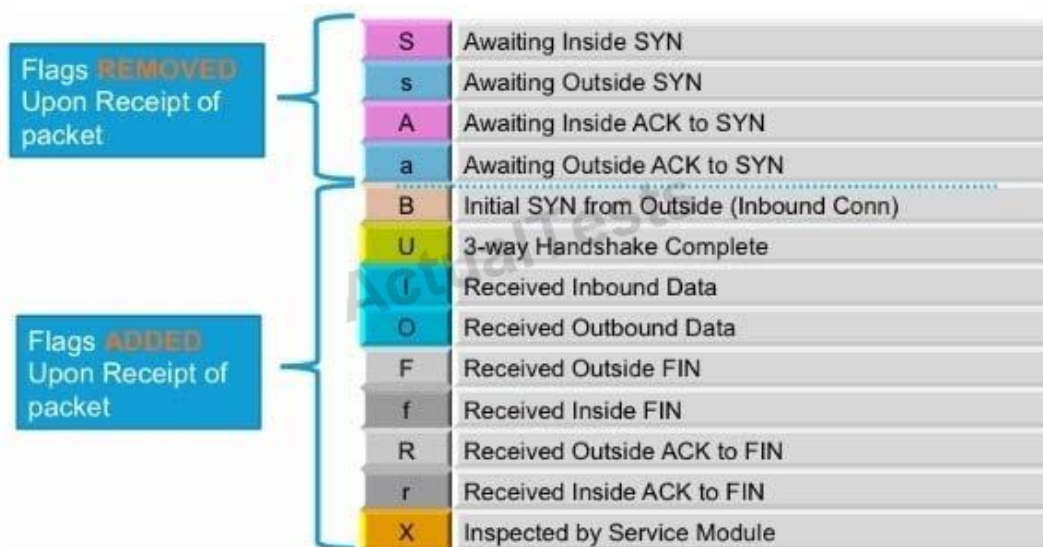
QUESTION NO: 117

Refer to the exhibit.

```
ASA5520# show conn
29 in use, 63 most used
TCP out 10.10.49.10:23 in 10.1.1.15:1026 idle 0:00:22 bytes 1774 flags UIO
<output omitted>
```

Which two statements are true? (Choose two.)

- A. The connection is awaiting outside ACK to SYN.
- B. The connection is initiated from the inside.
- C. The connection is active and has received inbound and outbound data.
- D. The connection is an incomplete TCP connection.
- E. The connection is a DNS connection.

Answer: B,C**Explanation:**

C:\Documents and Settings\user-nwz\Desktop\1.JPG

Flags: A - awaiting inside ACK to SYN, a - awaiting outside ACK to SYN,
 B - initial SYN from outside, b - TCP state-bypass or nailed, C - CTIQBE media,
 D - DNS, d - dump, E - outside back connection, F - outside FIN, f - inside FIN,
 G - group, g - MGCP, H - H.323, h - H.225.0, I - inbound data,
 i - incomplete, J - GTP, j - GTP data, K - GTP t3-response
 k - Skinny media, M - SMTP data, m - SIP media, n - GUP
 O - outbound data, P - inside back connection, p - Phone-proxy TFTP connection,
 q - SQL*Net data, R - outside acknowledged FIN,
 R - UDP SUNRPC, r - inside acknowledged FIN, S - awaiting inside SYN,
 s - awaiting outside SYN, T - SIP, t - SIP transient, U - up,
 V - VPN orphan, W - WAAS,
 X - inspected by service module

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 118

The Cisco ASA is configured in multiple mode and the security contexts share the same outside physical interface. Which two packet classification methods can be used by the Cisco ASA to determine which security context to forward the incoming traffic from the outside interface? (Choose two.)

- A. unique interface IP address
- B. unique interface MAC address
- C. routing table lookup
- D. MAC address table lookup
- E. unique global mapped IP addresses

Answer: B,E

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/contexts.html>

Unique Interfaces

If only one context is associated with the ingress interface, the ASA classifies the packet into that context. Intransparent firewall mode, unique interfaces for contexts are required, so this method is used to classify packets at all times.

Unique MAC Addresses

If multiple contexts share an interface, then the classifier uses the interface MAC address. The ASA lets you assign a different MAC address in each context to the same shared interface, whether it is a shared physical interface or a shared subinterface. By default, shared interfaces do not have unique MAC addresses; the interface uses the physical interface burned-in MAC address

in every context. An upstream router cannot route directly to a context without unique MAC addresses. You can set the MAC addresses manually when you configure each interface (see the "Configuring the MAC Address" section), or you can automatically generate MAC addresses (see the "Automatically Assigning MAC Addresses to Context Interfaces" section).

NAT Configuration

If you do not have unique MAC addresses, then the classifier intercepts the packet and performs a destination IP address lookup. All other fields are ignored; only the destination IP address is used. To use the destination address for classification, the classifier must have knowledge about the subnets located behind each security context. The classifier relies on the NAT configuration to determine the subnets in each context. The classifier matches the destination IP address to either a static command or a global command. In the case of the global command, the classifier does not need a matching nat command or an active NAT session to classify the packet. Whether the packet can communicate with the destination IP address after classification depends on how you configure NAT and NAT control.

For example, the classifier gains knowledge about subnets 10.10.10.0, 10.20.10.0 and 10.30.10.0 when the context administrators configure static commands in each context:

•Context A:

```
static (inside,shared) 10.10.10.0 10.10.10.0 netmask 255.255.255.0
```

•Context B:

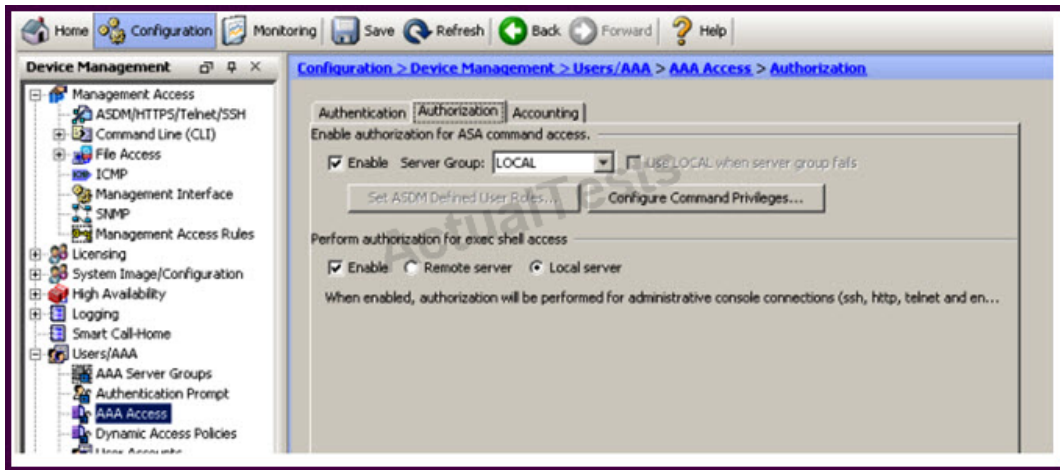
```
static (inside,shared) 10.20.10.0 10.20.10.0 netmask 255.255.255.0
```

•Context C:

```
static (inside,shared) 10.30.10.0 10.30.10.0 netmask 255.255.255.0
```

QUESTION NO: 119

Refer to the exhibit.



Which two CLI commands result from this configuration? (Choose two.)

- A. aaa authorization network LOCAL
- B. aaa authorization network default authentication-server LOCAL
- C. aaa authorization command LOCAL
- D. aaa authorization exec LOCAL
- E. aaa authorization exec authentication-server LOCAL
- F. aaa authorization exec authentication-server

Answer: C,D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/access_management.html#wp1145888

QUESTION NO: 120

Which three statements are the default security policy on a Cisco ASA appliance? (Choose three.)

- A. Traffic that goes from a high security level interface to a lower security level interface is allowed.
- B. Outbound TCP and UDP traffic is statefully inspected and returning traffic is allowed to traverse the Cisco ASA appliance.
- C. Traffic that goes from a low security level interface to a higher security level interface is allowed.
- D. Traffic between interfaces with the same security level is allowed by default.
- E. Traffic can enter and exit the same interface by default.
- F. When the Cisco ASA appliance is accessed for management purposes, the access must be made to the nearest Cisco ASA interface.
- G. Inbound TCP and UDP traffic is statefully inspected and returning traffic is allowed to traverse

the Cisco ASA appliance.

Answer: A,B,F

Explanation:

The security algorithm is responsible for implementing and enforcing your security policies.

The algorithm uses a tiered hierarchy that allows you to implement multiple levels of security. To accomplish this, each interface on the appliance is assigned a security level number from 0 to 100, where 0 is the least secure and 100 is the most secure. The algorithm uses these security levels to enforce its default policies.

Here are the four default security policy rules for traffic as it flows through the appliance:

Traffic flowing from a higher-level security interface to a lower one is permitted by default.

Traffic flowing from a lower-level security interface to a higher one is denied by default.

Traffic flowing from one interface to another with the same security level is denied by default.

Traffic flowing into and then out of the same interface is denied by default

http://www.cisco.com/en/US/docs/security/asa/asa83/configuration/guide/access_rules.html#wp1120072

Implicit Permits

For routed mode, the following types of traffic are allowed through by default:

- IPv4 traffic from a higher security interface to a lower security interface.
- IPv6 traffic from a higher security interface to a lower security interface.

For transparent mode, the following types of traffic are allowed through by default:

- IPv4 traffic from a higher security interface to a lower security interface.
- IPv6 traffic from a higher security interface to a lower security interface.
- ARPs in both directions.

Implicit Deny

Interface-specific access rules do not have an implicit deny at the end, but global rules on inbound traffic do have an implicit deny at the end of the list, so unless you explicitly permit it, traffic cannot pass. For example, if you want to allow all users to access a network through the adaptive security appliance except for particular addresses, then you need to deny the particular addresses and then permit all others.

When you have no global access rules in your configuration, the implicit deny rule is applied at the end of interface access rules. When you configure both an interface access rule and a global access rule, the implicit deny (any any) is no longer located at the end of the interface-based access rule. The implicit deny (any any) is enforced at the end of the global access rule. Logically, the entries on the interface-based access rule are processed first, followed by the entries on the global access rule, and then finally the implicit deny (any any) at the end of the global access rule.

For example, when you have an interface-based access rule and a global access rule in your

configuration, the following processing logic applies:

1. interface access control rules
2. global access control rules
3. default global access control rule (deny any any)

When only interface-based access rules are configured, the following processing logic applies:

1. interface access control rules
2. default interface access control rule (deny any any)

For EtherType rules, the implicit deny does not affect IPv4 or IPv6 traffic or ARPs; for example, if you allow

EtherType 8037 (the EtherType for IPX), the implicit deny at the end of the list does not block any IP traffic that you previously allowed with an access rule (or implicitly allowed from a high security interface to a low security interface). However, if you explicitly deny all traffic with an EtherType rule, then IP and ARP traffic is denied.

Management access to an interface other than the one from which you entered the adaptive security appliance is not supported. For example, if your management host is located on the outside interface, you can only initiate a management connection directly to the outside interface. The only exception to this rule is through a VPN connection, and entering the management-access command. For more information about the management-access command, see the Cisco ASA 5500 Series Command Reference.

QUESTION NO: 121

Which two configurations are the minimum needed to enable EIGRP on the Cisco ASA appliance? (Choose two.)

- A. Enable the EIGRP routing process and specify the AS number.
- B. Define the EIGRP default-metric.
- C. Configure the EIGRP router ID.
- D. Use the neighbor command(s) to specify the EIGRP neighbors.
- E. Use the network command(s) to enable EIGRP on the Cisco ASA interface(s).

Answer: A,E

Explanation:

http://www.cisco.com/en/US/products/ps6120/products_configuration_example09186a008086ebd2.shtml

!EIGRP Configuration - the CLI configuration is very similar to the
!Cisco IOS router EIGRP configuration.

QUESTION NO: 122

Refer to the exhibit and to the four HTTP inspection requirements and the Cisco ASA configuration.

```
!ASA partial config:
!
regex BASIC-SQL-INJECTION "[Ss][Ee][Ll][Ee][Cc][Tt].+[Ff][Rr][Oo][Mm]"
regex MY-URI "^/myapp"
!
policy-map type inspect http MY-HTTP-POLICY
parameters
  protocol-violation action drop-connection log
  match not request method get
    drop-connection log
  match not request uri regex MY-URI
    drop-connection log
  match request args regex BASIC-SQL-INJECTION
    drop-connection log
!
access-list WEB-SERVER-ACL permit tcp any eq http host 10.10.10.10
!
class-map WEB-SERVER-PROTECTION
match access-list WEB-SERVER-ACL
!
policy-map global_policy
class WEB-SERVER-PROTECTION
  inspect http MY-HTTP-POLICY
!
service-policy global_policy global
```

Which two statements about why the Cisco ASA configuration is not meeting the specified HTTP inspection requirements are true? (Choose two.)

1. All outside clients can use only the HTTP GET method on the protected 10.10.10.10 web server.
2. All outside clients can access only HTTP URIs starting with the "/myapp" string on the protected 10.10.10.10 web server.
3. The security appliance should drop all requests that contain basic SQL injection attempts (the string "SELECT" followed by the string "FROM") inside HTTP arguments.

4. The security appliance should drop all requests that do not conform to the HTTP protocol.

- A. Both instances of match not request should be changed to match request.
- B. The policy-map type inspect http MY-HTTP-POLICY configuration is missing the references to the class maps.
- C. The BASIC-SQL-INJECTION regular expression is not configured correctly.
- D. The MY-URI regular expression is not configured correctly.
- E. The WEB-SERVER-ACL ACL is not configured correctly.

Answer: D,E

Explanation:

QUESTION NO: 123 DRAG DROP

Based on this NAT command, drag the IP address network on the left to the correct NAT address type on the right.

Nat(inside, outside) source dynamic 10.0.1.0_obj 192.168.1.7_obj destination static 209.165.200.226_Server 209.265.201.21_Server

10.0.1.0_obj	Outside Local
192.168.1.7_obj	Inside Local
209.165.200.226_server	Outside global
209.165.201.21_server	Inside global

Answer:

10.0.1.0_obj	209.165.201.21_server
192.168.1.7_obj	10.0.1.0_obj
209.165.200.226_server	209.165.200.226_server
209.165.201.21_server	192.168.1.7_obj

Explanation:

209.165.201.21_server
10.0.1.0_obj
209.165.200.226_server
192.168.1.7_obj

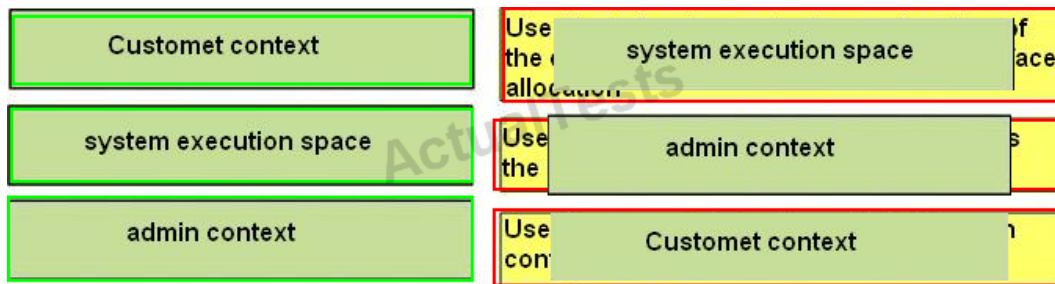
C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 124 DRAG DROP

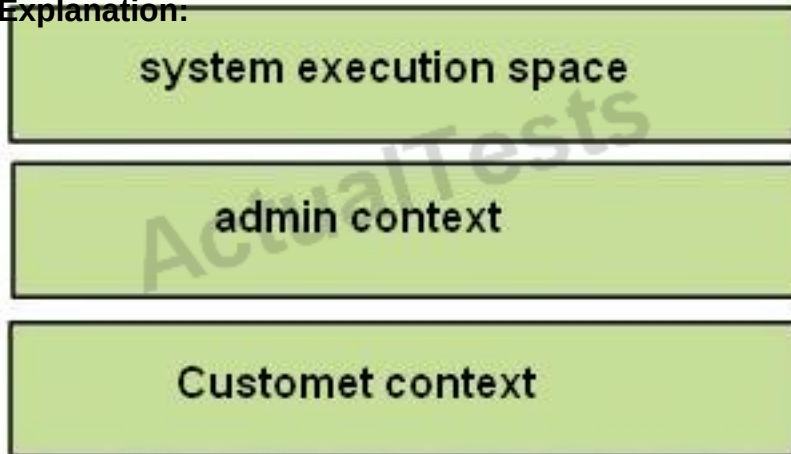
Drag the Cisco ASR modes from the left to the correct description on the right.

Customet context	Used to define the context name, location of the context startup configuration and interface allocation
system execution space	Used by the Cisco ASA appliance to access the required network resources
admin context	Used to support virtual firewall with its own configuration

Answer:



Explanation:



C:\Documents and Settings\user-nwz\Desktop\1.JPG

Systems Execution Space Used to define the context name, location of the context startup configuration and interface allocation Admin Context Used by the Cisco ASA appliance to access the required network resources Customer context Used to support virtual firewall with its own configuration

Context Configurations The security appliance includes a configuration for each context that identifies the security policy, interfaces, and almost all the options you can configure on a standalone device. You can store context configurations on the internal Flash memory or the external Flash memory card, or you can download them from a TFTP, FTP, or HTTP(S) server.

System Configuration

The system administrator adds and manages contexts by configuring each context configuration location, allocated interfaces, and other context operating parameters in the system configuration, which, like a single mode configuration, is the startup configuration. The system configuration identifies basic settings for the security appliance. The system configuration does not include any network interfaces or network settings for itself; rather, when the system needs to access network resources (such as downloading the contexts from the server), it uses one of the contexts that is designated as the admin context. The system configuration does include a specialized failover interface for failover traffic only.

Admin Context Configuration

The admin context is just like any other context, except that when a user logs in to the admin context, then that user has system administrator rights and can access the system and all other contexts. The admin context is not restricted in any way, and can be used as a regular context. However, because logging into the admin context grants you administrator privileges over all contexts, you might need to restrict access to the admin context to appropriate users. The admin context must reside on Flash memory, and not remotely.

If your system is already in multiple context mode, or if you convert from single mode, the admin context is created automatically as a file on the internal Flash memory called admin.cfg. This context is named "admin." If you do not want to use admin.cfg as the admin context, you can change the admin context.

QUESTION NO: 125 DRAG DROP

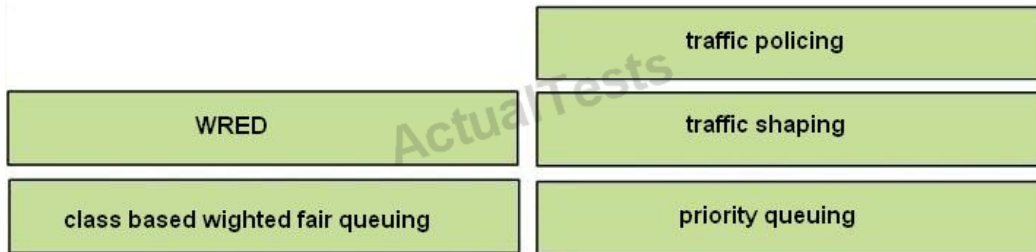
Click and drag the supported ASA QoS option on the left to the correct description on the right.
(Some of the options on the left are not used)

traffic shaping	can be applied in the input or output direction
WRED	can be applied only to the class-default traffic class
class based wighted fair queuing	can optionally define the queue limit and the transaction nng limit
priority queuing	
traffic policing	

Answer:

traffic shaping	traffic policing
WRED	traffic shaping
class based wighted fair queuing	priority queuing
priority queuing	
traffic policing	

Explanation:



C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 126 DRAG DROP

Drag the correct three access list entries (from the left) and drop them (on the right) in the order that is used when the interface ACL and global ACL are configured. Not all access list entries are required.



Answer:

implicit deny ip any any global access list entry

interface access list entries

implicit deny ip any any interface access list rule entry

global access list entries

implicit permit ip any any interface access list rule entry

implicit deny ip any any interface access list rule entry

implicit permit ip any any global access list rule entry

interface access list entries

global access list entries

implicit permit ip any any from higher security level interface to lower security level interface

Explanation:

implicit deny ip any any global access list entry

interface access list entries

global access list entries

implicit permit ip any any interface access list rule entry

implicit deny ip any any interface access list rule entry

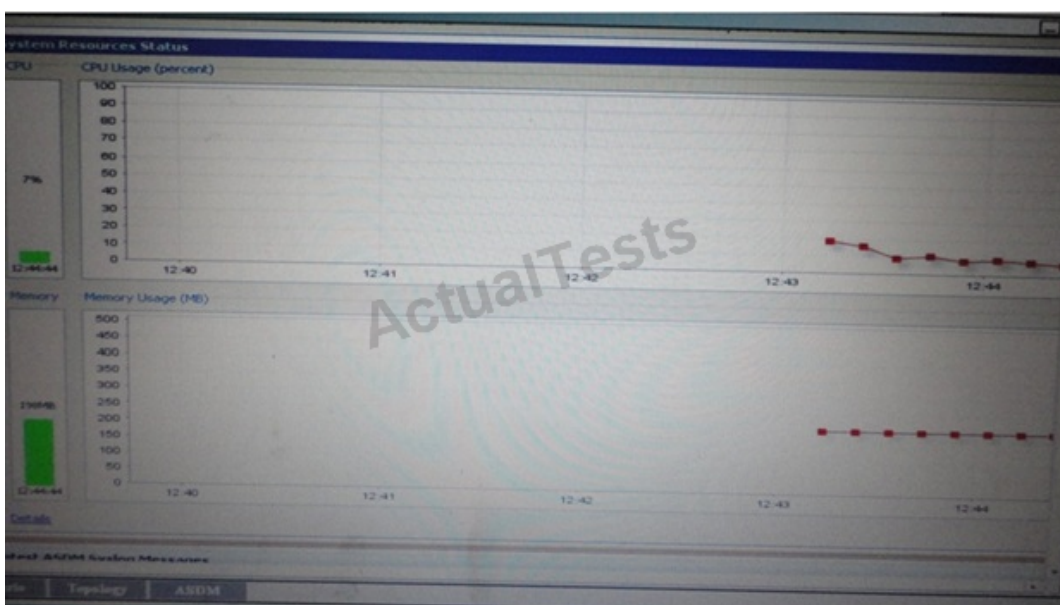
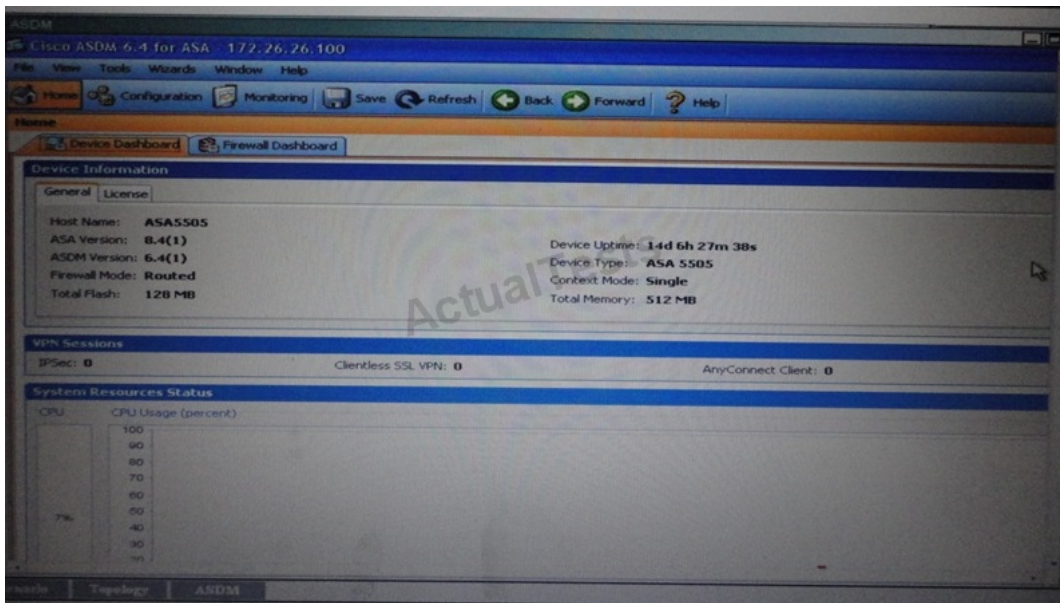
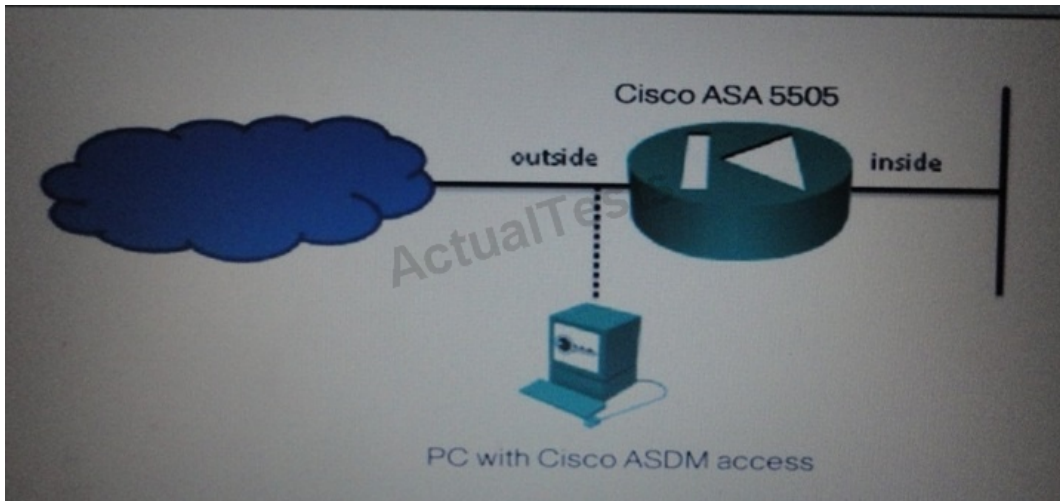
implicit permit ip any any global access list rule entry

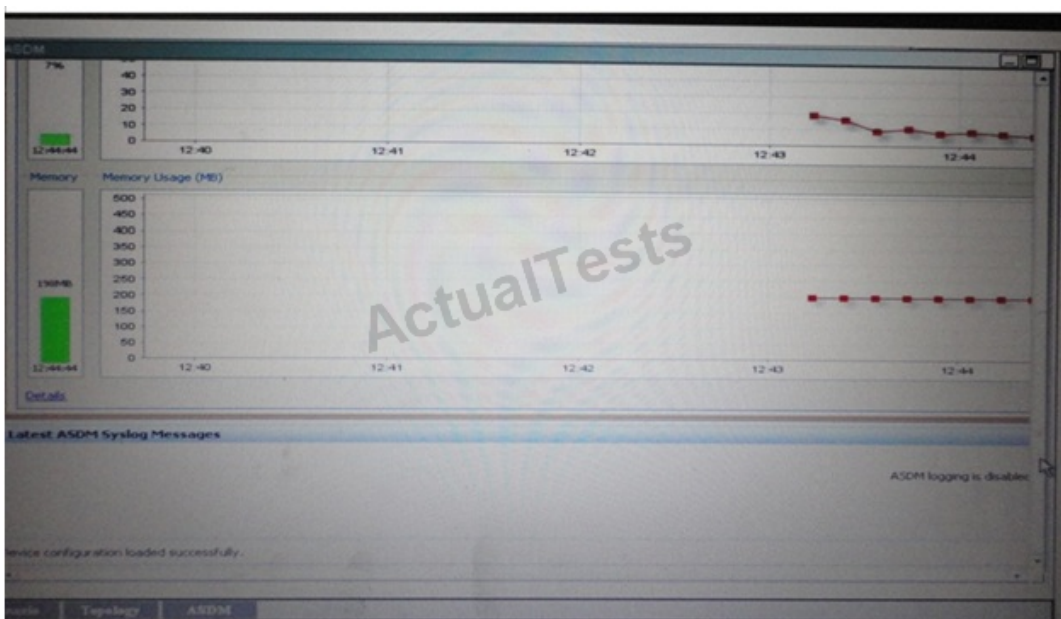
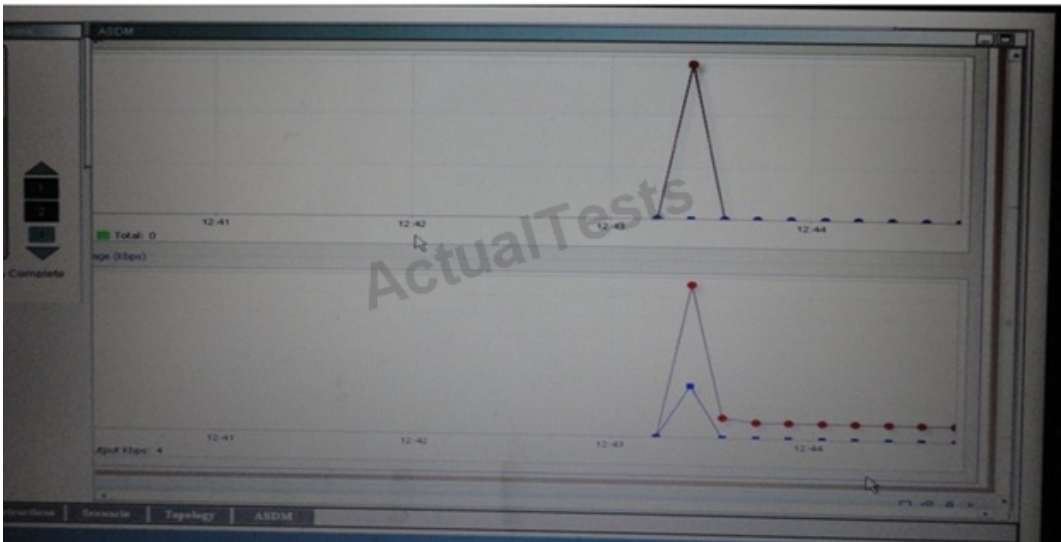
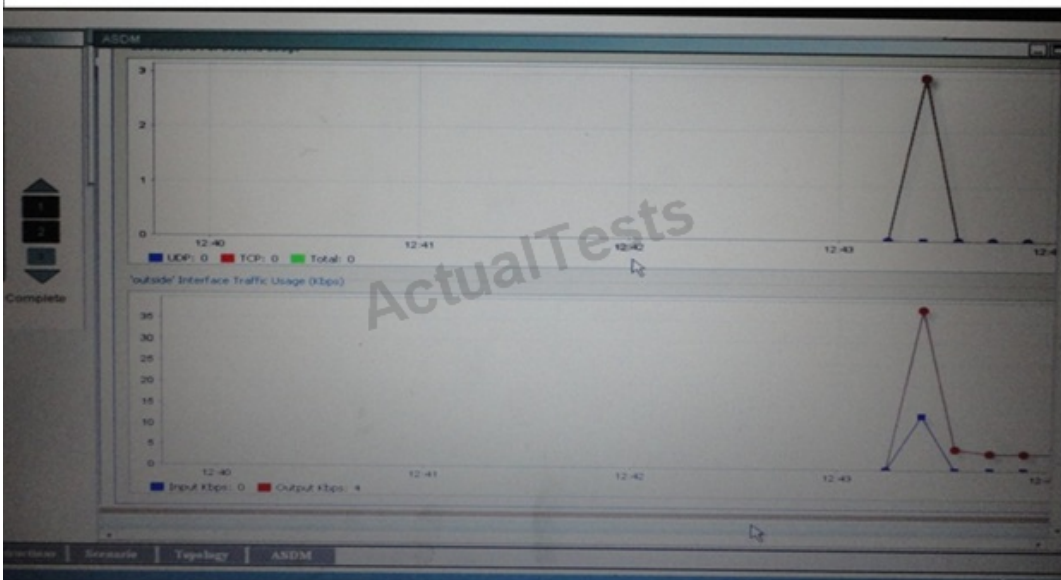
implicit permit ip any any from higher security level interface to lower security level interface

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 127

Scenario: To access Cisco ASDM, click the PC icon in the Topology window, ASDM and answer the following question as:



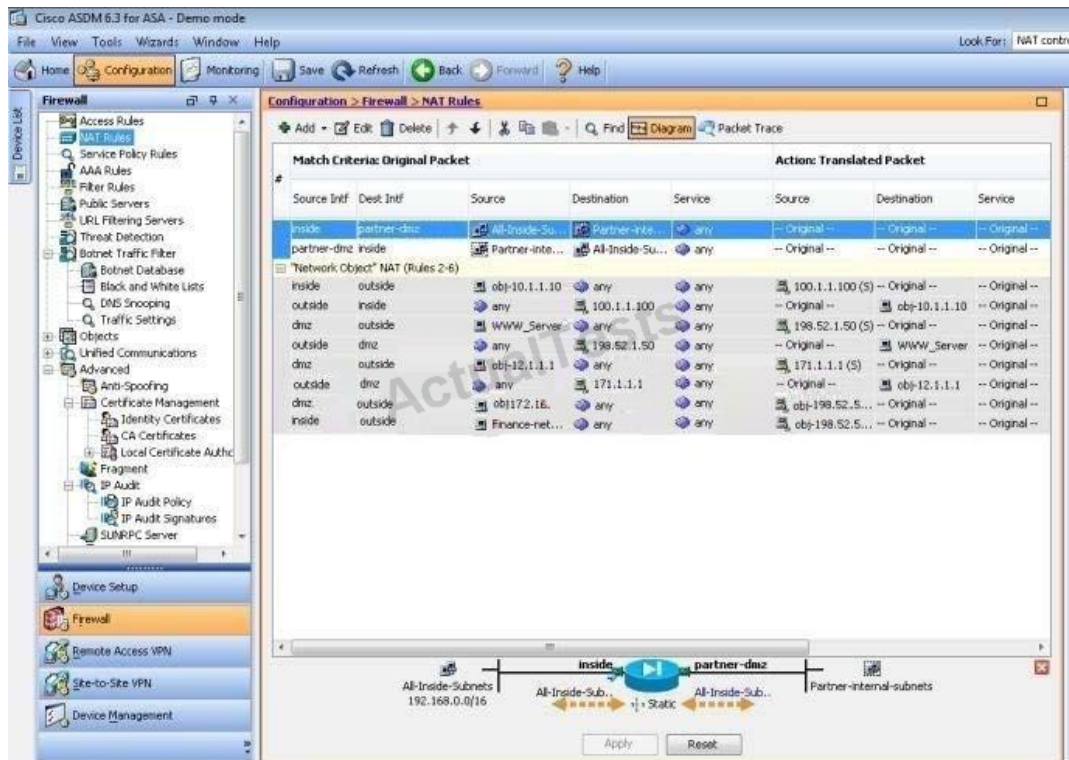


Which statement about the Cisco ASA configuration is true?

- A. All input traffic on the inside interface is denied by the global ACL.
- B. All input and output traffic on the outside interface is denied by the global ACL.
- C. ICMP echo-request traffic is permitted from the inside to the outside, and ICMP echo-reply will be permitted from the outside back to inside.
- D. HTTP inspection is enabled in the global policy.
- E. Traffic between two hosts connected to the same interface is permitted.

Answer: B

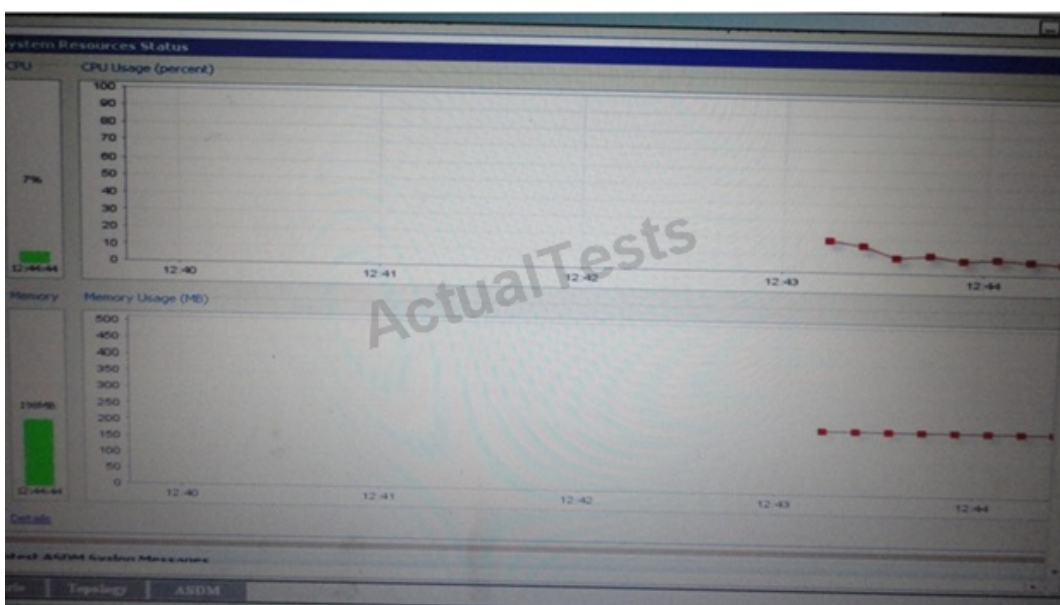
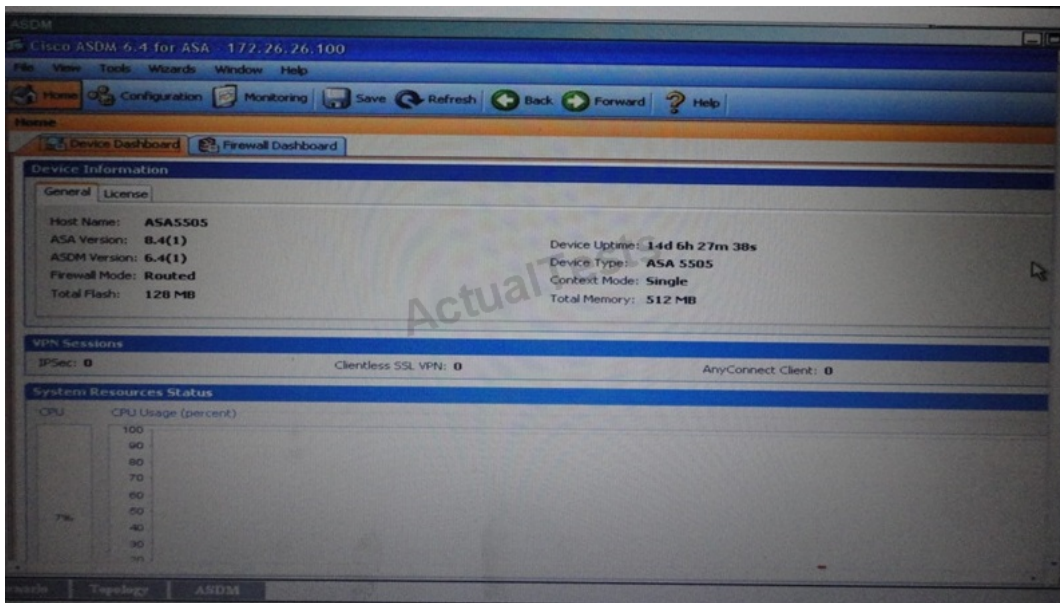
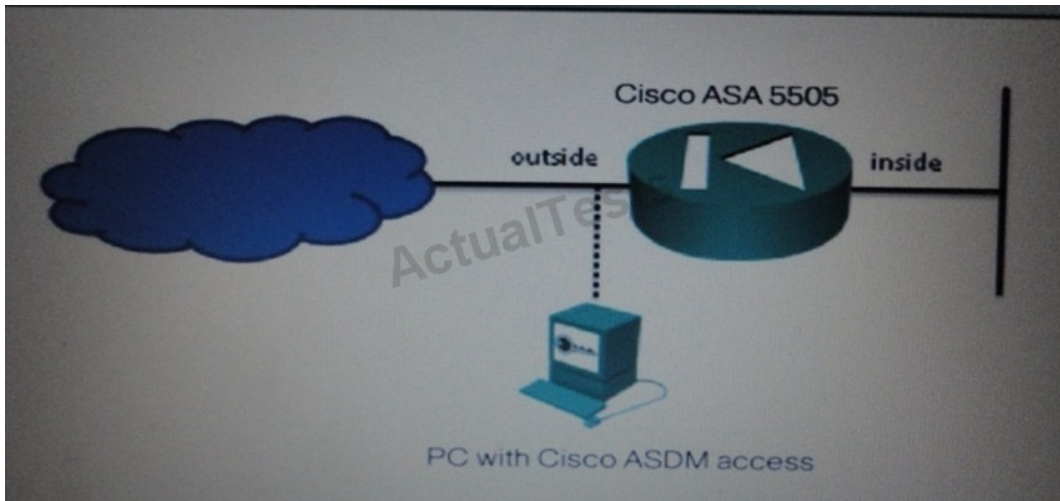
Explanation:

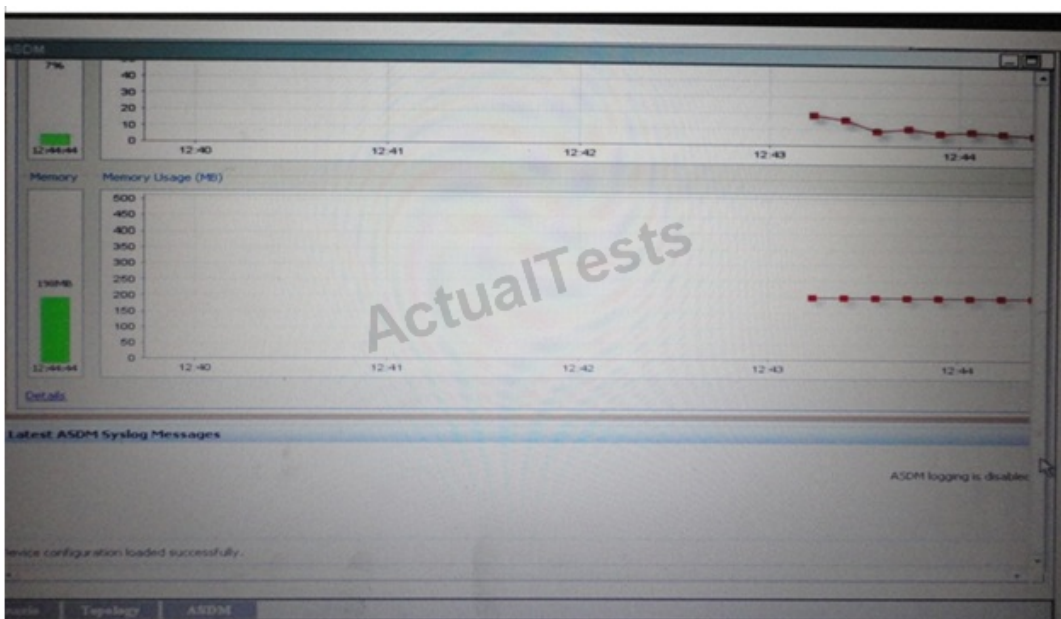
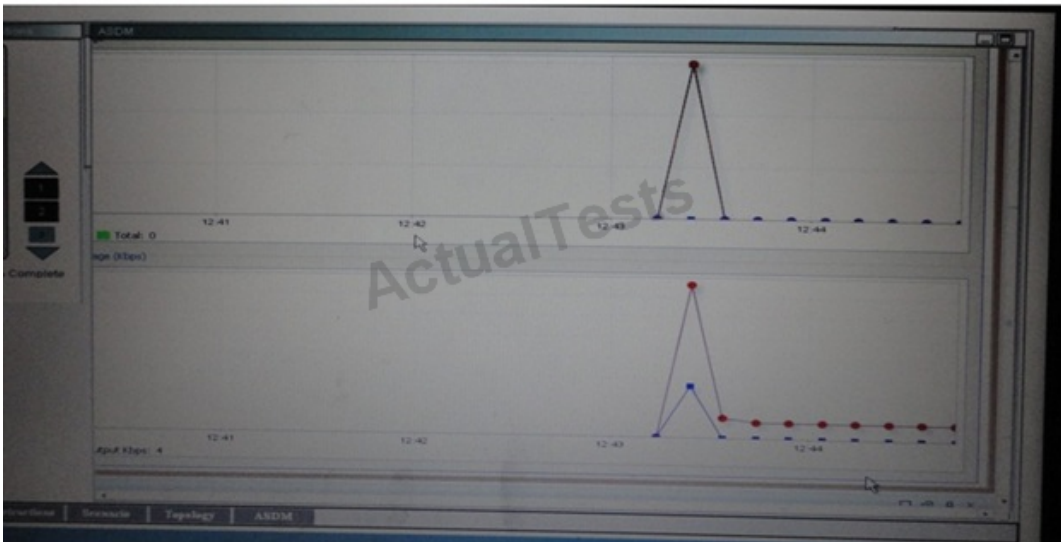
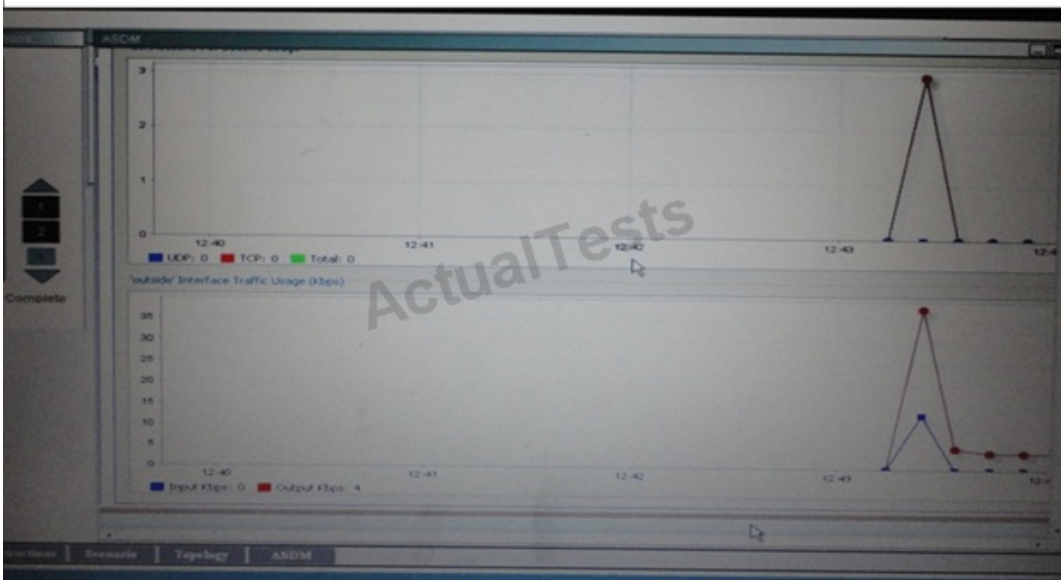


C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 128

Scenario: To access Cisco ASDM, click the PC icon in the Topology window, ASDM and answer the following question as:





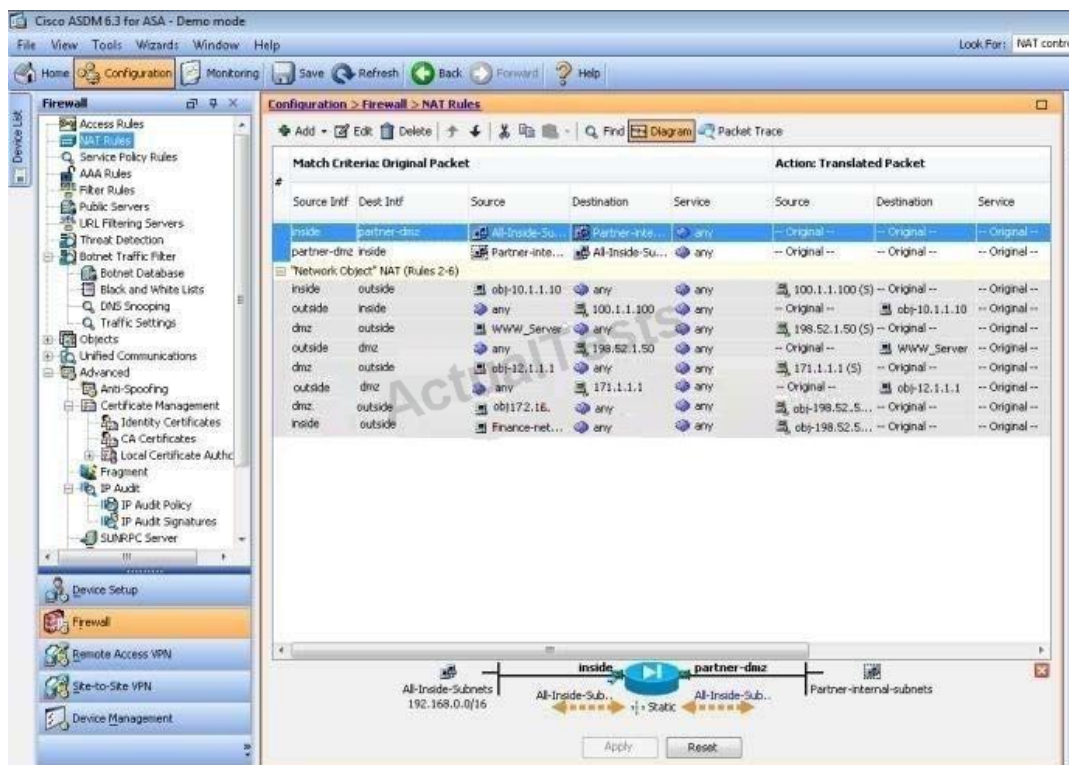
Which two statements about the running configuration of the Cisco ASA are true? (Choose Two)

- A.** The auto NAT configuration causes all traffic arriving on the inside interface destined to any outside destinations to be translated with dynamic port address transmission using the outside interface IP address.
- B.** The Cisco ASA is using the Cisco ASDM image from disk1:/asdm-642.bin
- C.** The Cisco ASA is setup as the DHCP server for hosts that are on the inside and outside interfaces.
- D.** SSH and Cisco ASDM access to the Cisco ASA requires AAA authentication using the LOCAL user database.
- E.** The Cisco ASA is using a persistent self-signed certified so users can authenticate the Cisco ASA when accessing it via ASDM

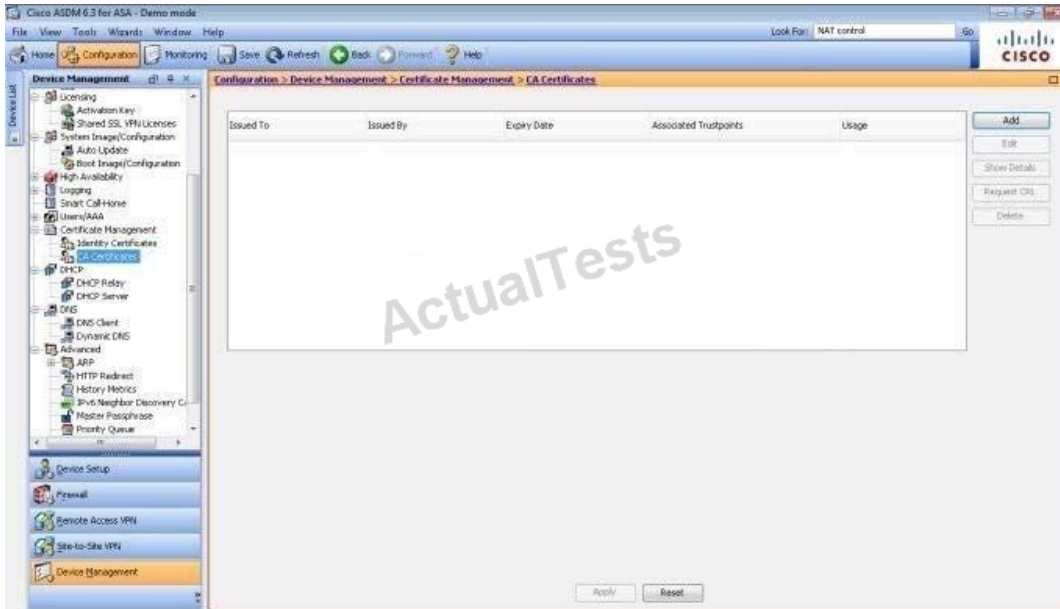
Answer: A,E

Explanation:

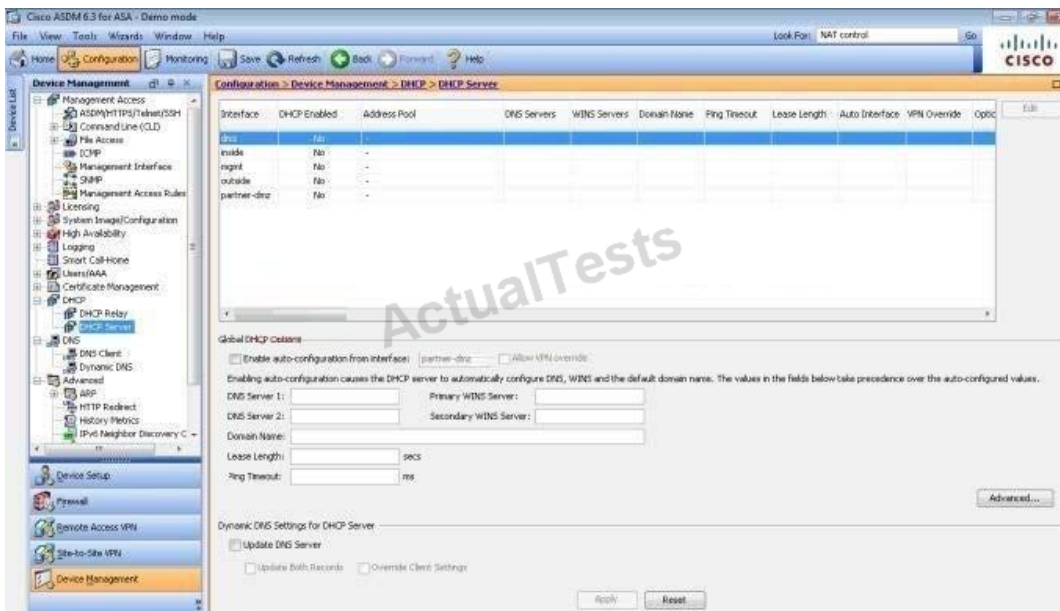
Have to check each and every setting -- expect different results for different exams



C:\Documents and Settings\user-nwz\Desktop\1.JPG



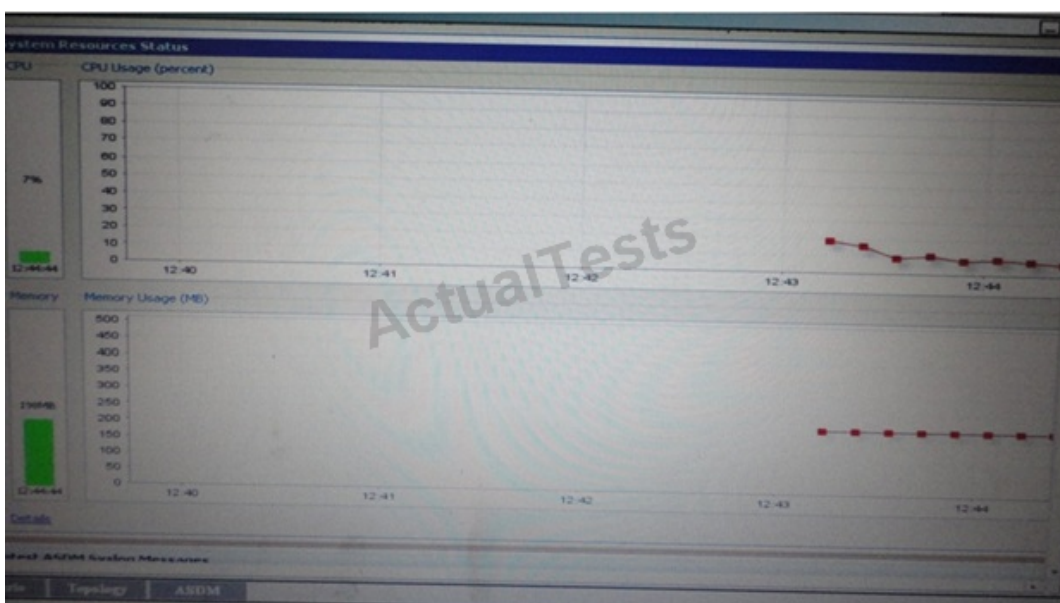
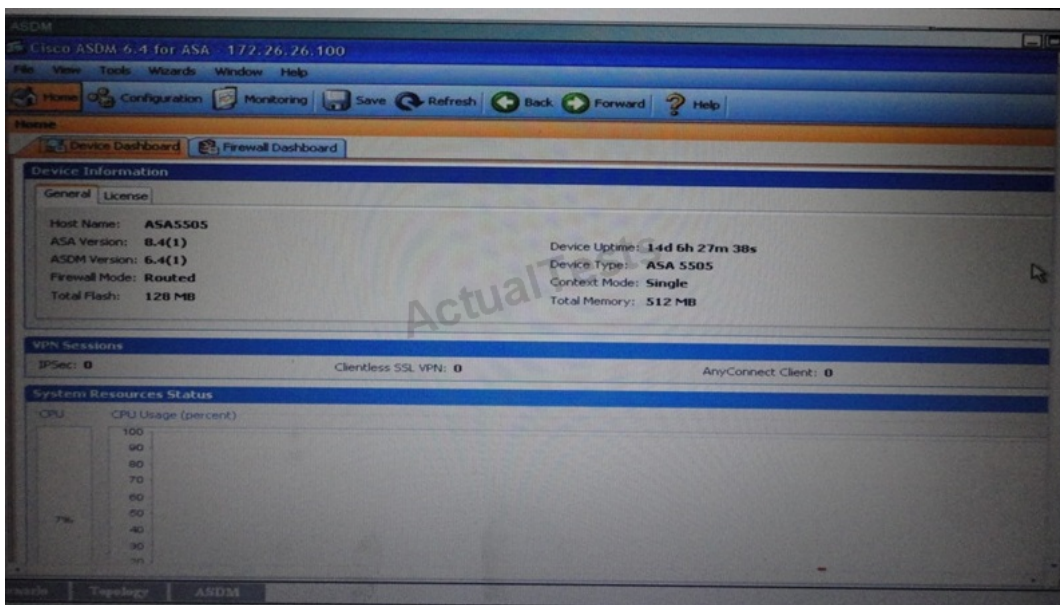
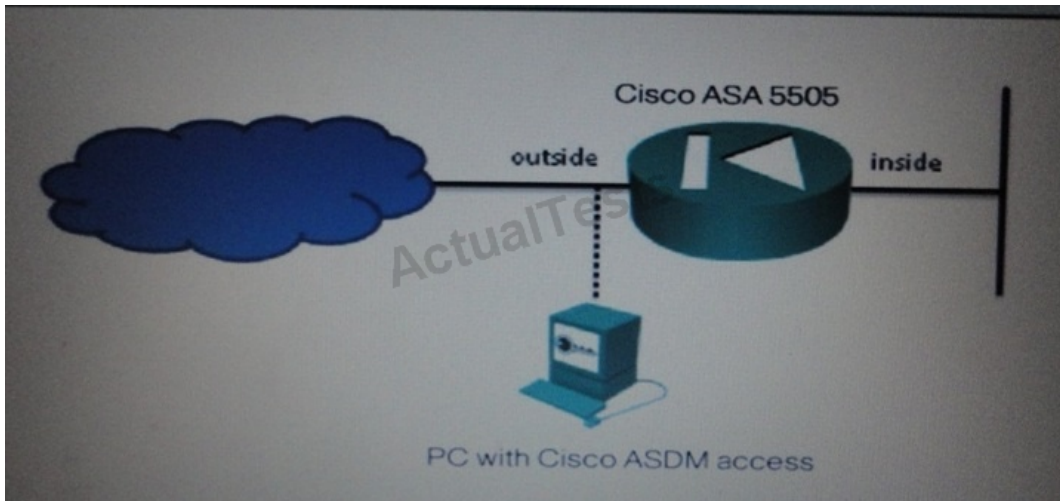
C:\Documents and Settings\user-nwz\Desktop\1.JPG

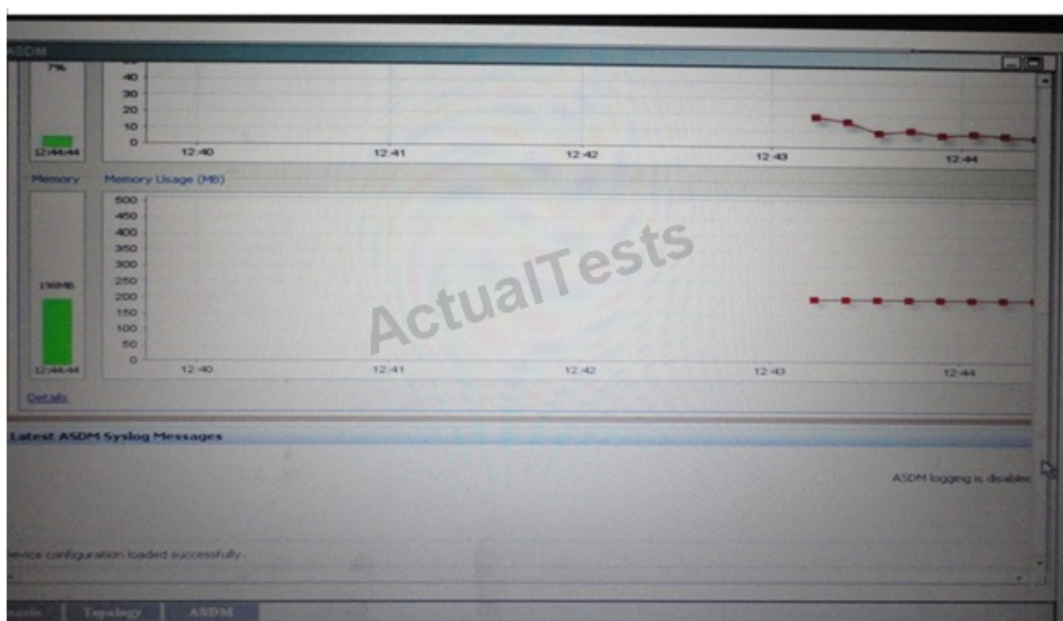
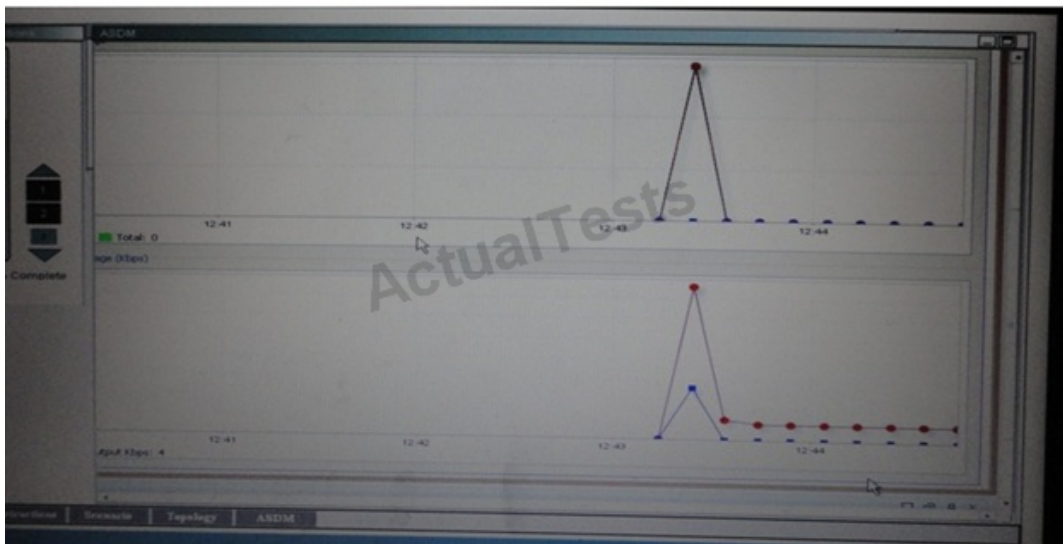
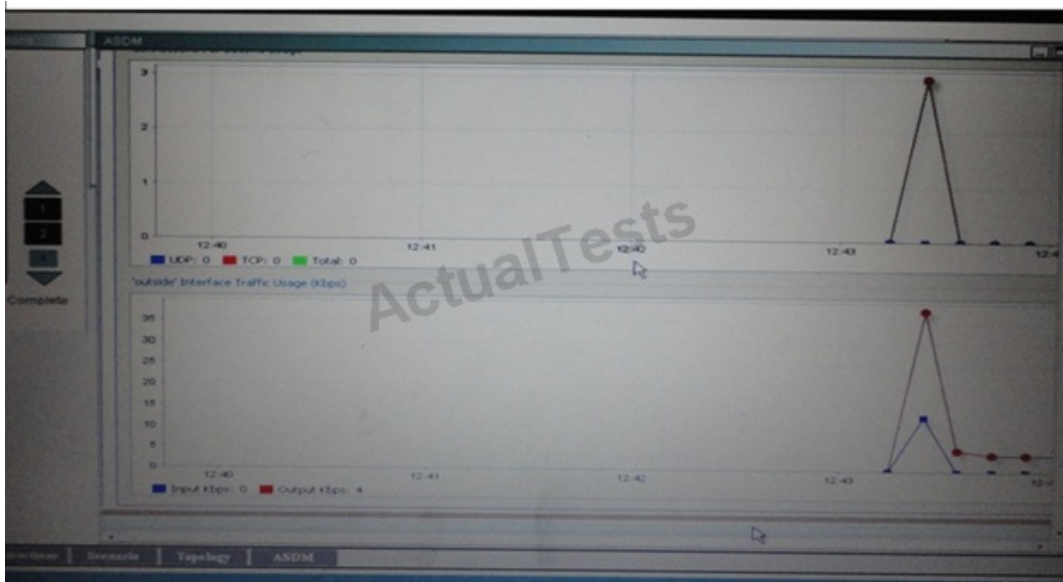


C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 129

Scenario: To access Cisco ASDM, click the PC icon in the Topology window, ASDM and answer the following question as:





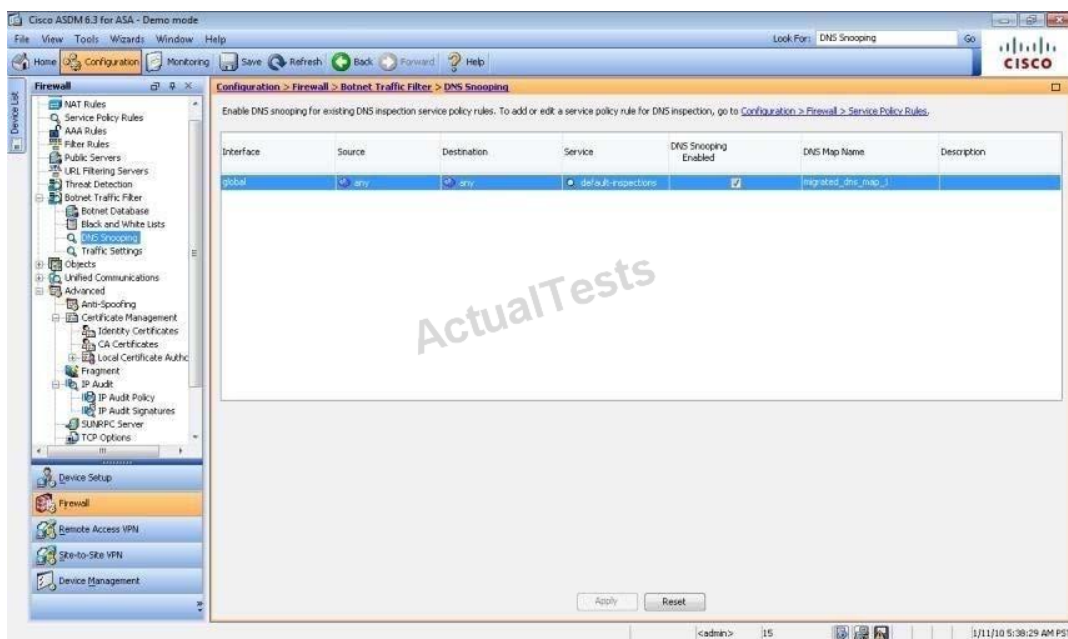
The Cisco ASA administration must enable the Cisco ASA to automatically drop suspicious botnet traffic. After the Cisco ASA administrator entered the initial configuration, the Cisco ASA is not automatically dropping the suspicious botnet traffic. What else must be enabled in order to make it work?

- A. DNS snooping
- B. Botnet traffic filtering on atleast one of the Cisco ASA interface.
- C. Periodic download of the dynamic botnet database from Cisco.
- D. DNS inspection in the global policy.
- E. Manual botnet black and white lists.

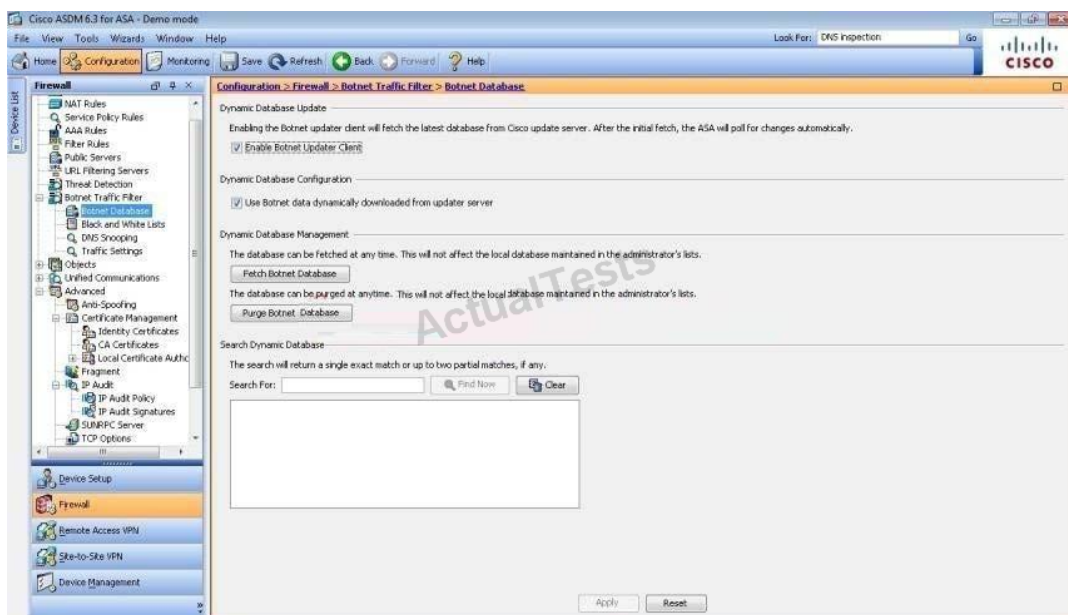
Answer: A

Explanation:

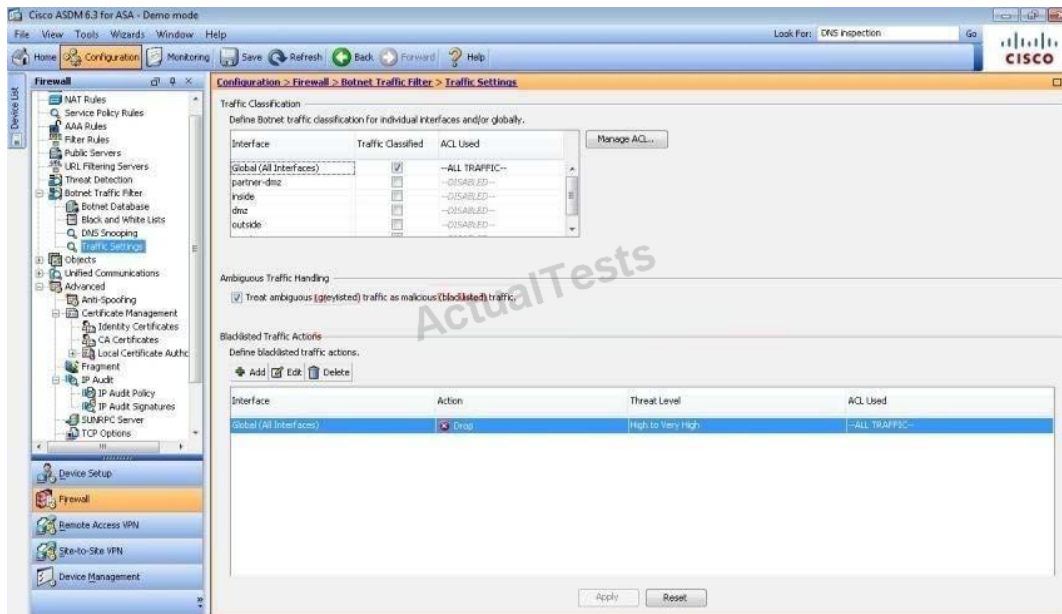
Just check all the following settings - certain they will change from time to time



C:\Documents and Settings\user-nw\Desktop\1.JPG



C:\Documents and Settings\user-nwz\Desktop\1.JPG



C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 130 CORRECT TEXT**Instructions**

This item contains a simulation task. Refer to the scenario and topology before you start. When you are ready, open the Topology window and click the required device to open the GUI window on a virtual terminal. Scroll to view all parts of the Cisco ASDM screens.

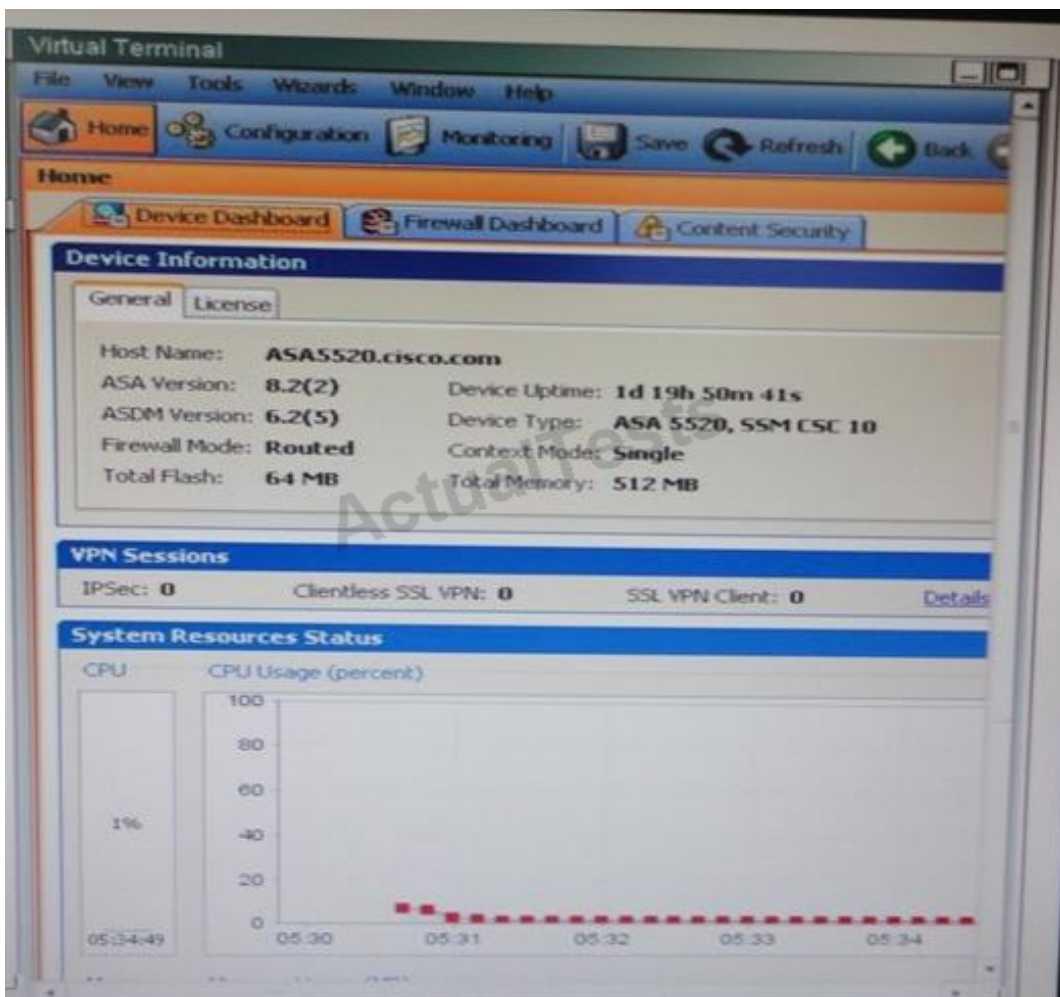
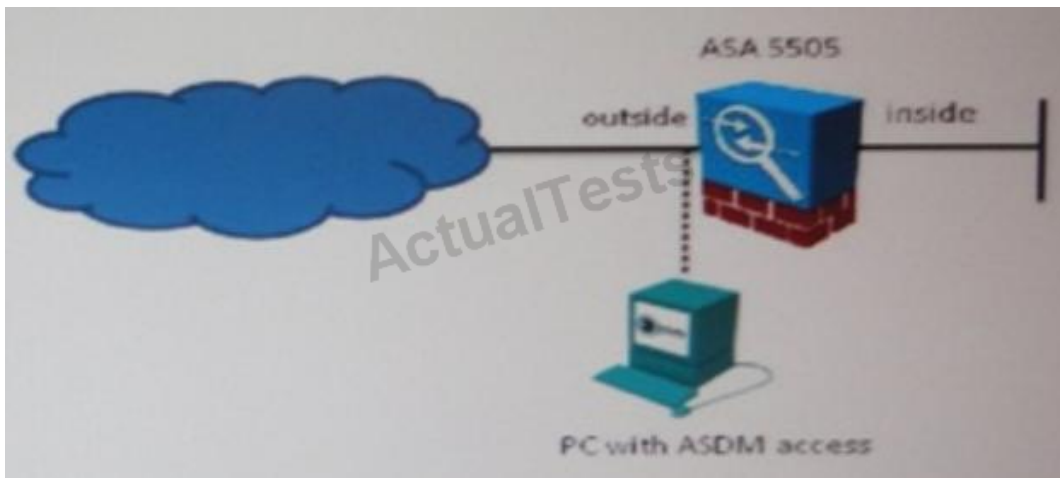
Scenario

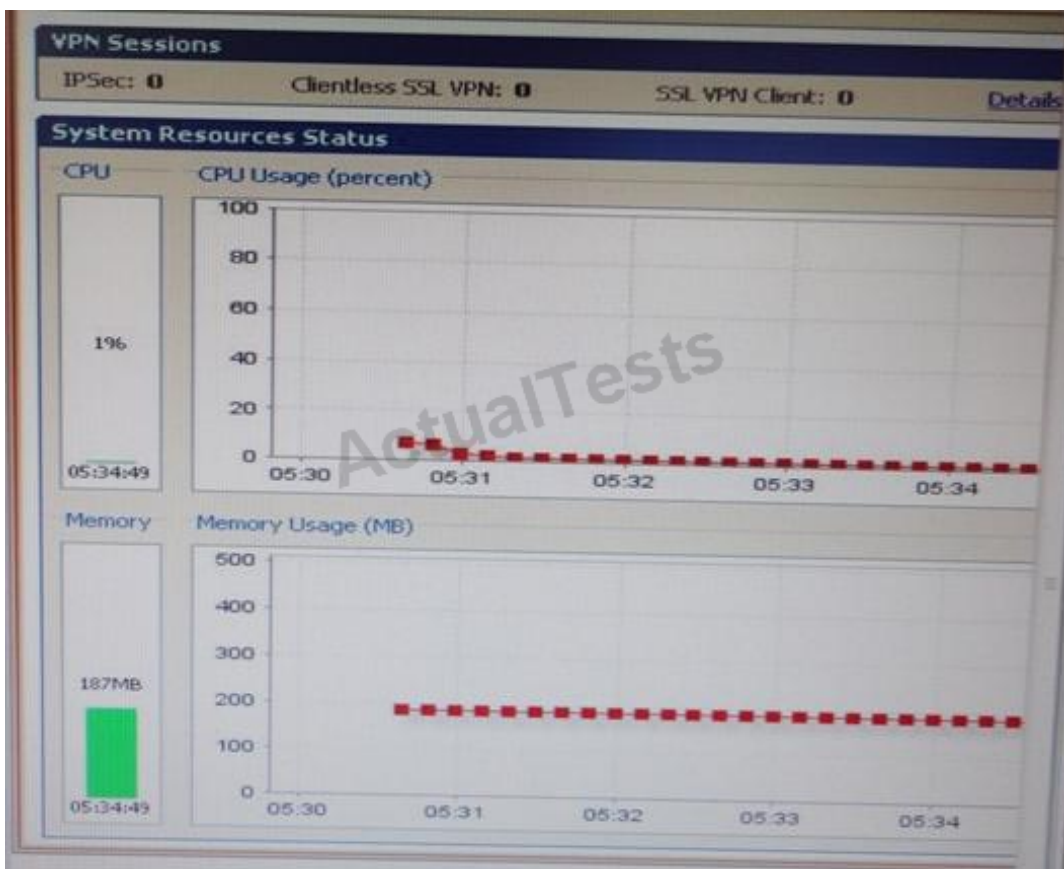
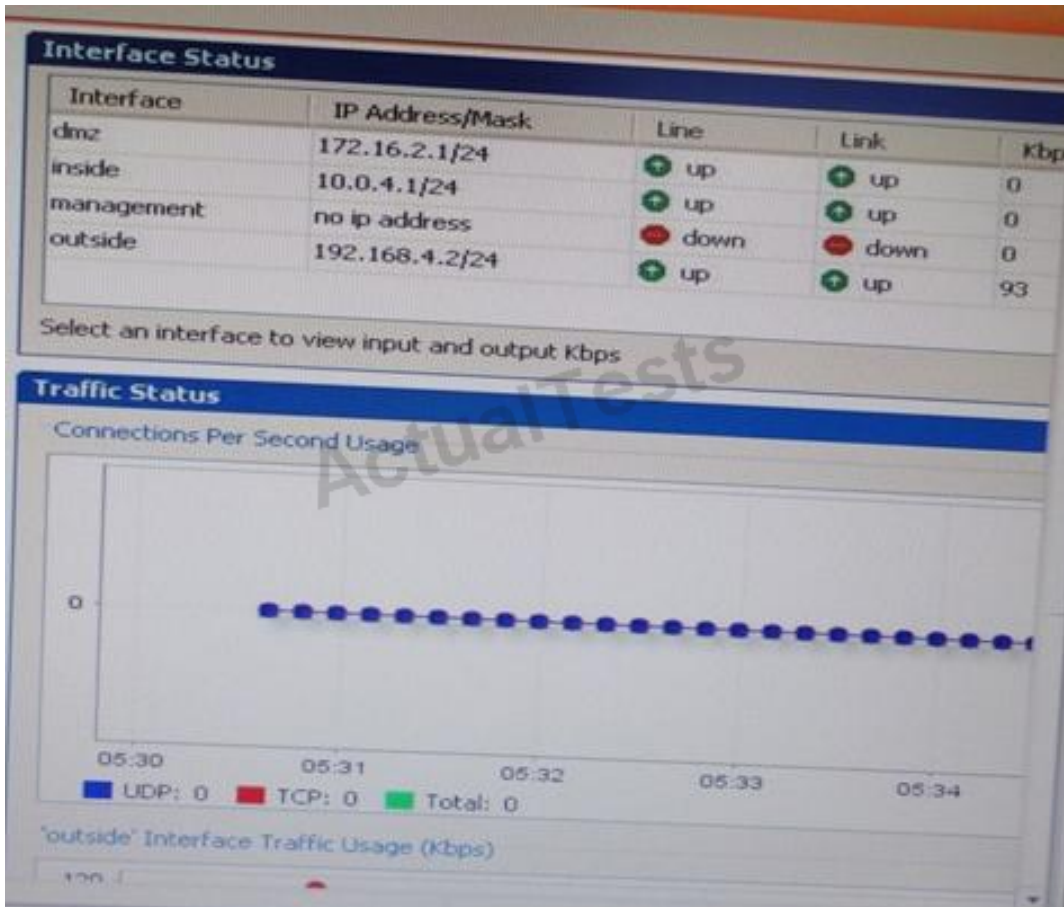
Click the PC icon to launch Cisco ASDM. You have access to a Cisco ASA 5505 via Cisco ASDM. Use Cisco ASDM to edit the Cisco ASA 5505 configurations to enable Advanced HTTP Application inspection by completing the following tasks:

1. Enable HTTP inspection globally on the Cisco ASA
2. Create a new HTTP inspect Map named: http-inspect-map to:
 - a. Enable the dropping of any HTTP connections that encounter HTTP protocol violations
 - b. Enable the dropping and logging of any HTTP connections when the content type in the HTTP response does not match one of the MIME types in the accept filed of the HTTP request

Note: In the simulation, you will not be able to test the HTTP inspection policy after you complete your configuration. Not all Cisco ASDM screens are fully functional.

After you complete the configuration, you do not need to save the running configuration to the start-up config, you will not be able to test the HTTP inspection policy that is created after you complete your configuration. Also not all the ASDM screens are fully functional.







Answer: Here are the step by step Solution for this:

Explanation:

- 1.>Go to Configuration>>Firewall>>Objects>>Inspect Maps>>HTTP>>Add>>Add name "http-inspect-map">>click on detail>>
 - a. select "check for protocol violations"
 - b. Action: Drop connection
 - c. Log: Enable
 - d. Click on Inspection: Click Add
 - e. Select Single Match>>Match type: No Match
 - f. Criterion: response header field
 - g. Field: Predefined: Content type
 - h. value: Content type
 - i. Action: Drop connection
 - j. Log: Enable
 - h. ok>>>ok>>>Apply

HTTP inspection is disabled in global policy by default - we need to enable and use this Inspect Map

Achieve this through command line:

policy-map type inspect http http-inspect-map
parameters

protocol-violation action drop-connection

match req-resp content-type mismatch

drop-connection log

policy-map global_policy

class inspaection_default

inspect http http-inspect-map

also you have to edit the global policy to apply this inspection into it.

Add/Edit HTTP Map

The Add/Edit HTTP Map dialog box is accessible as follows:

Configuration > Global Objects > Inspect Maps > HTTP > HTTP Inspect Map > Advanced View > Add/Edit

HTTP Inspect

The Add/Edit HTTP Inspect dialog box lets you define the match criterion and value for the HTTP inspect map.

Fields

- Single Match—Specifies that the HTTP inspect has only one match statement.

- Match Type—Specifies whether traffic should match or not match the values.

For example, if No Match is selected on the string "example.com," then any traffic that contains "example.com" is excluded from the class map.

- Criterion—Specifies which criterion of HTTP traffic to match.

- Request/Response Content Type Mismatch—Specifies that the content type in the response must match one of the MIME types in the accept field of the request.

- Request Arguments—Applies the regular expression match to the arguments of the request.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

- Request Body Length—Applies the regular expression match to the body of the request with field length greater than the bytes specified.

Greater Than Length—Enter a field length value in bytes that request field lengths will be matched against.

- Request Body—Applies the regular expression match to the body of the request.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Request Header Field Count—Applies the regular expression match to the header of the request with a maximum number of header fields.

Predefined—Specifies the request header fields: accept, accept-charset, accept-encoding, accept-language, allow, authorization, cache-control, connection, content-encoding, content-language, content-length, contentlocation, content-md5, content-range, content-type, cookie, date, expect, expires, from, host, if-match, ifmodified-since, if-none-match, if-range, if-unmodified-since, last-modified, max-forwards, pragma, proxyauthorization, range, referer, te, trailer, transfer-encoding, upgrade, user-agent, via, warning.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Greater Than Count—Enter the maximum number of header fields.

–Request Header Field Length—Applies the regular expression match to the header of the request with field length greater than the bytes specified.

Predefined—Specifies the request header fields: accept, accept-charset, accept-encoding, accept-language, allow, authorization, cache-control, connection, content-encoding, content-language, content-length, contentlocation, content-md5, content-range, content-type, cookie, date, expect, expires, from, host, if-match, ifmodified-since, if-none-match, if-range, if-unmodified-since, last-modified, max-forwards, pragma, proxyauthorization, range, referer, te, trailer, transfer-encoding, upgrade, user-agent, via, warning.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Greater Than Length—Enter a field length value in bytes that request field lengths will be matched against.

–Request Header Field—Applies the regular expression match to the header of the request.

Predefined—Specifies the request header fields: accept, accept-charset, accept-encoding, accept-language, allow, authorization, cache-control, connection, content-encoding, content-language, content-length, contentlocation, content-md5, content-range, content-type, cookie, date, expect, expires, from, host, if-match, ifmodified-since, if-none-match, if-range, if-unmodified-since, last-modified, max-forwards, pragma, proxyauthorization, range, referer, te, trailer, transfer-encoding, upgrade, user-agent, via, warning.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Request Header Count—Applies the regular expression match to the header of the request with a maximum number of headers.

Greater Than Count—Enter the maximum number of headers.

–Request Header Length—Applies the regular expression match to the header of the request with length greater than the bytes specified.

Greater Than Length—Enter a header length value in bytes.

–Request Header non-ASCII—Matches non-ASCII characters in the header of the request.

–Request Method—Applies the regular expression match to the method of the request.

Method—Specifies to match on a request method: bcopy, bdelete, bmove, bpropfind, bproppatch, connect, copy, delete, edit, get, getattribute, getattributenames, getproperties, head, index, lock, mkcol, mkdir, move, notify, options, poll, post, propfind, proppatch, put, revadd, revlabel, revlog, revnum, save, search, setattribute, startrev, stoprev, subscribe, trace, unedit, unlock, unsubscribe.

Regular Expression—Specifies to match on a regular expression.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Request URI Length—Applies the regular expression match to the URI of the request with length greater than the bytes specified.

Greater Than Length—Enter a URI length value in bytes.

–Request URI—Applies the regular expression match to the URI of the request.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Response Body—Applies the regex match to the body of the response.

ActiveX—Specifies to match on ActiveX.

Java Applet—Specifies to match on a Java Applet.

Regular Expression—Specifies to match on a regular expression.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular

expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Response Body Length—Applies the regular expression match to the body of the response with field length greater than the bytes specified.

Greater Than Length—Enter a field length value in bytes that response field lengths will be matched against.

–Response Header Field Count—Applies the regular expression match to the header of the response with a maximum number of header fields.

Predefined—Specifies the response header fields: accept-ranges, age, allow, cache-control, connection, content-encoding, content-language, content-length, content-location, content-md5, content-range, content-type, date, etag, expires, last-modified, location, pragma, proxy-authenticate, retry-after, server, set-cookie, trailer, transfer-encoding, upgrade, vary, via, warning, www-authenticate.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Greater Than Count—Enter the maximum number of header fields.

–Response Header Field Length—Applies the regular expression match to the header of the response with field length greater than the bytes specified.

Predefined—Specifies the response header fields: accept-ranges, age, allow, cache-control, connection, content-encoding, content-language, content-length, content-location, content-md5, content-range, content-type, date, etag, expires, last-modified, location, pragma, proxy-authenticate, retry-after, server, set-cookie, trailer, transfer-encoding, upgrade, vary, via, warning, www-authenticate.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Greater Than Length—Enter a field length value in bytes that response field lengths will be matched against.

–Response Header Field—Applies the regular expression match to the header of the response.

Predefined—Specifies the response header fields: accept-ranges, age, allow, cache-control, connection, content-encoding, content-language, content-length, content-location, content-md5, content-range, content-type, date, etag, expires, last-modified, location, pragma, proxy-authenticate, retry-after, server, set-cookie, trailer, transfer-encoding, upgrade, vary, via, warning, www-authenticate.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

–Response Header Count—Applies the regular expression match to the header of the response with a maximum number of headers.

Greater Than Count—Enter the maximum number of headers.

–Response Header Length—Applies the regular expression match to the header of the response with length greater than the bytes specified.

Greater Than Length—Enter a header length value in bytes.

–Response Header non-ASCII—Matches non-ASCII characters in the header of the response.

–Response Status Line—Applies the regular expression match to the status line.

Regular Expression—Lists the defined regular expressions to match.

Manage—Opens the Manage Regular Expressions dialog box, which lets you configure regular expressions.

Regular Expression Class—Lists the defined regular expression classes to match.

Manage—Opens the Manage Regular Expression Class dialog box, which lets you configure regular expression class maps.

- Multiple Matches—Specifies multiple matches for the HTTP inspection.

–H323 Traffic Class—Specifies the HTTP traffic class match.

–Manage—Opens the Manage HTTP Class Maps dialog box to add, edit, or delete HTTP Class Maps.

- Action—Drop connection, reset, or log.

- Log—Enable or disable.

NOTE:

http://www.cisco.com/en/US/docs/security/asa/asa83/asdm63/configuration_guide/inspect_basic.html#wp1144259

and/or

http://www.cisco.com/en/US/products/ps6120/products_tech_note09186a0080b84568.shtml

Through achieve this command line:

```
policy-map type inspect http http-inspect-map
```

```
parameters
```

```
protocol-violation action drop-connection log
```

```
policy-map type inspect http http-inspect-map
```

```
match not response header content-type application/msword
```

```
drop-connection log
```

New Questions

QUESTIONNO:131

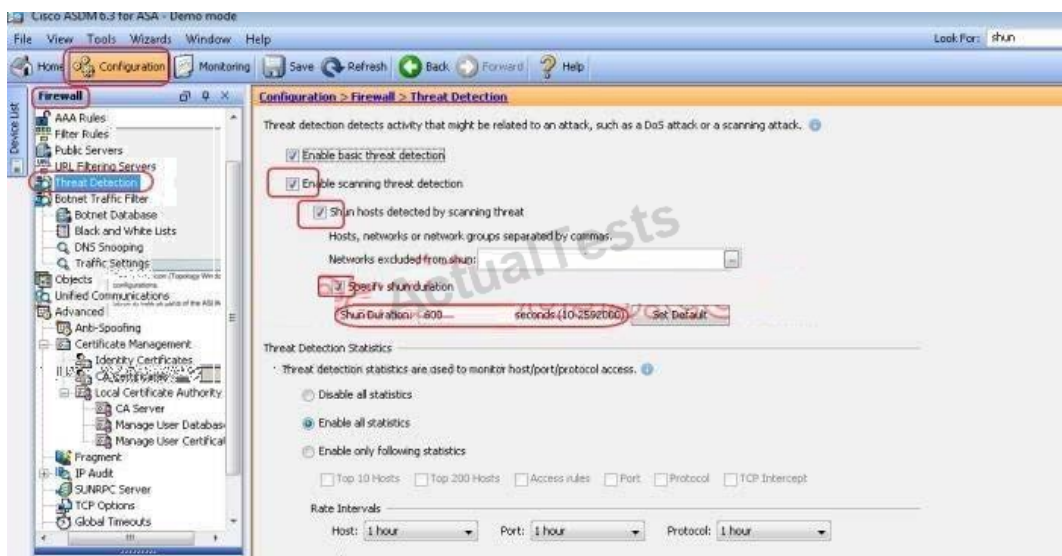
When the Cisco ASA detects scanning attacks, how long is the attacker who is performing the scan shunned?



C:\Documents and Settings\user-nwz\Desktop\1.JPG

- A. 120 seconds
- B. 600 seconds
- C. 1200 seconds
- D. 3600 seconds
- E. 6000 seconds

Answer: B



C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 131

By default, which access rule is applied inbound to the inside interface?

- A. All IP traffic is denied.
- B. All IP traffic is permitted.
- C. All IP traffic sourced from any source to any less secure network destinations is permitted.
- D. All IP traffic sourced from any source to any more secure network destinations is permitted

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/access_rules.html#wp1083496

Implicit Permits

For routed mode, the following types of traffic are allowed through by default:

- IPv4 traffic from a higher security interface to a lower security interface.
- IPv6 traffic from a higher security interface to a lower security interface.

Note These defaults might not be true if you have configured a global access rule.

For transparent mode, the following types of traffic are allowed through by default:

- IPv4 traffic from a higher security interface to a lower security interface.
- IPv6 traffic from a higher security interface to a lower security interface.
- ARPs in both directions

QUESTION NO: 132

Refer to the Exhibit.

#	Type	Original Source	Destination	Service	Translated Interface	Address	Service
dmz_email (1 Static rules, 1 Dynamic rules)							
1	Static	dmz_emailserver		smtp	outside	public_email_address	smtp
2	Dynamic	dmz_emailserver			outside	public_email_address	
dmz_web (1 Static rules, 1 Dynamic rules)							
1	Static	dmz_webserver		http	outside	public_web_address	http
2	Dynamic	dmz_webserver			outside	public_web_address	
guest (1 Dynamic rules)							
1	Dynamic	guest-network/24			outside	outside	
inside (1 Dynamic rules)							
1	Dynamic	inside-network/24			outside	outside	

Which statement about the NAT/PAT configuration is true?

- A. Dynamic PAT is used for any traffic that is sourced from the dmz_emailserver to the outside
- B. Dynamic PAT is used for any traffic that is sourced from any host on the inside network to the outside
- C. Static NAT is used for any traffic that is sourced from the dmz_emailserver to the outside
- D. Static PAT is used for any traffic that is sourced from the dmz_emailserver to the outside
- E. Dynamic NAT is used for any traffic that is sourced from the dmz_emailserver to the outside
- F. Dynamic NAT is used for any traffic that is sourced from and host on the guest-network to the outside

Answer: B

Explanation:

QUESTION NO: 133

Which Cisco ASA platform should be selected if the requirements are to support 35,000 connections per second, 600,000 maximum connections, and traffic shaping?

- A. 5540
- B. 5550
- C. 5580-20
- D. 5580-40

Answer: B

Explanation:

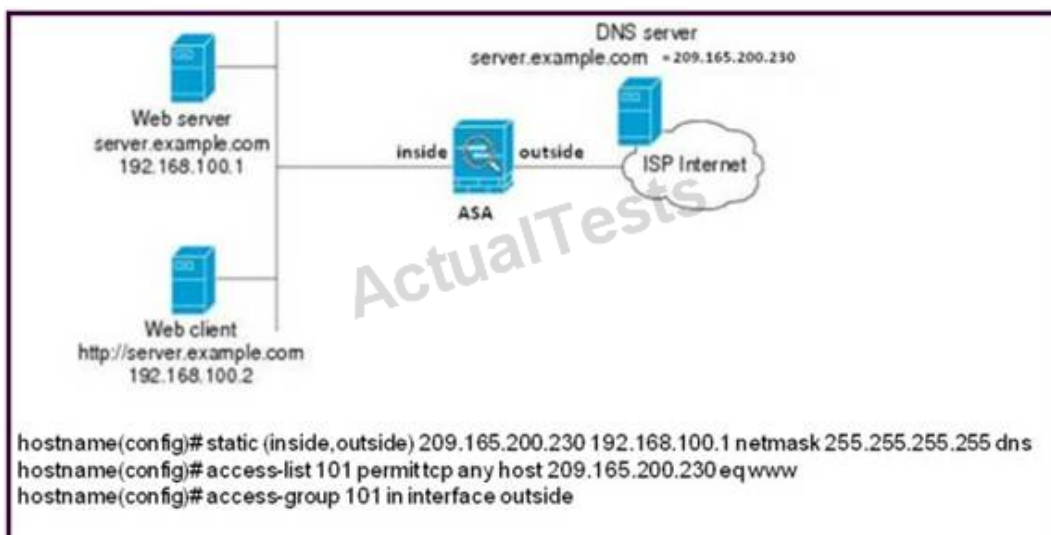
http://www.cisco.com/en/US/prod/collateral/vpndevc/ps6032/ps6094/ps6120/product_data_sheet0900aecd802930c5.pdf

Feature	Cisco ASA 5540	Cisco ASA 5550	Cisco ASA 5580-20	Cisco ASA 5580-40
				
Maximum Firewall and IPS Throughput	- Up to 500 Mbps with AIP-SSM-20 - Up to 650 Mbps with AIP-SSM-40	Not available	Unlimited	Unlimited
3DES/AES VPN Throughput	Up to 325 Mbps	Up to 425 Mbps	10 Gbps (Max); 5 Gbps (Real-world HTTP)	20 Gbps (Max); 10 Gbps (Real-world HTTP)
IPsec VPN Peers	5000	5000	10,000	10,000
Premium AnyConnect VPN Peers (Included/Maximum)	2/2500	2/5000	2/10,000	2/10,000
Concurrent Connections	400,000	650,000	2,000,000	4,000,000
New Connections/Second	25,000	33,000	90,000	150,000
Integrated Network Ports	4 Gigabit Ethernet, 1 Fast Ethernet	8 Gigabit Ethernet, 4 SFP Fiber, 1 Fast Ethernet	2,500,000	4,000,000
			2-port 10/100/1000	2-port 10/100/1000

C:\Documents and Settings\user-nwz\Desktop\1.JPG

QUESTION NO: 134

Refer to the exhibit.



Which two options will result from the Cisco ASA configuration? (Choose two.)

A. The outside hosts can use the 192.168.100.1 IP address to reach the web server on the inside

network.

B. The global IP address of the web server is 209.165.200.230.

C. The inside web client will use the 209.165.200.230 IP address to reach the web server and the Cisco ASA will translate the 209.165.200.230 IP address to the 192.168.100.1 IP address.

D. The Cisco ASA will translate the DNS A-Record reply from the DNS server to any inside client for the web server (web server IP = 192.168.100.1).

E. The web server will be reachable only from the inside.

F. The web server will be reachable only from the outside.

Answer: B,D

Explanation:

http://www.cisco.com/en/US/docs/security/asa/asa84/configuration/guide/inspect_basic.html#wp1350877

When DNS inspection is enabled, DNS rewrite provides full support for NAT of DNS messages originating from any interface.

If a client on an inside network requests DNS resolution of an inside address from a DNS server on an outside interface, the DNS A-record is translated correctly. If the DNS inspection engine is disabled, the A-record is not translated.

As long as DNS inspection remains enabled, you can configure DNS rewrite using the `alias`, `static`, or `nat` commands.

DNS Rewrite performs two functions:

- Translating a public address (the routable or "mapped" address) in a DNS reply to a private address (the "real" address) when the DNS client is on a private interface.
- Translating a private address to a public address when the DNS client is on the public interface.

QUESTION NO: 135

Where in the ACS are the individual downloadable ACL statements configured to achieve the most scalable deployment?

- A.** Group Setup
- B.** User Setup
- C.** Shared Profile Components
- D.** Network Access Profiles
- E.** Network Configuration
- F.** Interface Configuration

Answer: C

Explanation:

The Shared Profile Components section enables you to develop and name reusable, shared sets of authorization components which may be applied to one or more users or groups of users and referenced by name within their profiles. These include network access restrictions (NARs), command authorization sets, and downloadable PIX ACLs.

The Shared Profile Components section of Cisco Secure ACS addresses the scalability of selective authorization. Shared profile components can be configured once and then applied to many users or groups.

Without this ability, flexible and comprehensive authorization could only be accomplished by explicitly configuring the authorization of each user group for each possible command on each possible device. Creating and applying these named shared profile components (access restrictions, command sets, and ACLs) makes it unnecessary to repeatedly enter long lists of devices or commands when defining network access parameters.

Shared profile components also enable Cisco Secure ACS to authorize a command on behalf of another device or devices. Their scalability extends to the following capabilities:

A way to determine the list of commands a user could issue against one or more devices in the network.

A way to determine the list of devices on which a particular user may execute a particular command

http://www.cisco.com/en/US/products/sw/secursw/ps2086/products_user_guide_chapter09186a0080205a4a.html

QUESTION NO: 136

Which two methods can be used to access the Cisco AIP-SSM CLI? (Choose two.)

- A. initiating an SSH connection to the Cisco AIP-SSM external management Ethernet port
- B. connecting to the console port on the Cisco AIP-SSM
- C. using the setup command on the Cisco ASA CLI
- D. using the session 1 command on the Cisco ASA CLI
- E. using the hw-module command on the Cisco ASA CLI

Answer: A,D

Explanation:

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/ssm.html#wp1058664>

And

<http://www.cisco.com/en/US/docs/security/asa/asa72/configuration/guide/intrface.html#wpxref951>
29